



化学测量学—人工智能化学分析

Chapter 8

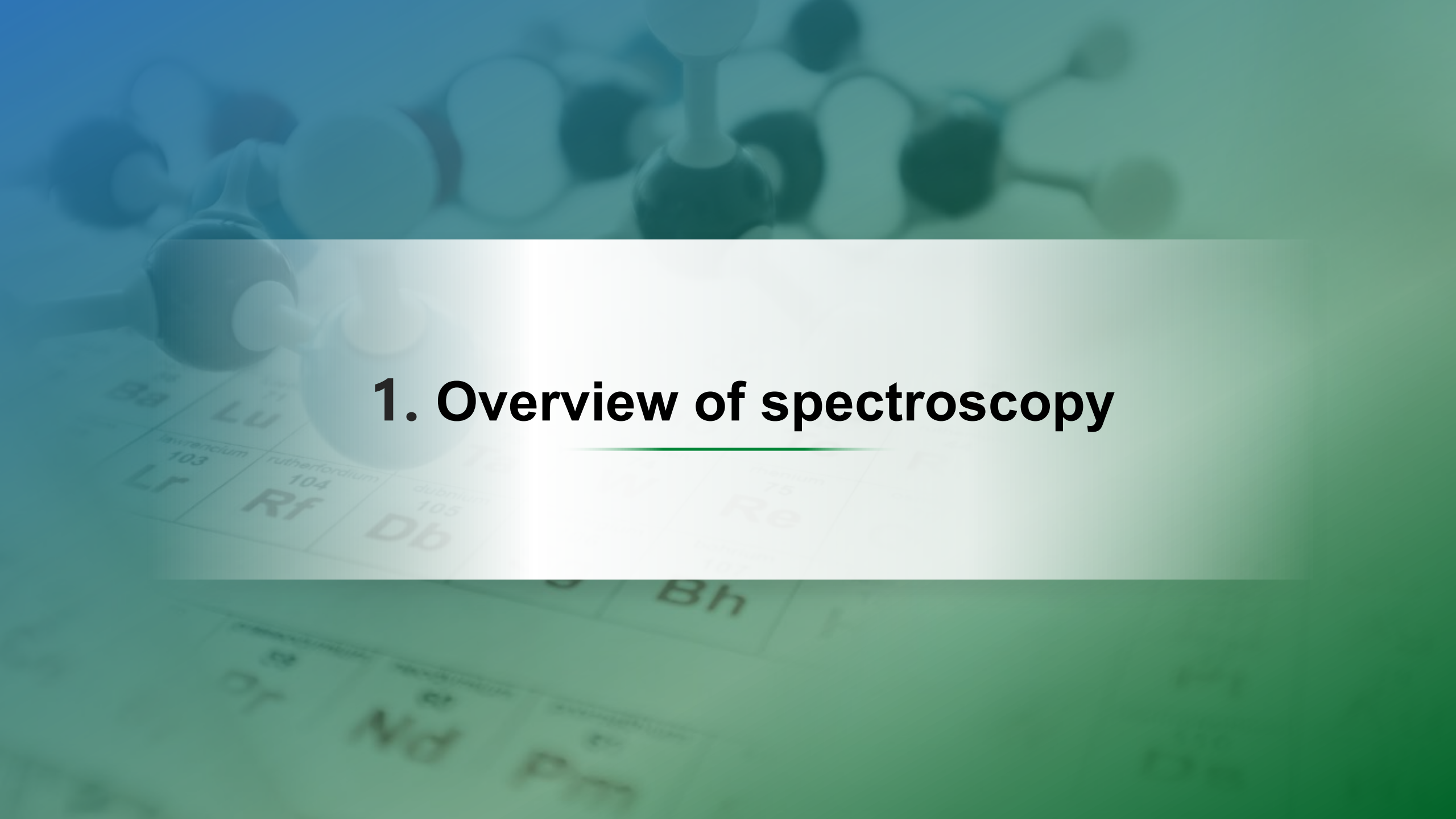
Spectroscopy and spectral analysis

人工智能化学分析

马昊 厦门大学

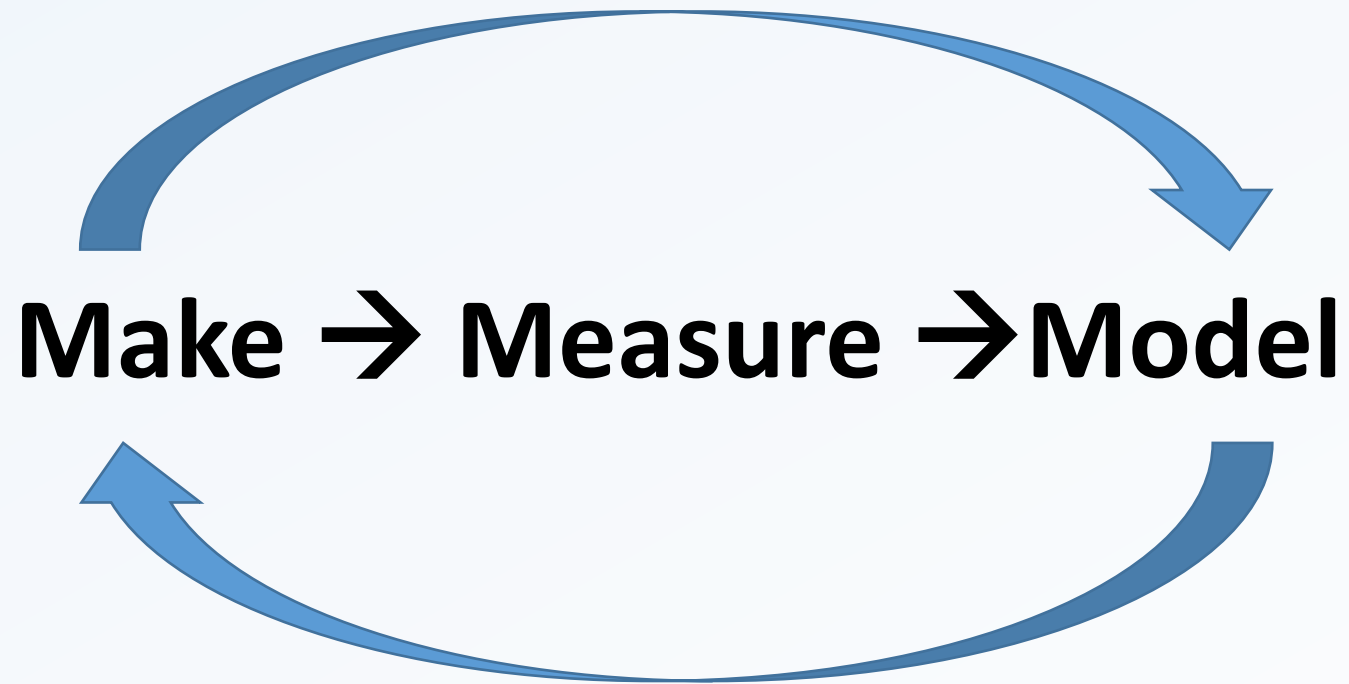
Contents

- 1. Overview of spectroscopy**
- 2. AI for molecular spectroscopy**
- 3. UV, IR, Raman...**
4. MS and NMR...
5. AI for Mapping

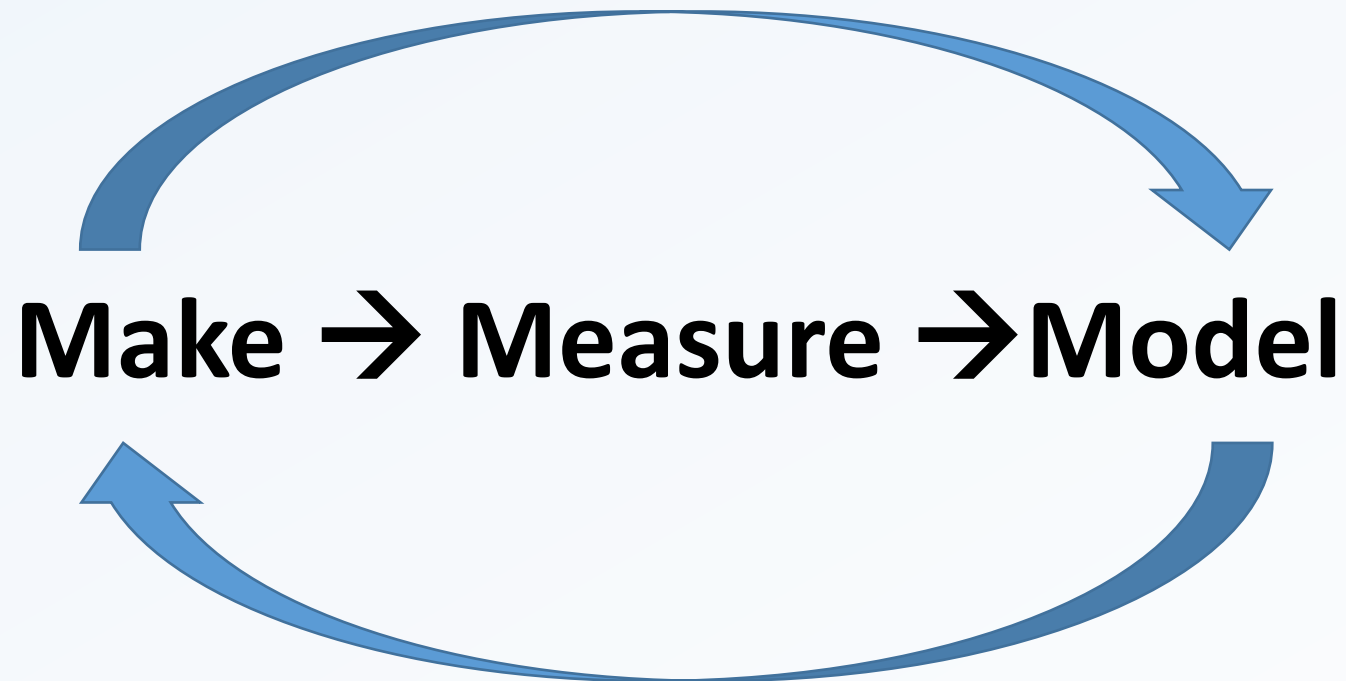


1. Overview of spectroscopy

Chemistry- 3M



Chemistry- 3M

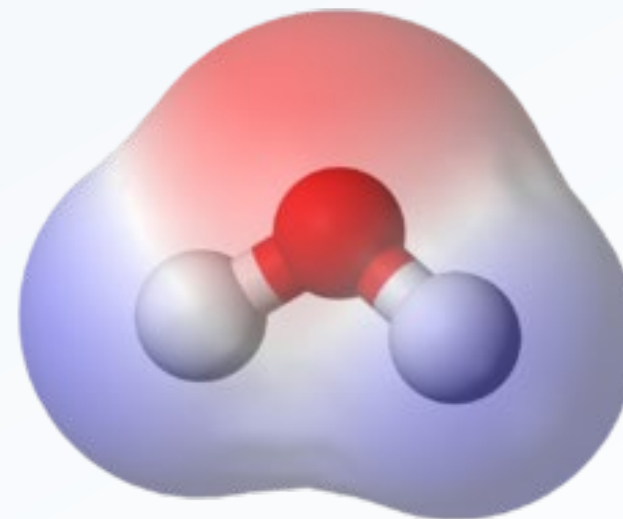
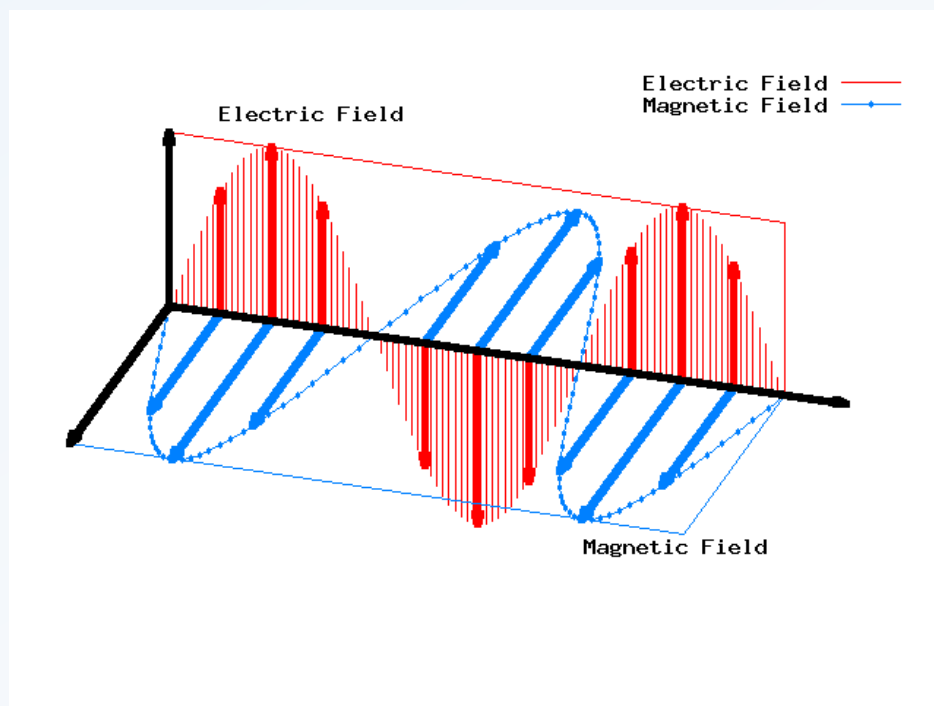


What it is and how many
Qualitative and Quantitative

Overview of spectroscopy

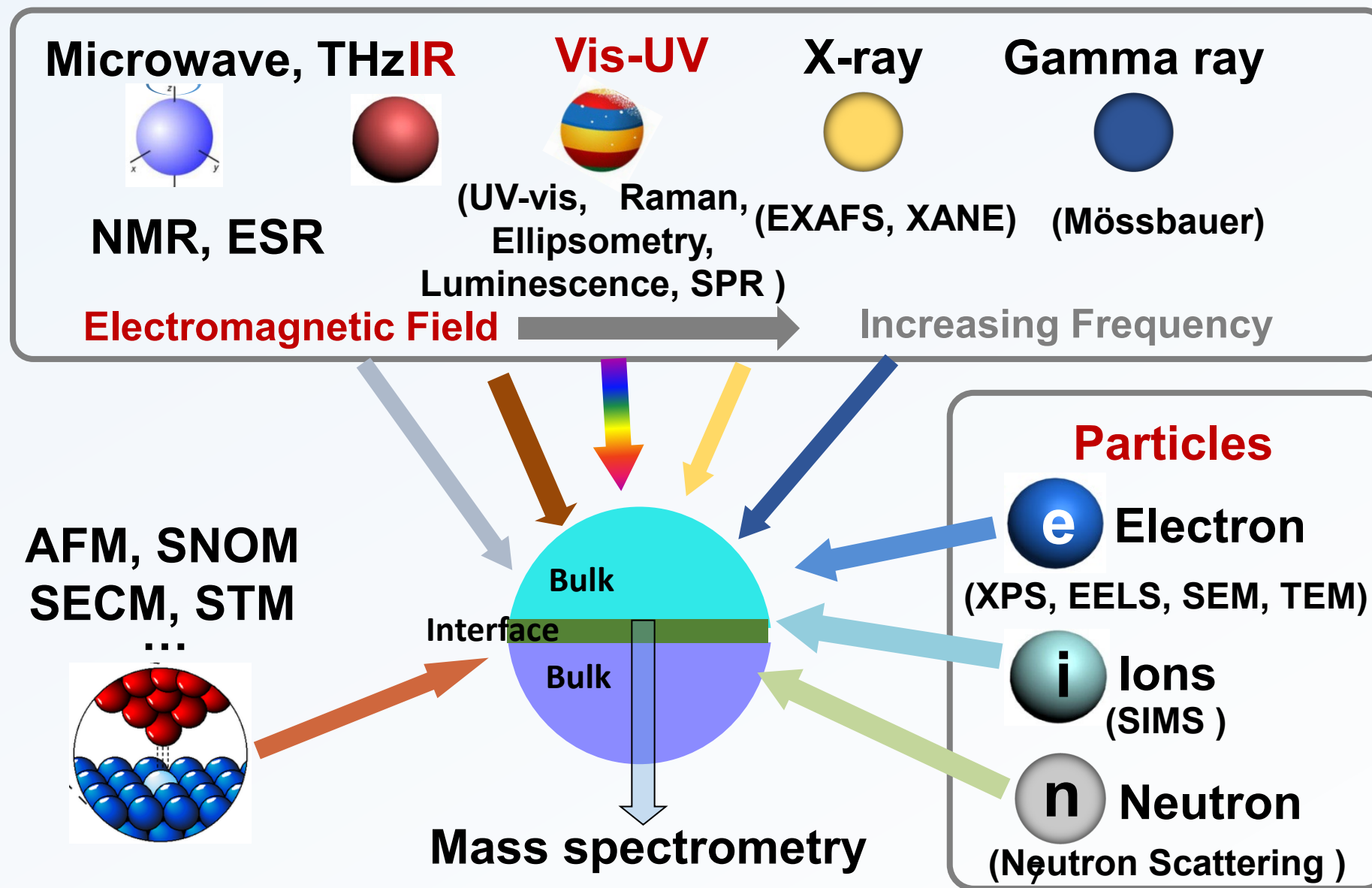
What is spectroscopy?

Latin: “spectron”—ghost or spirit
Greek: “σκοπειν”—to see

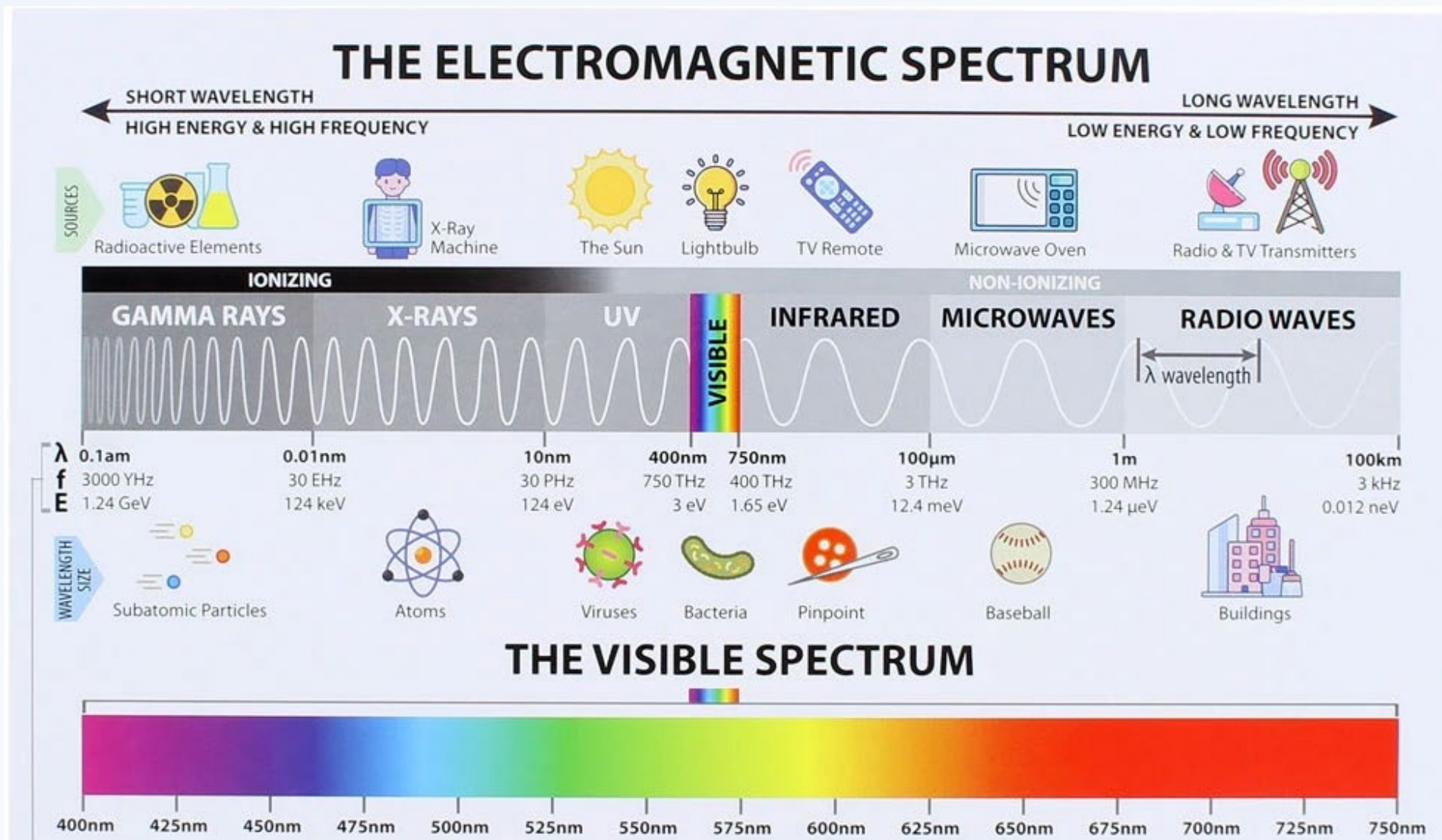


We utilize different spectroscopy to reveal different messages of the molecules, with the aid of interactions between molecules and light.

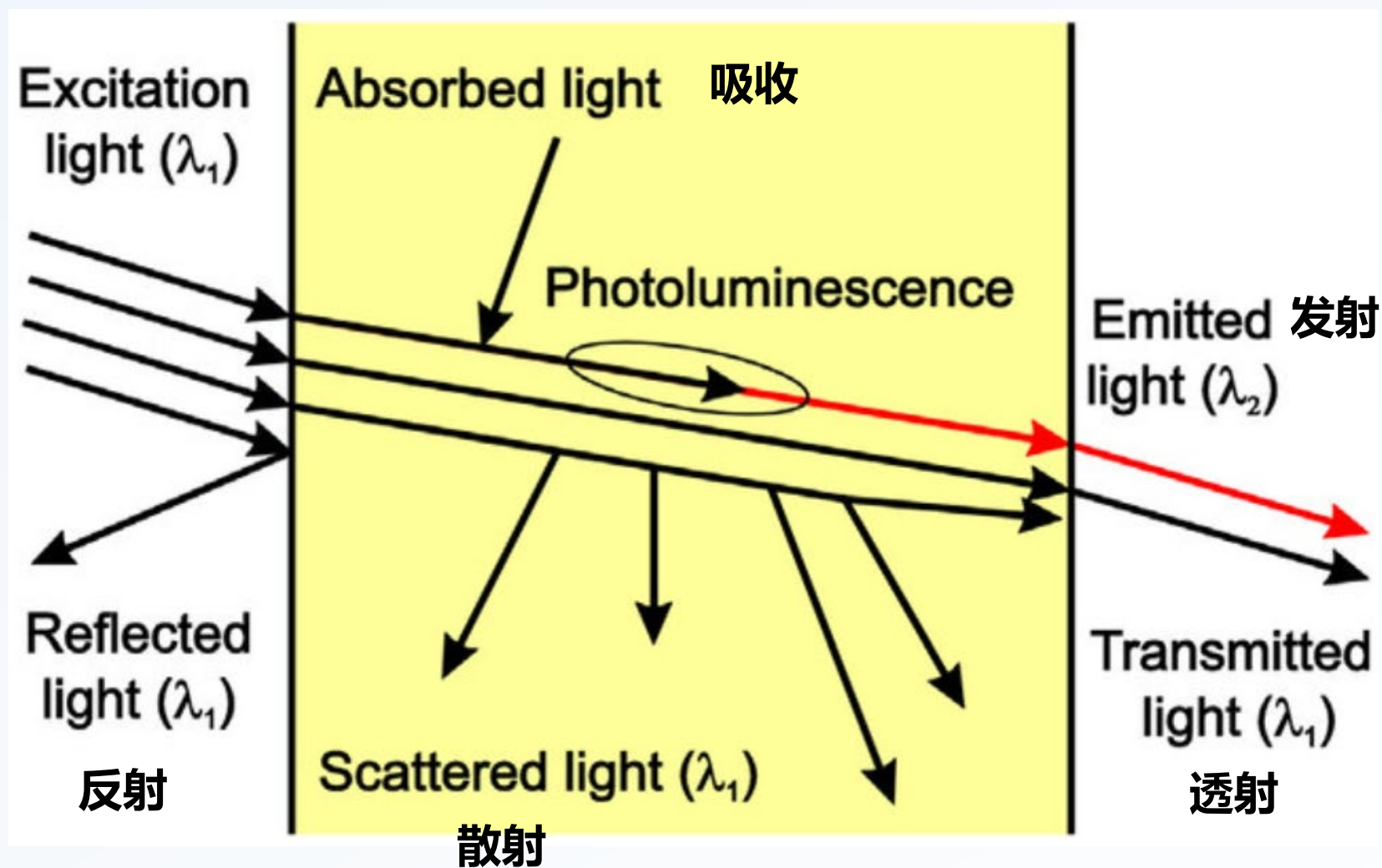
Overview of spectroscopy



Overview of spectroscopy



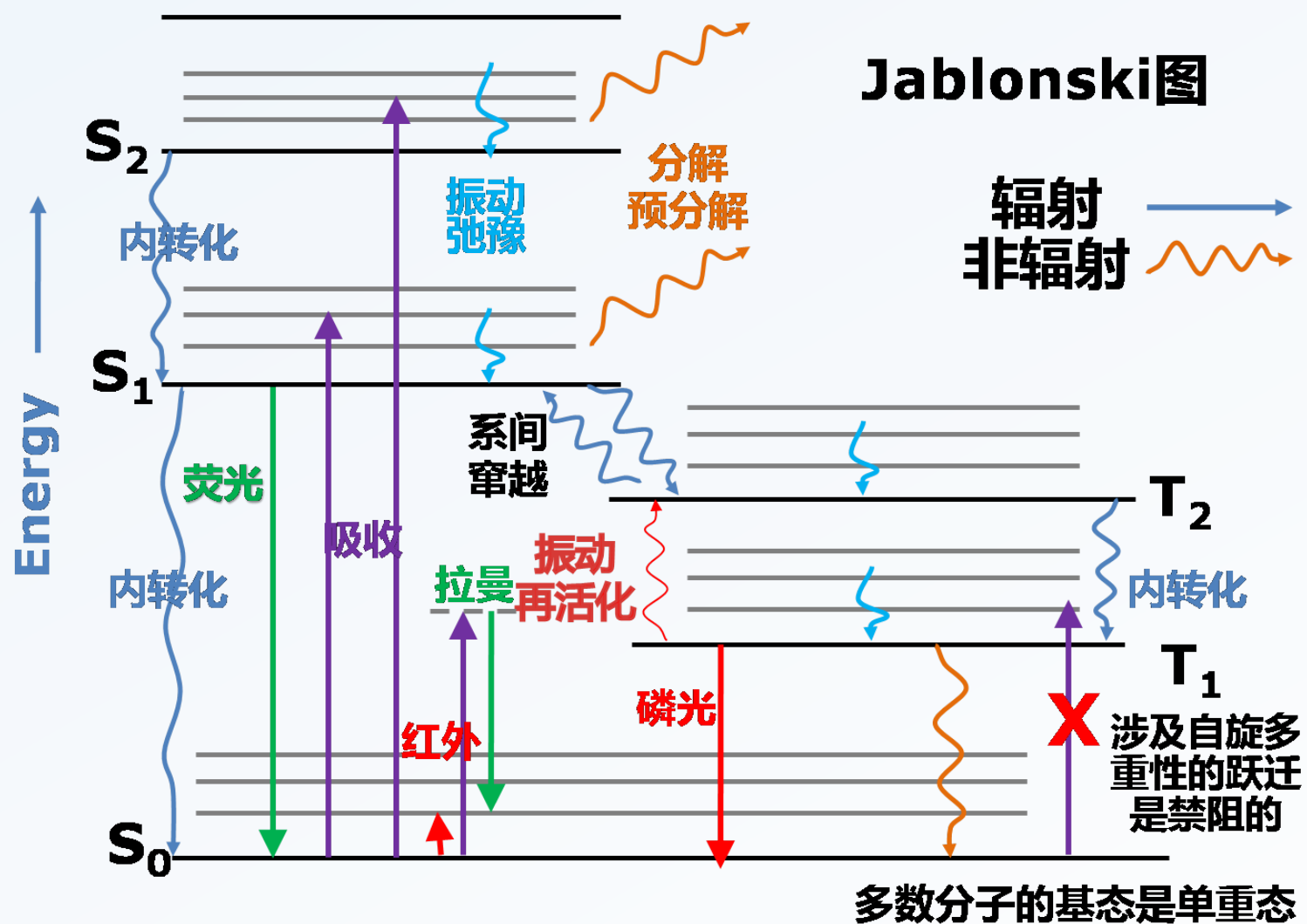
Overview of spectroscopy



Overview of spectroscopy

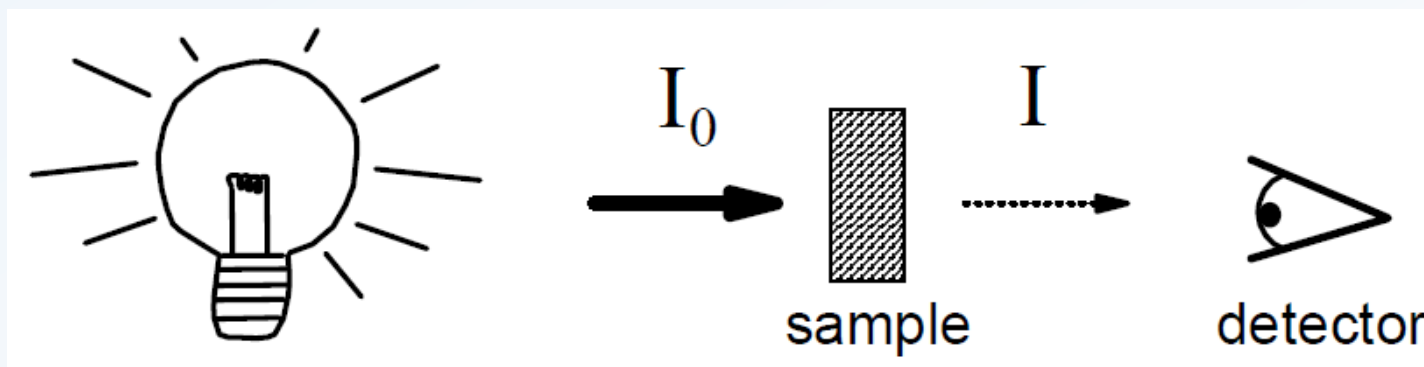
Change of spin		Change of orientation	Change of configuration	Change of electron distribution		Change of nuclear configuration
N.m.r.	E.s.r.	Microwave	Infra-red	Visible and ultra-violet	X-ray	γ -ray
10^{-2}	1	100	10^4	10^6	10^8	
cm ⁻¹						
10 m	100 cm	1 cm	100 μ m	1 μ m	10 nm	100 pm
3×10^6	3×10^8	3×10^{10}	3×10^{12}	3×10^{14}	3×10^{16}	3×10^{18}
Hz						
Frequency						
10^{-3}	10^{-1}	10	10^3	10^5	10^7	10^9
joules/mole						
Energy						

Overview of spectroscopy



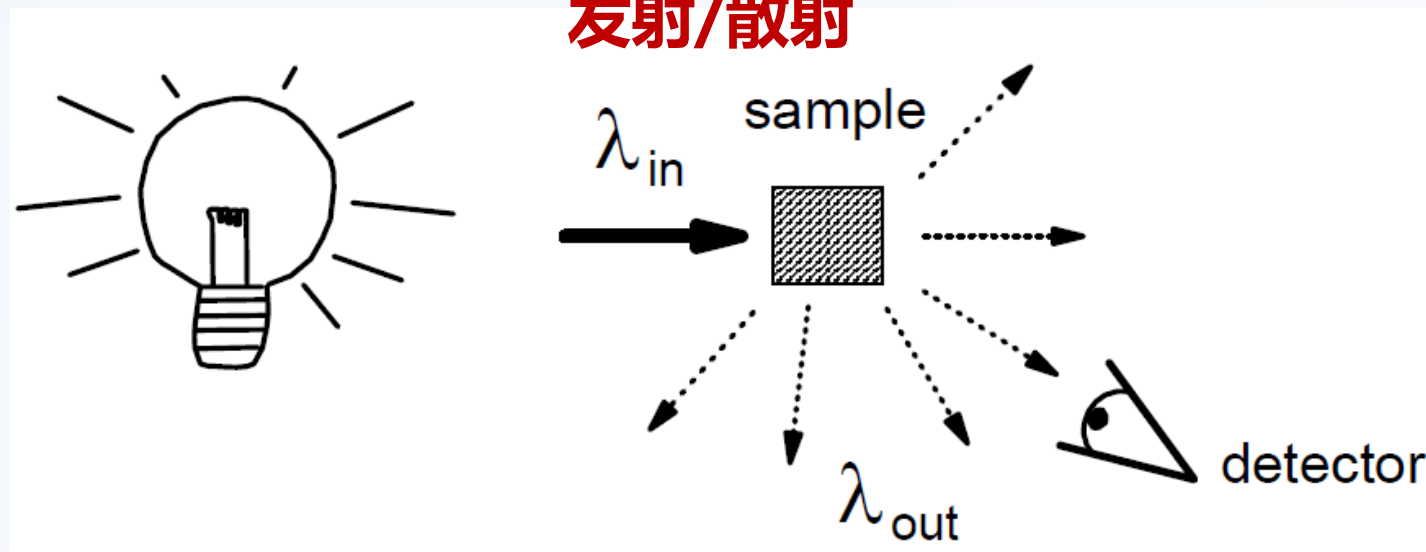
Overview of spectroscopy

吸收/透射/反射

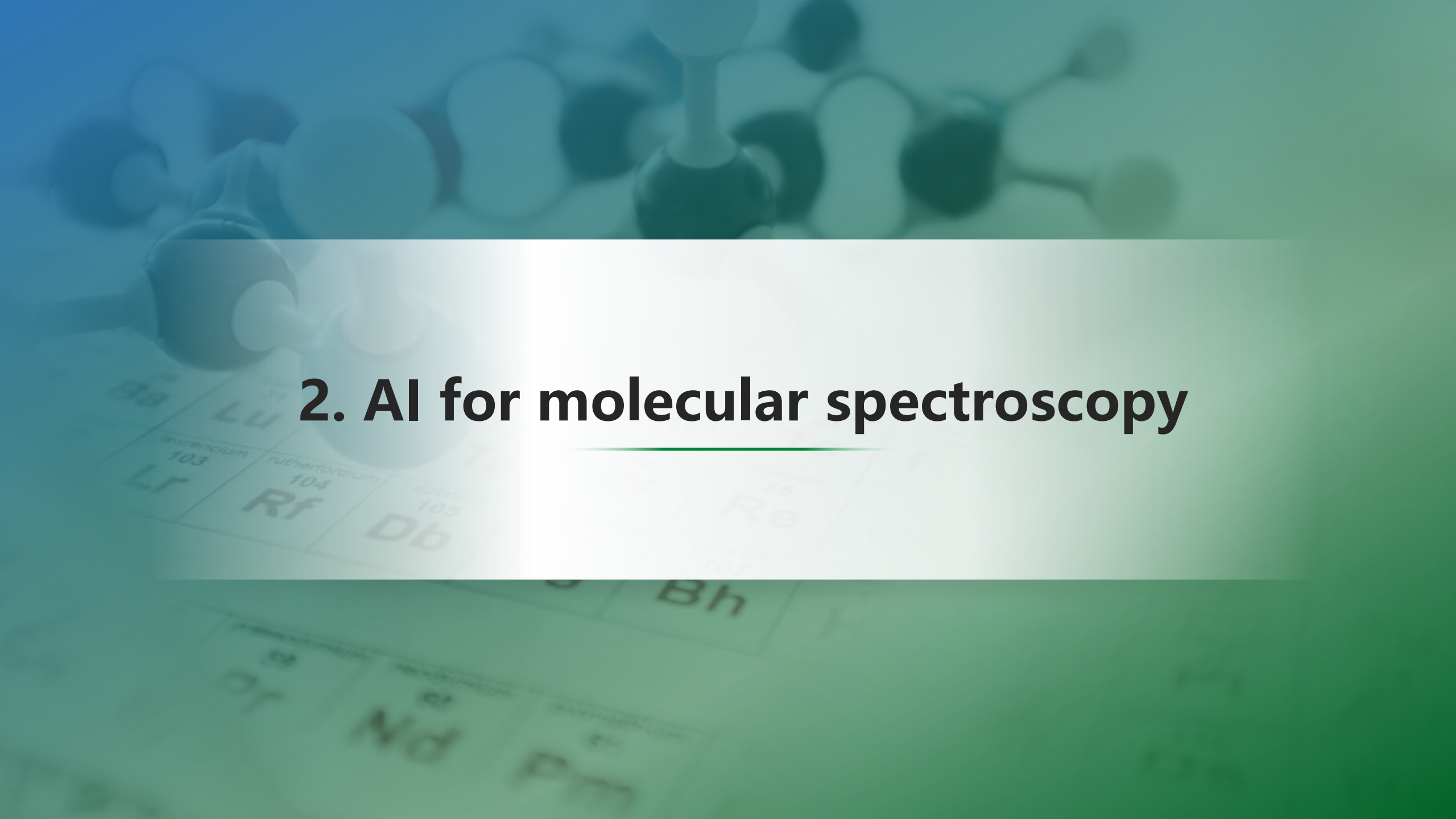


Excitation

发射/散射

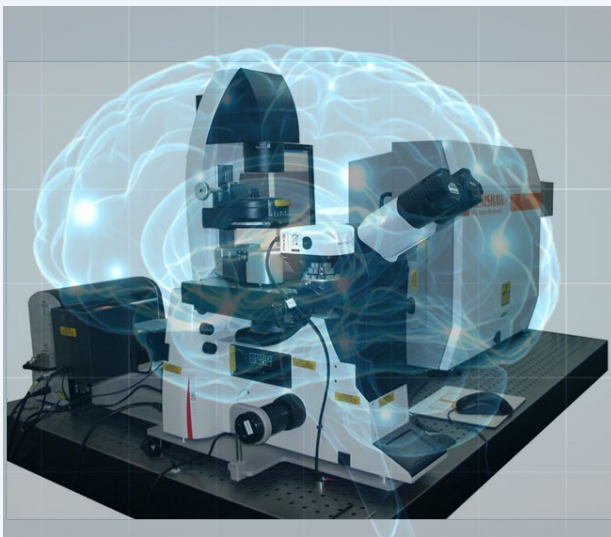


Detection

The background of the slide features a blurred molecular model with white and grey spheres connected by rods, positioned over a portion of a periodic table. The periodic table shows elements such as Lu, Lr, Rf, Db, Re, Bh, Nd, and Pt. A semi-transparent white rectangular box is centered over the text.

2. AI for molecular spectroscopy

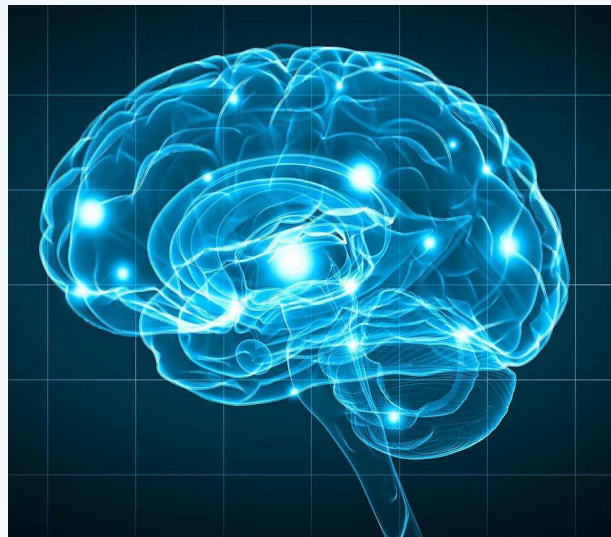
AI for molecular spectroscopy



Spectrometer

智慧的人

- Automatic detection
- Automatic calibration
- Enhancing performance



Algorithm

聪明的大脑

- Spectral processing
 - Denoising, baseline
- Spectral analysis
 - Classification, spectrum-structure relationship



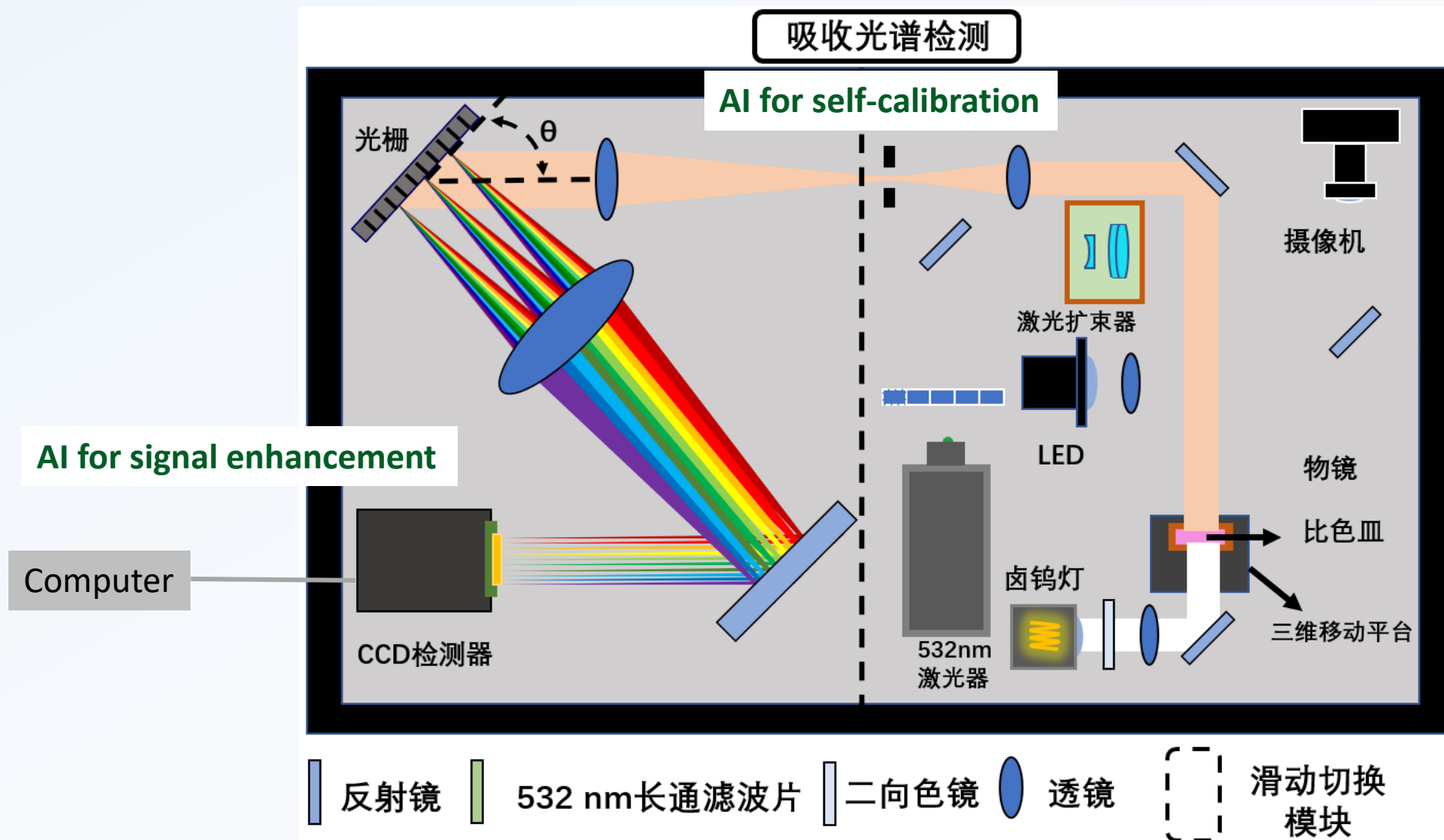
Data base

正确的书库

- ◆ Data and model can be transferred and reused.

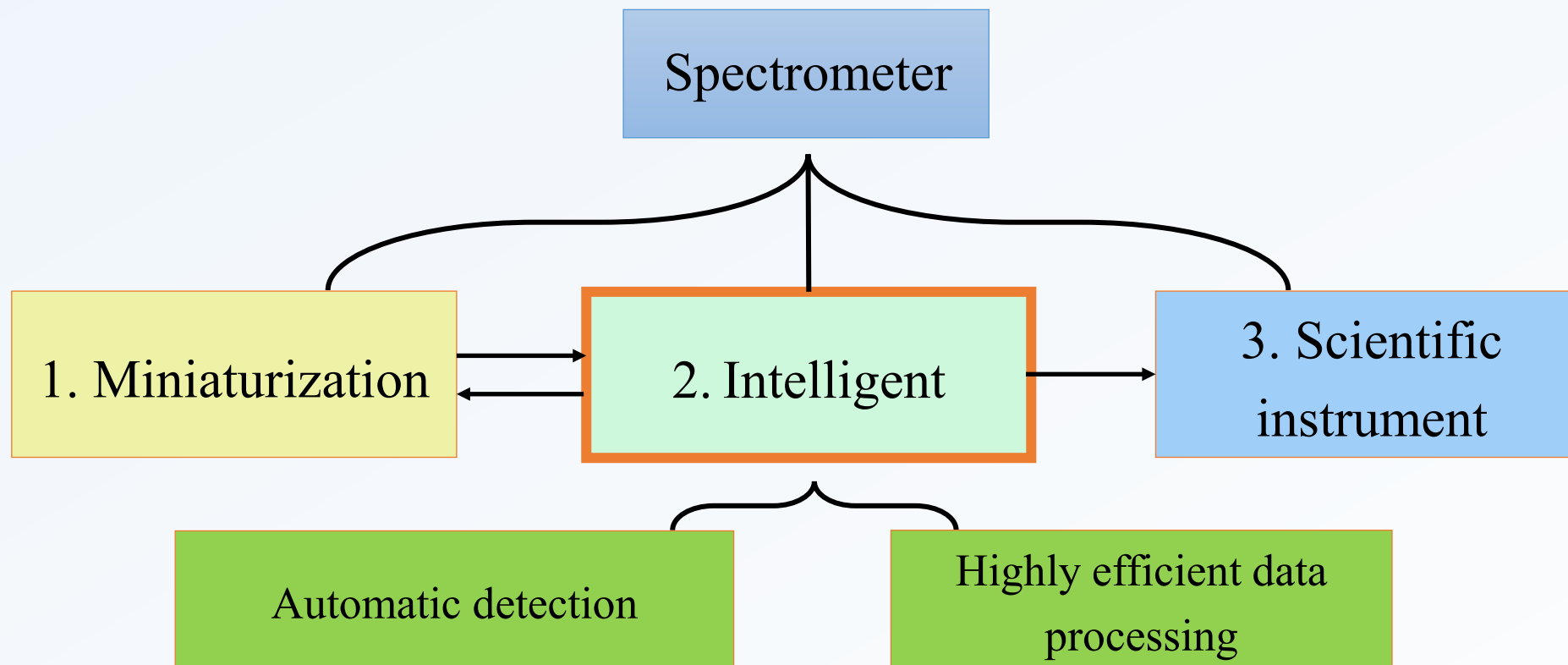
AI for molecular spectroscopy

1. Before we analyze the spectrum, how to obtain it?



AI for automatic detection

AI for molecular spectroscopy

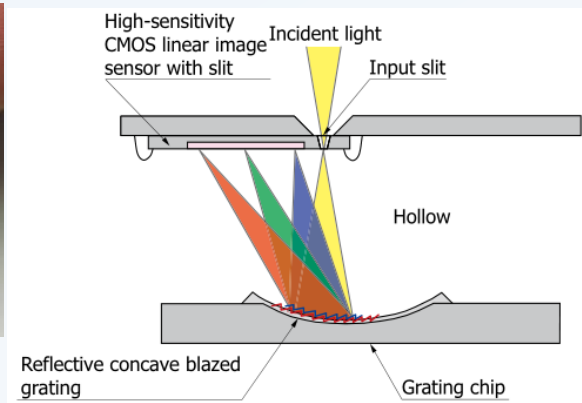


Sensitivity is of great importance,
which primarily depends on CCD and light path in the spectrometer.

AI for molecular spectroscopy



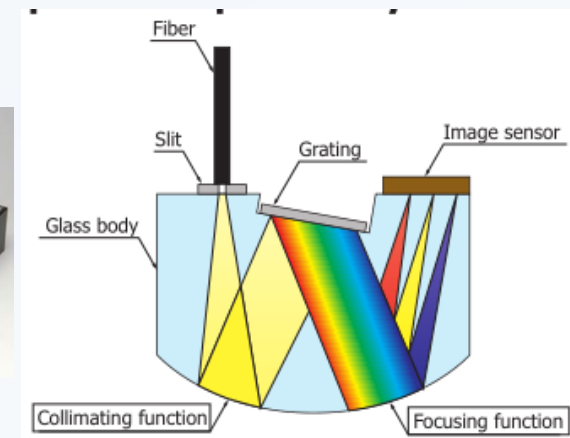
C12666MA



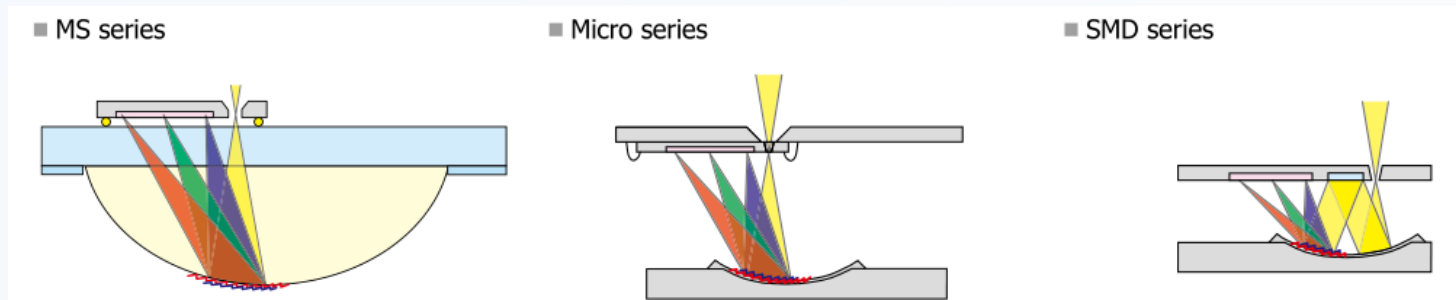
spectral region:340nm-780nm
spectral resolution:15nm
Size:20.1 × 12.5 × 10.1 mm



RC

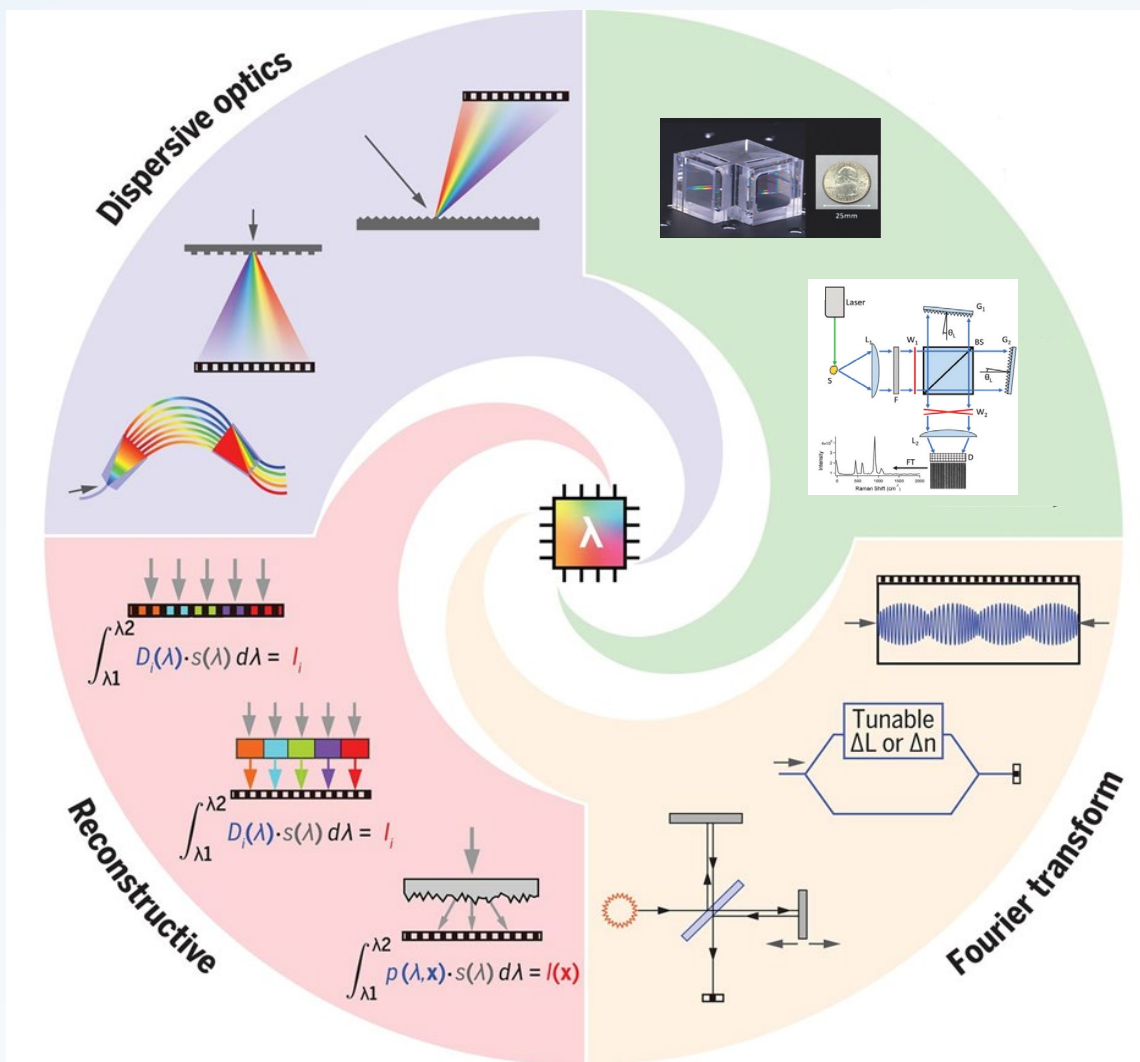


spectral region:340nm-780nm
spectral resolution:9nm
Size:55 × 100 × 48 mm

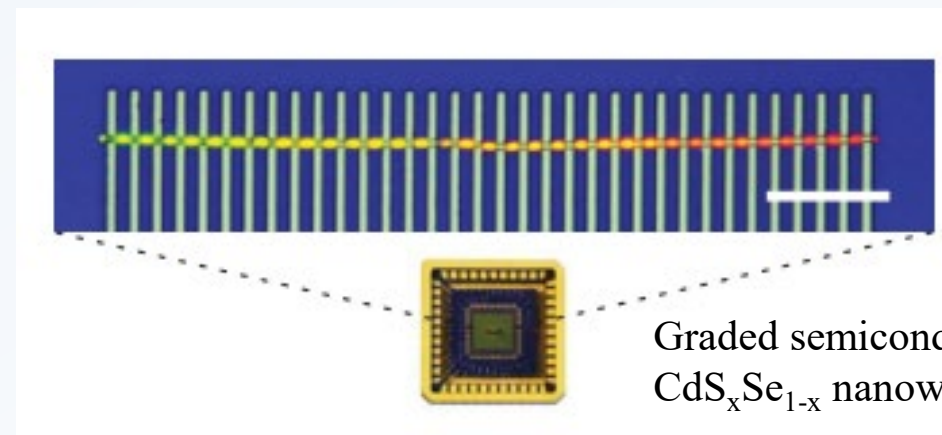


spectral region:640nm-1050nm
spectral resolution:5nm
Size:11.5 × 4 × 3.1 mm

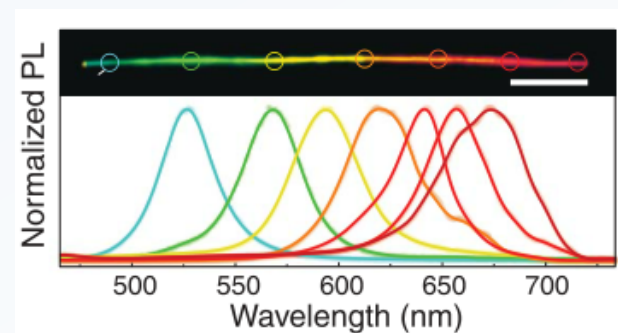
AI for molecular spectroscopy



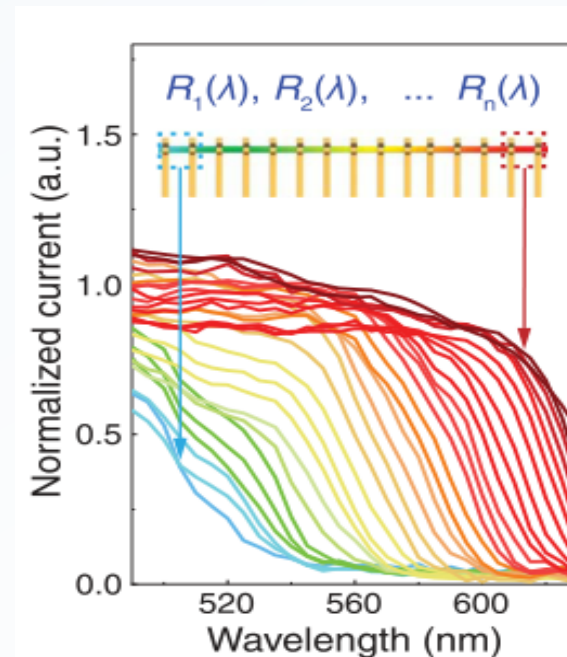
Zongyin Yang et al. *Science* **2021**;371:eabe0722



Graded semiconducting
 $\text{CdS}_x\text{Se}_{1-x}$ nanowire

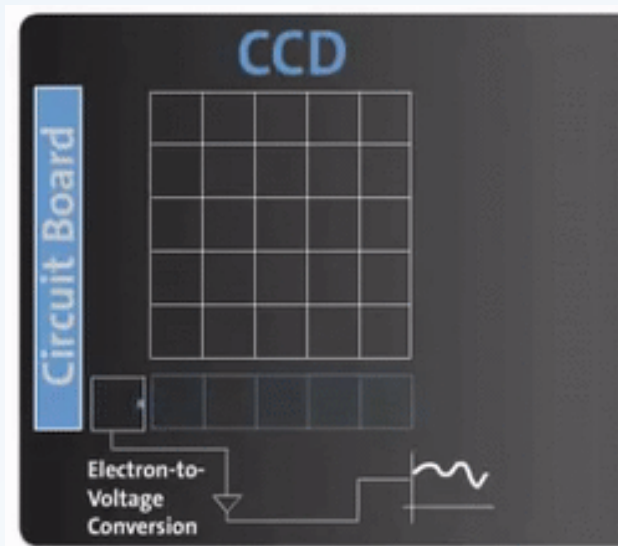


Spectral response
functions $R_i(\lambda)$



AI for molecular spectroscopy

CCD



Detector material: Si, InGaAs

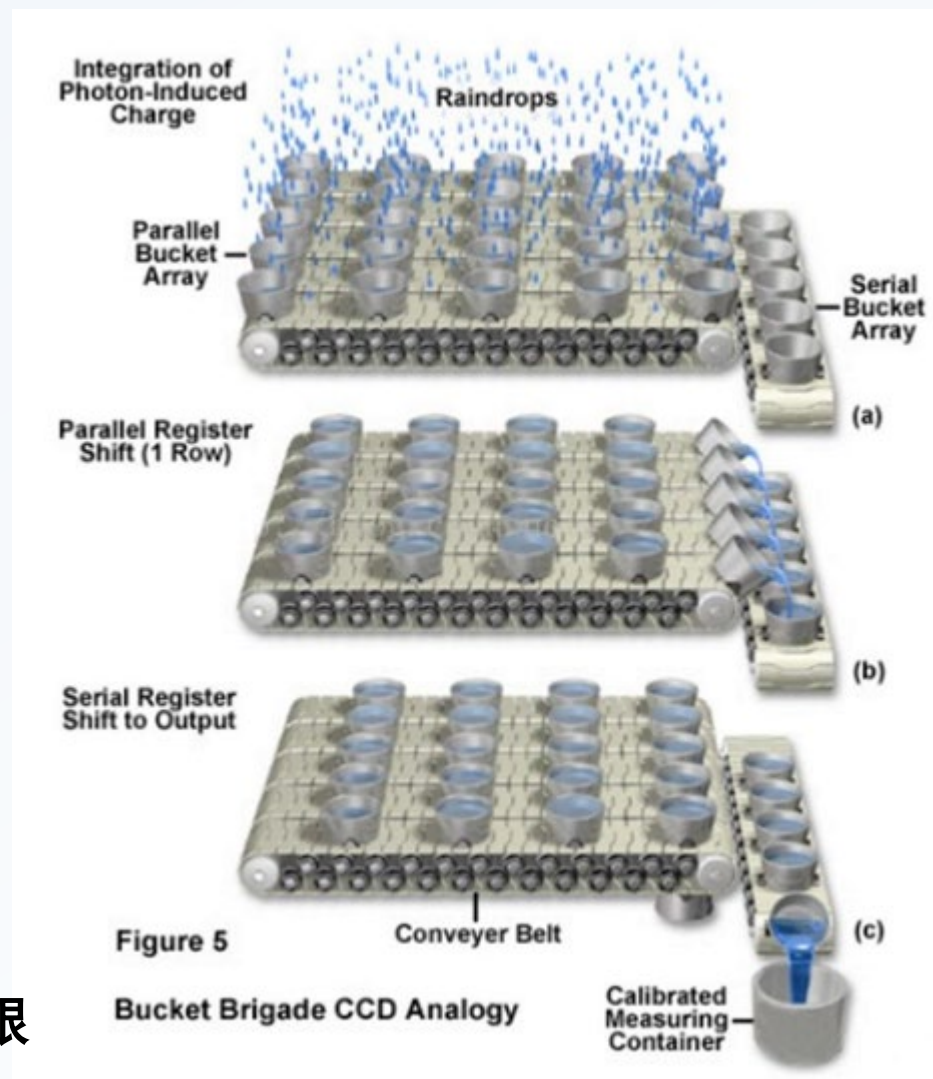
$$\lambda_{max} = \frac{hc}{E_{gap}}$$

E_{gap} : 带隙能量

λ_{max} : 该检测器可以检测到的波长上限

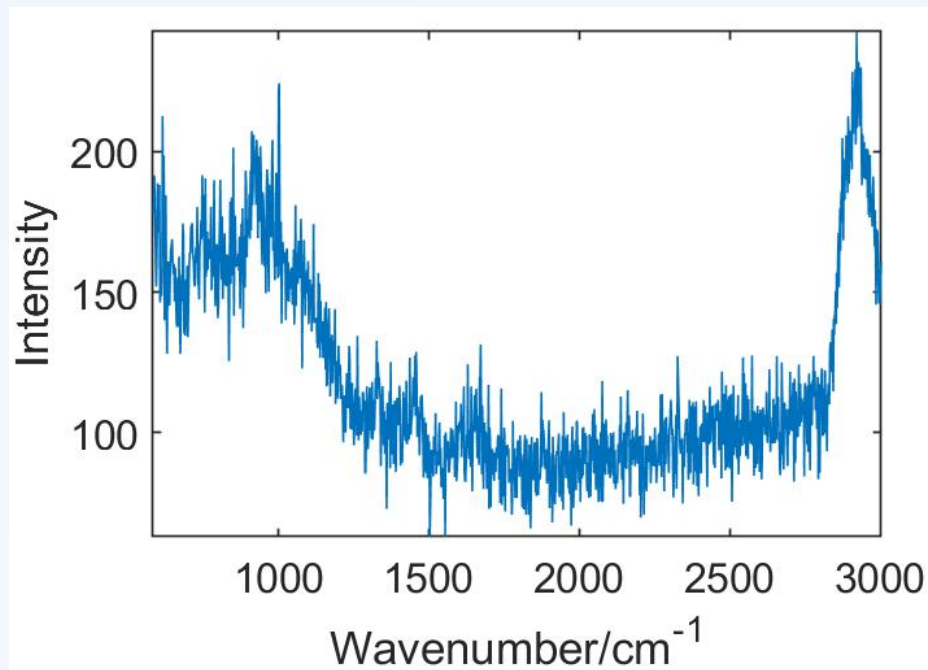
h : 普朗克常数

c : 光速

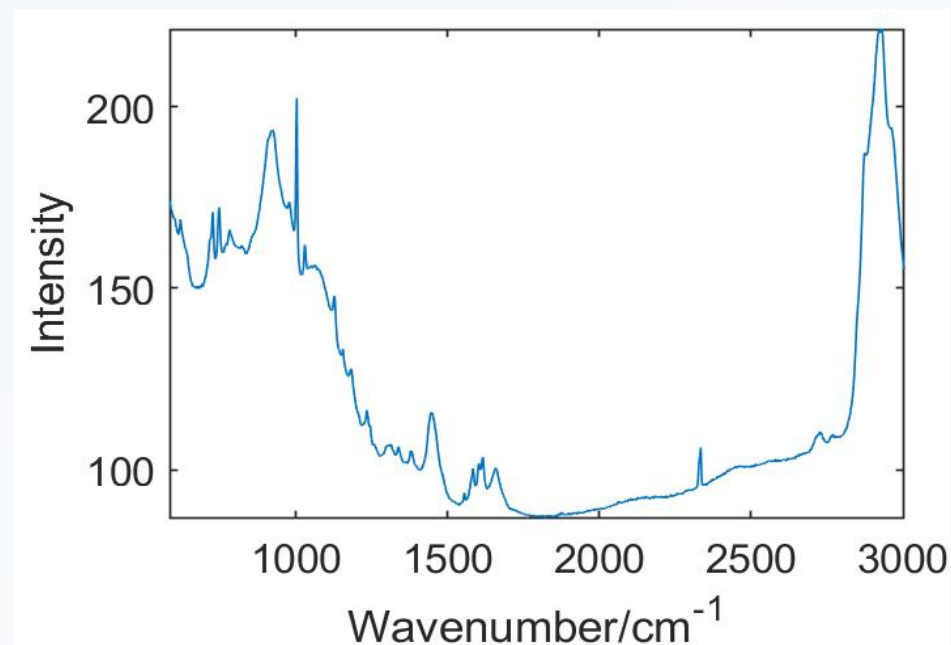


AI for molecular spectroscopy

Original spectrum



Expected spectrum

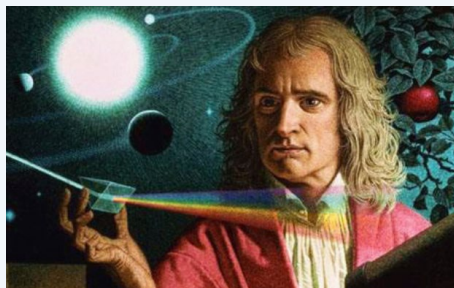


Sensitivity and temporal resolution!

AI for molecular spectroscopy

- Dispersive Spectrometer

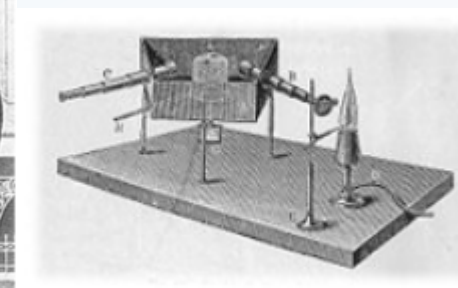
1666, Newton
Prism dispersion



1814, Fraunhofer



1859, Kirchhoff and (Bunsen)
First spectrometer



1989, 林中、范世福

Dispersion spectrometer

plane grating

Ebert-Fastie, Czerny-Turner

Littrow, Chupp-Graniz

concave grating

Paschen-Runge, Eagle

Wadsworth, Seya-Namioka

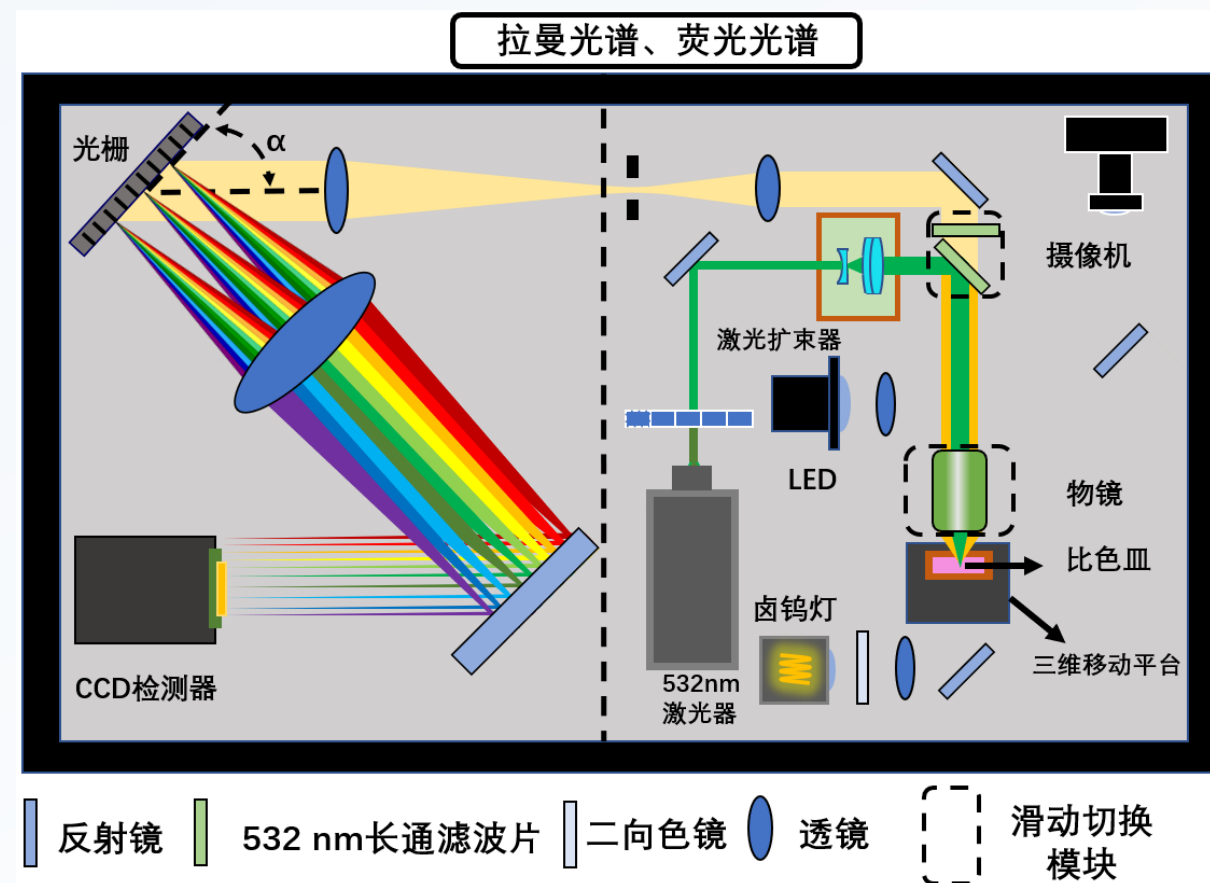
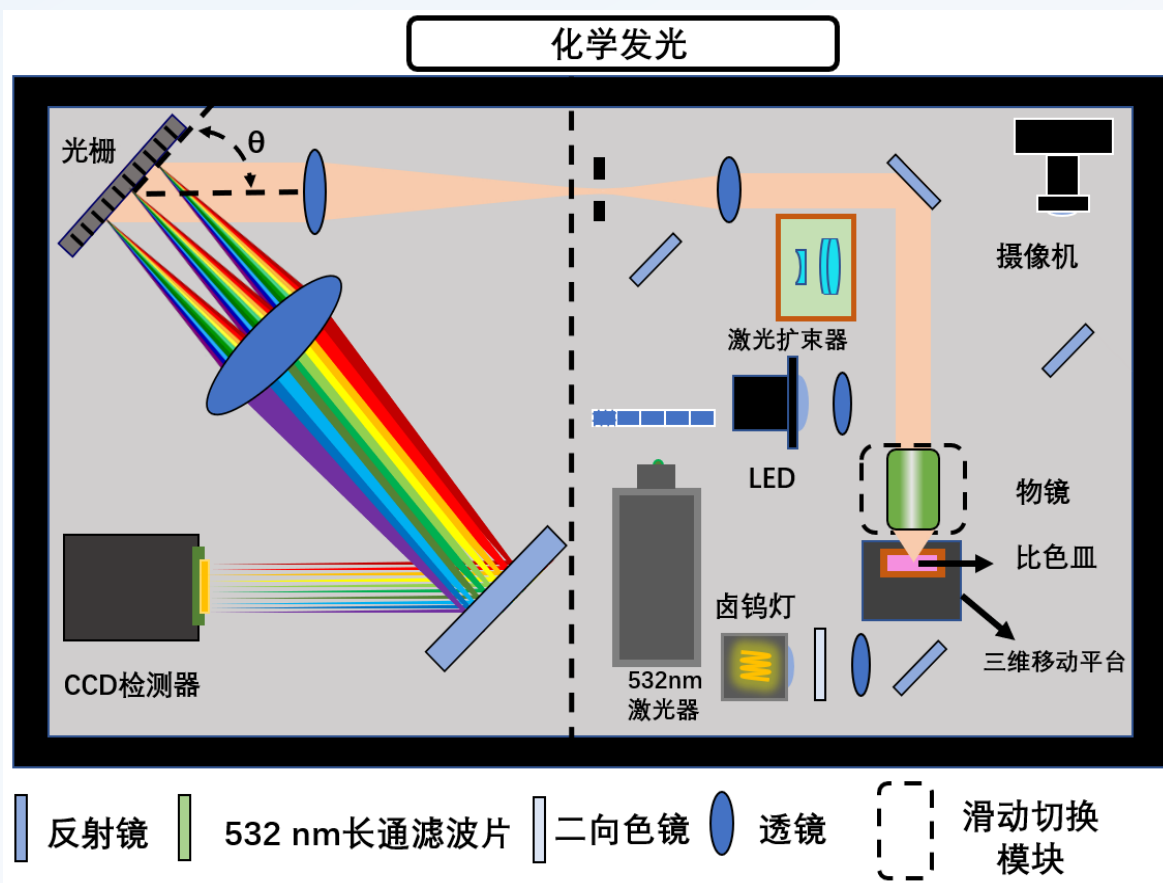
Photoelectric direct reading spectrometer

Fourier transform infrared spectrometer

Fabry-Perot interferometer

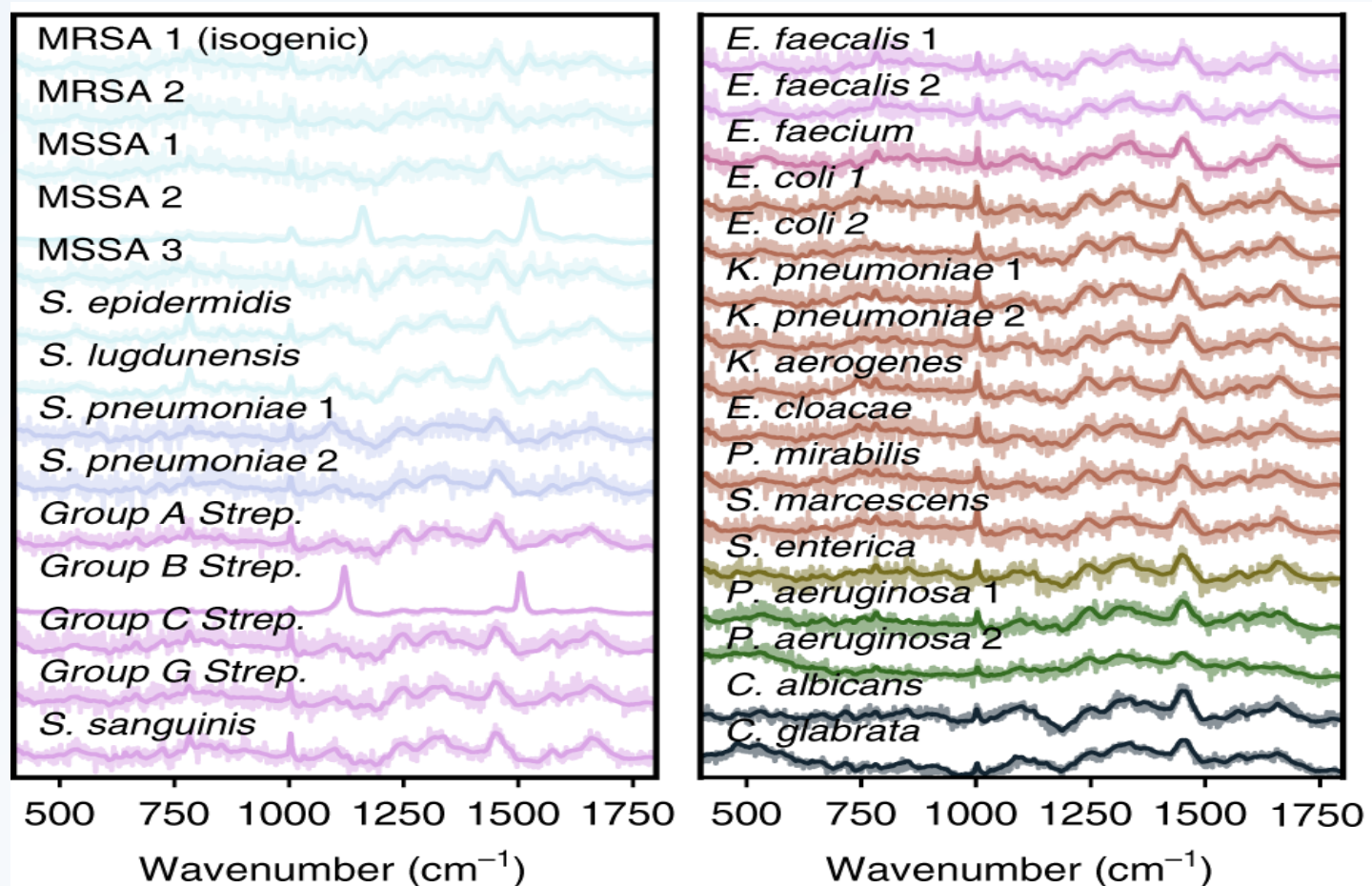
AI for molecular spectroscopy

Different systems feature different light path.



AI for molecular spectroscopy

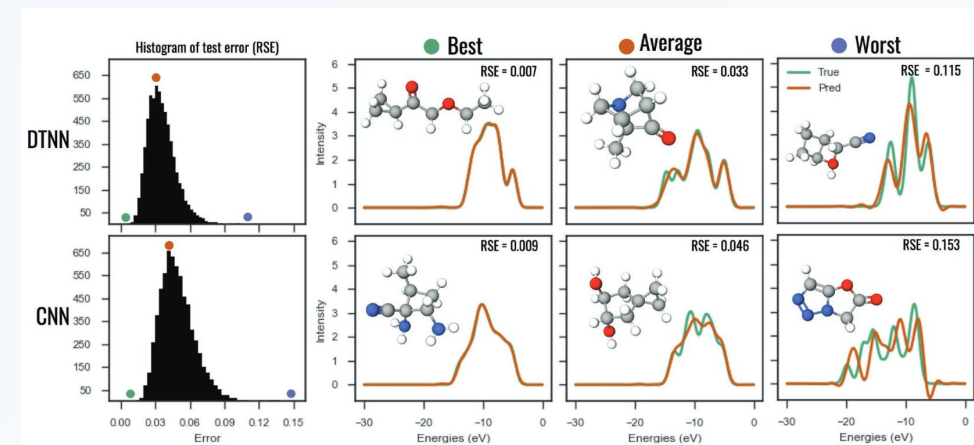
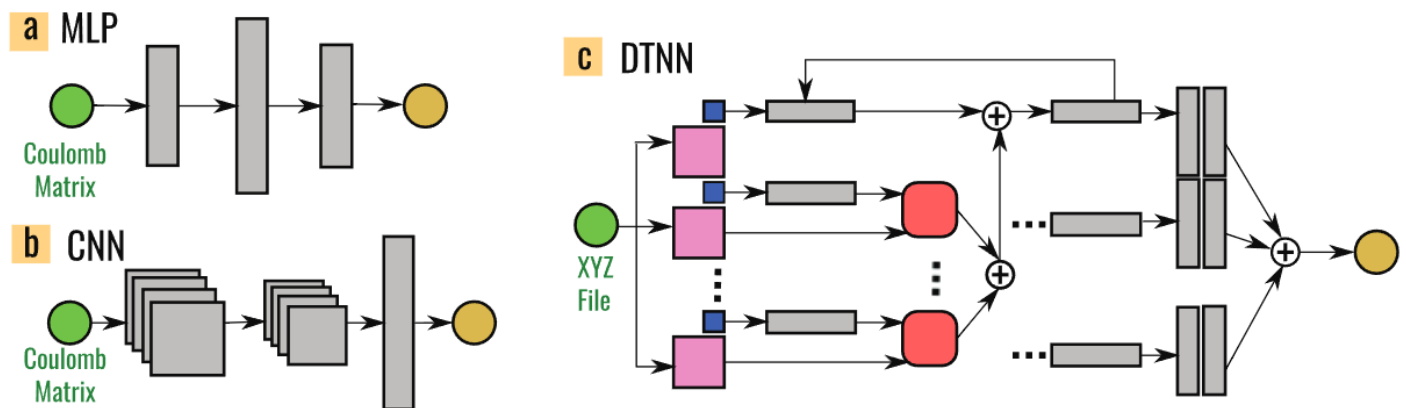
AI can be used for classification and regression.



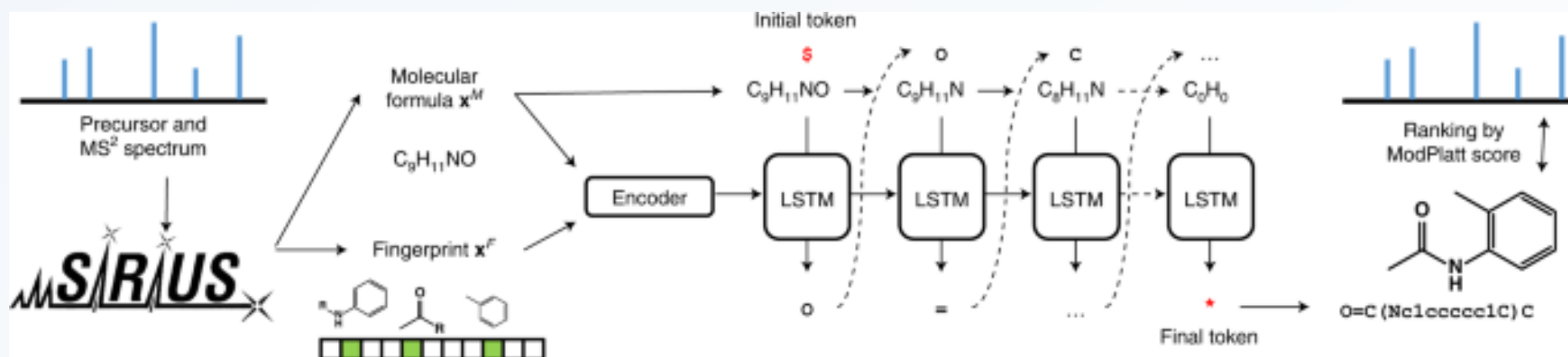
Dionne, et al., Nat. Commun. **2019**, 10, 4927

AI for molecular spectroscopy

AI can be used for constructing spectrum-structure relationship (prediction and generation).

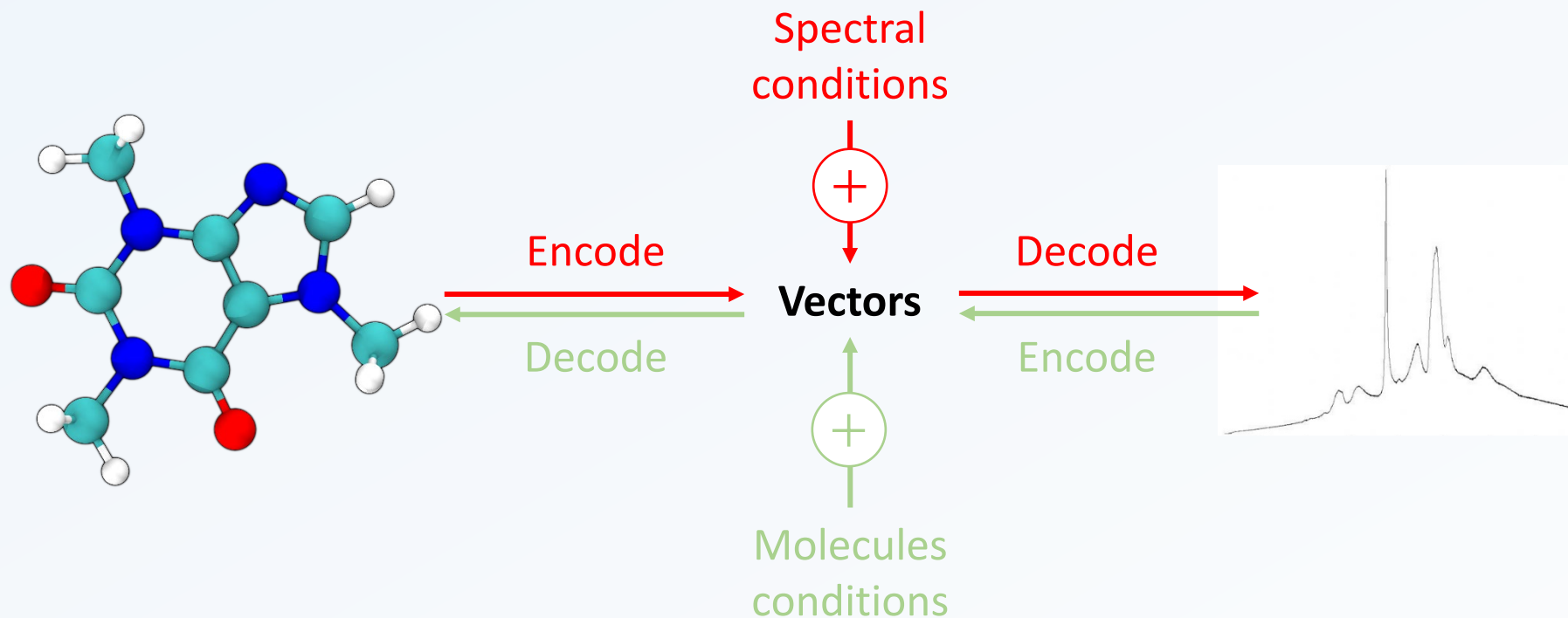


Adv. Sci. 2019, 6, 1801367



Nat Methods 2022, 19, 865–870

AI for molecular spectroscopy



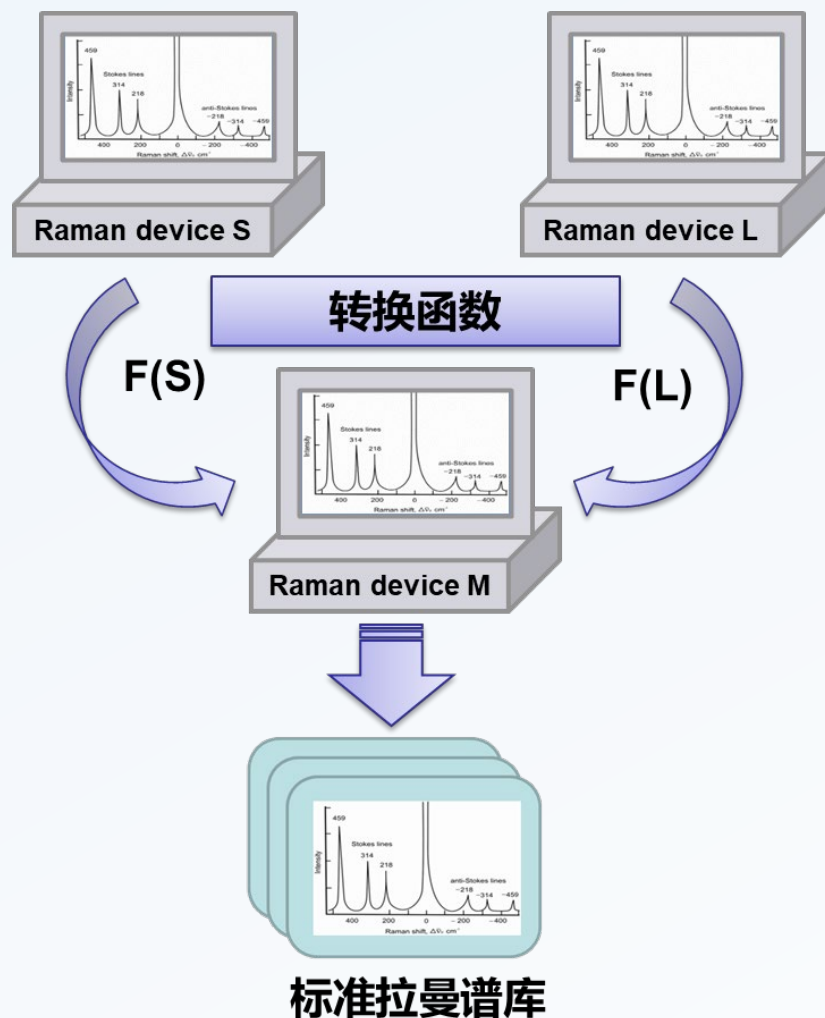
Thinking

Can the transformation of structure and spectrum be as simple as switching between Chinese and English?

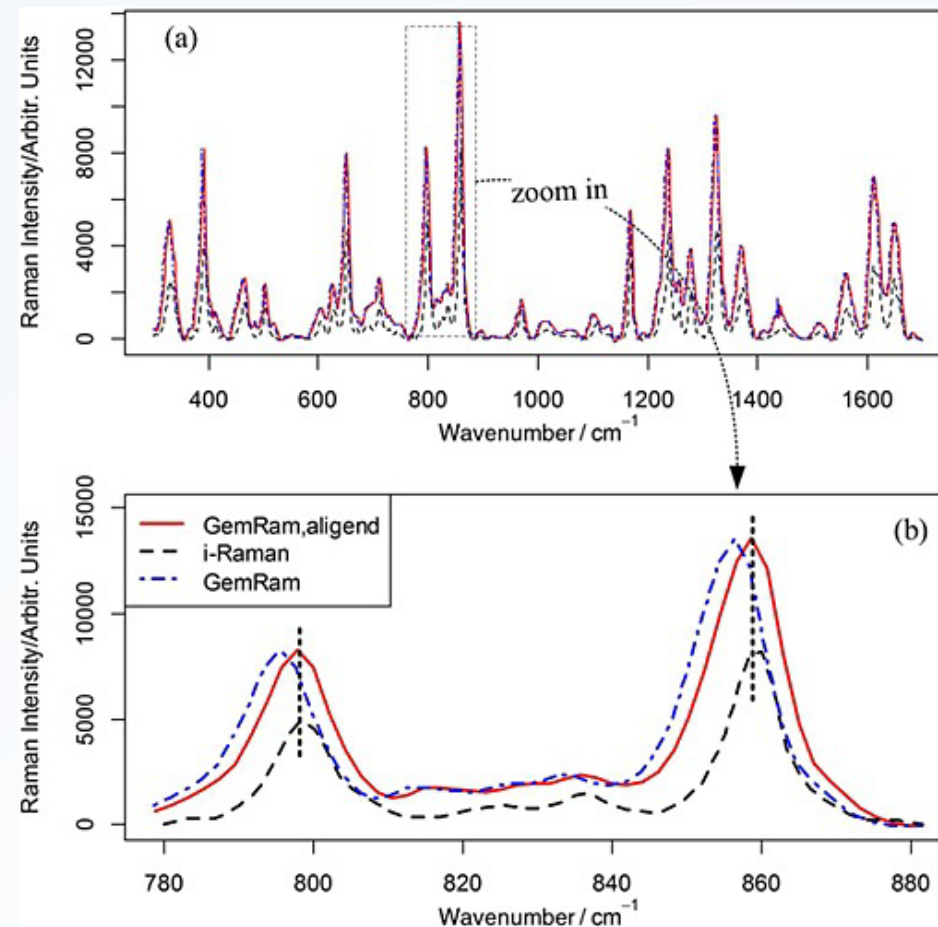
结构和光谱的转化能像中英文切换一样简单吗？

AI for molecular spectroscopy

Finally, the database

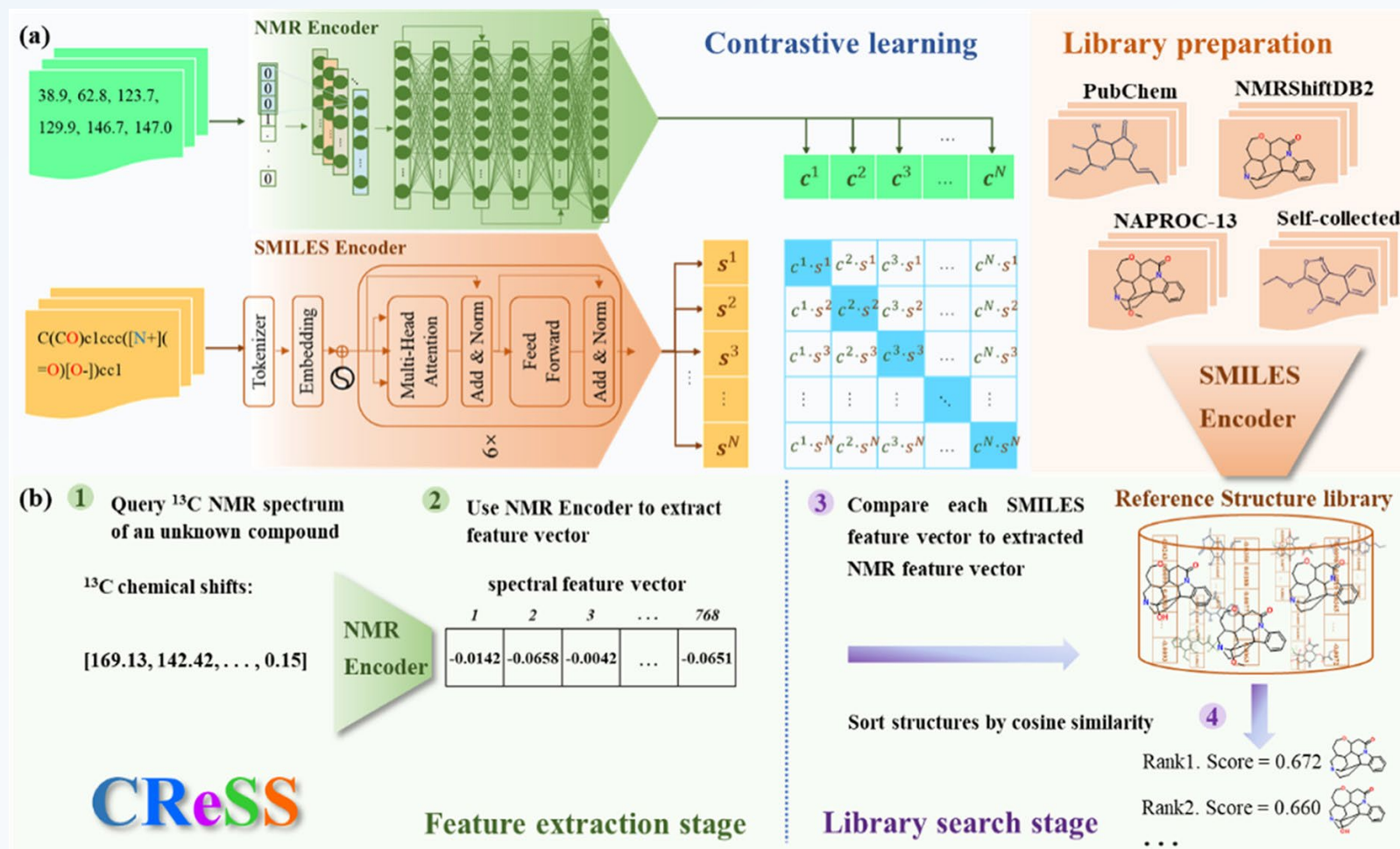


Artificial tuning → Automatic tuning

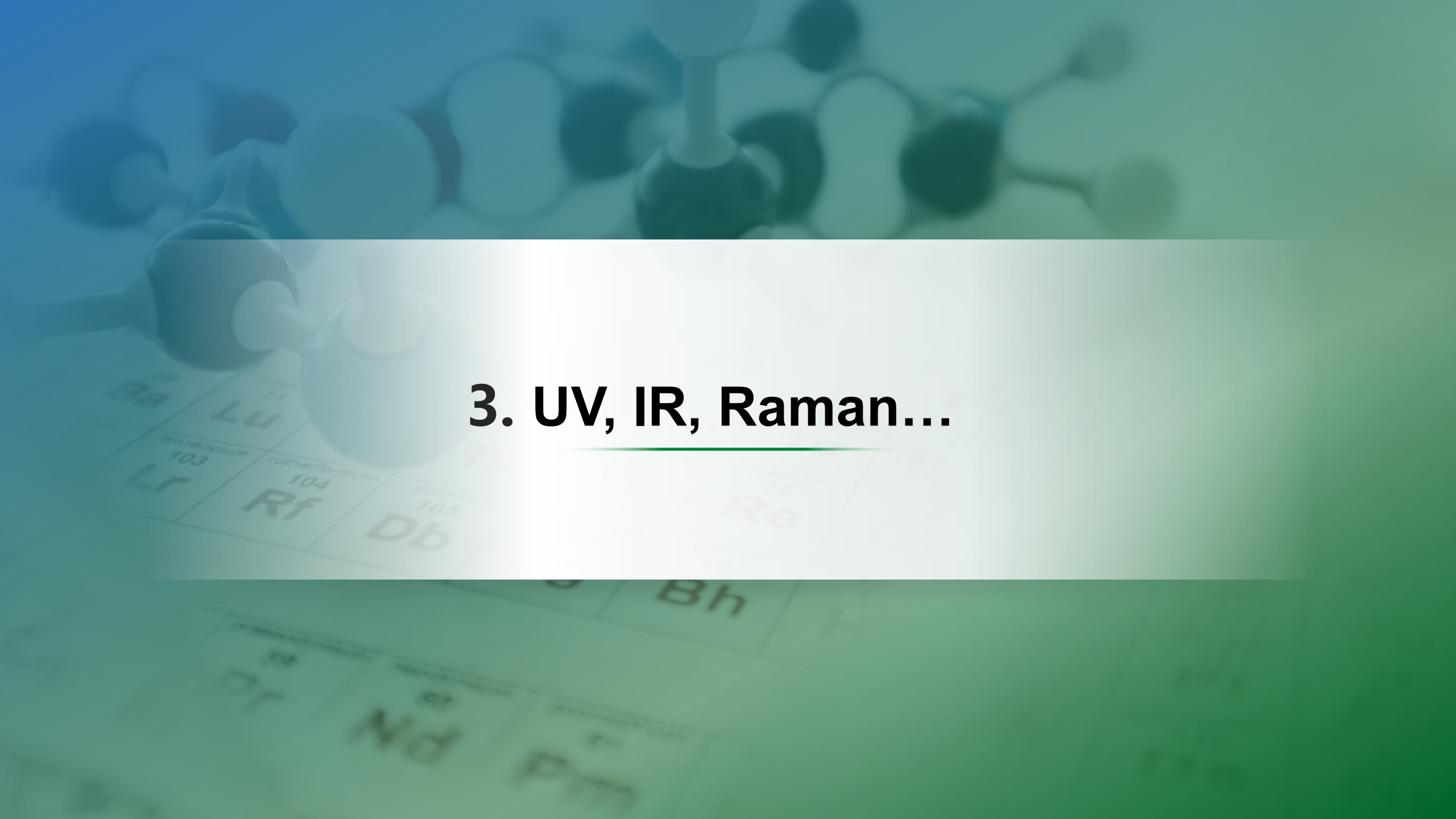


AI for molecular spectroscopy

AI can be used for library matching and searching.

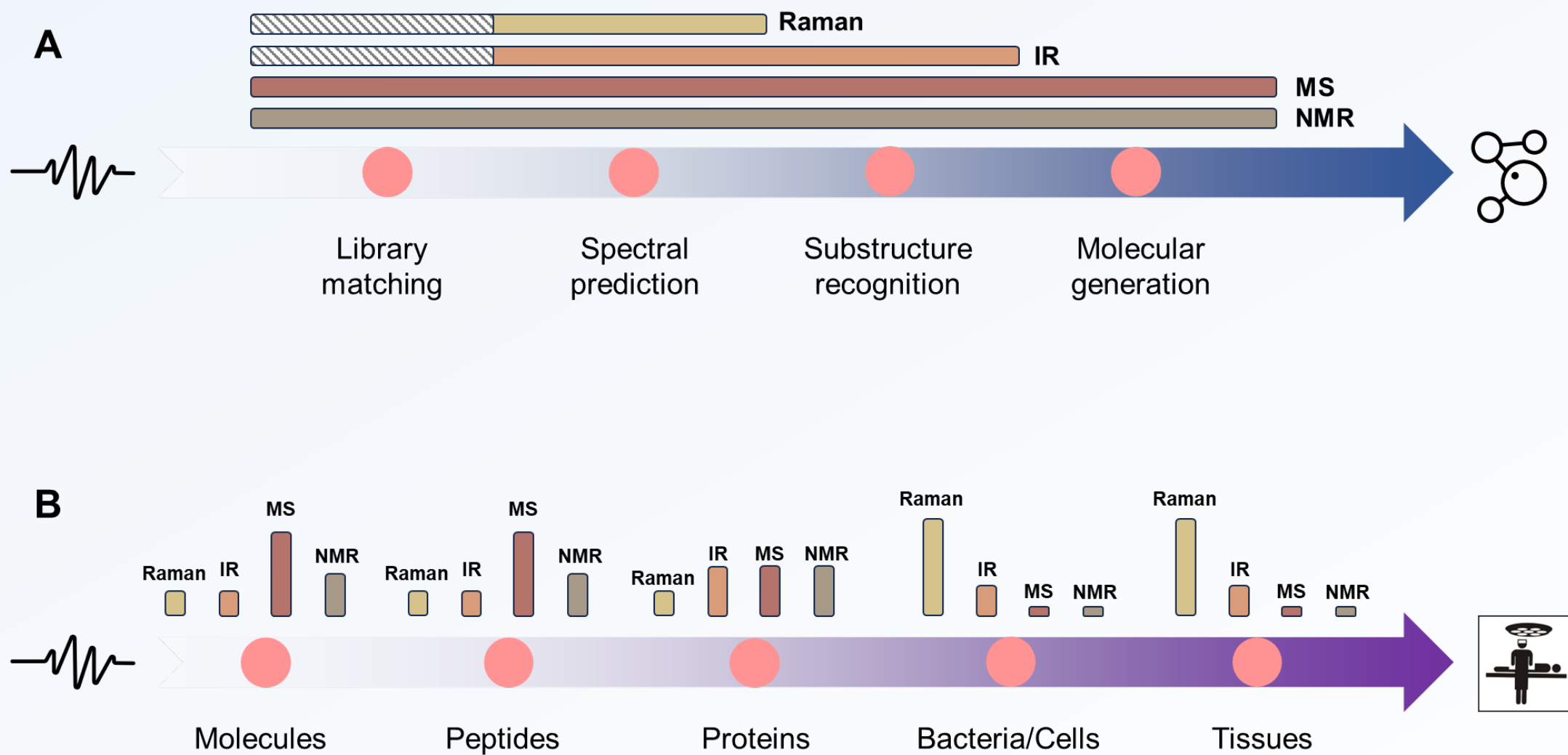


Anal. Chem. 2021, 93, 50, 16947–16955



3. UV, IR, Raman...

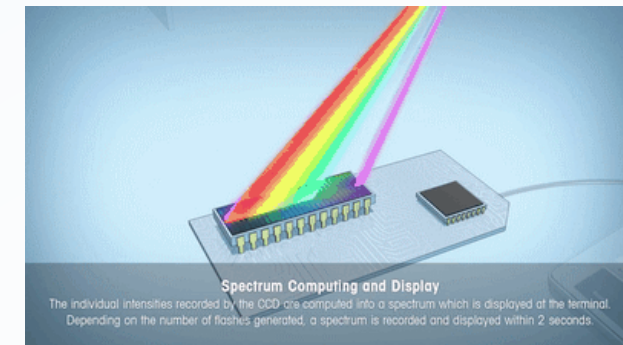
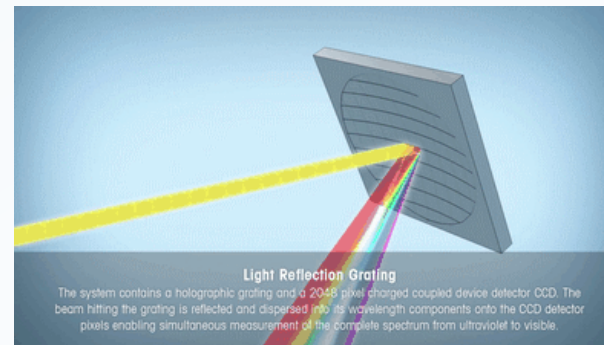
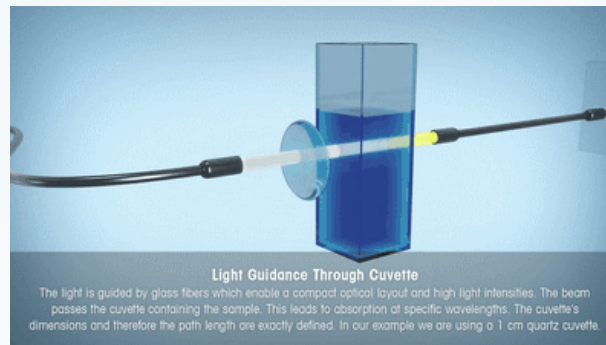
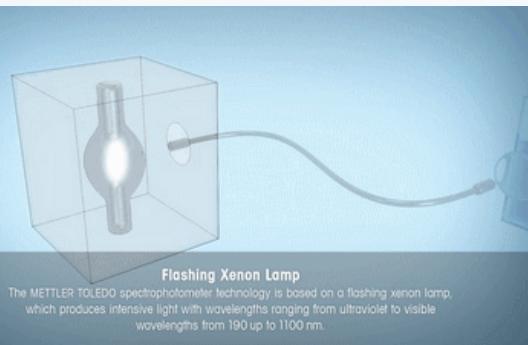
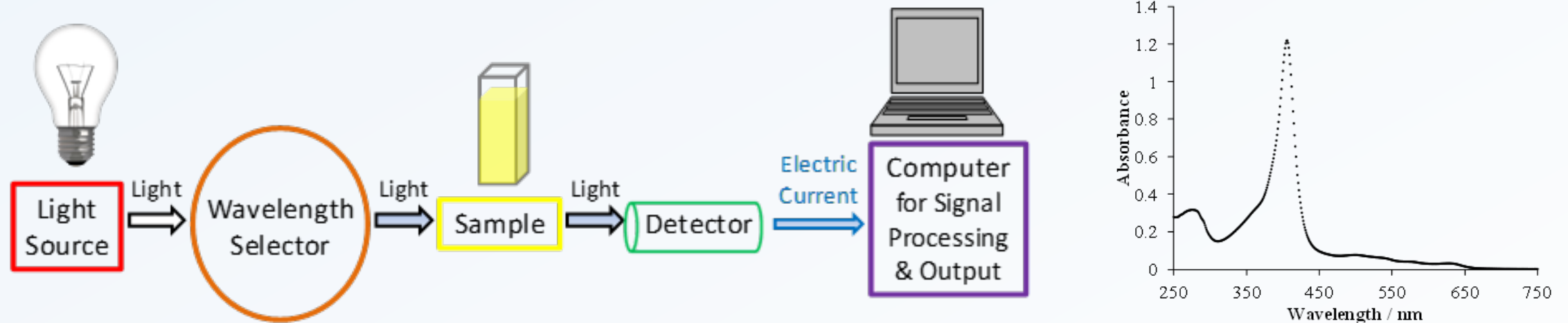
UV, IR, Raman...



UV, IR, Raman...

Ultraviolet-Visible Spectroscopy

UV-Vis absorption spectroscopy, also known as molecular electron transition spectroscopy, is produced by the electron transition in the outer layer of molecules after the absorption of UV or visible light.

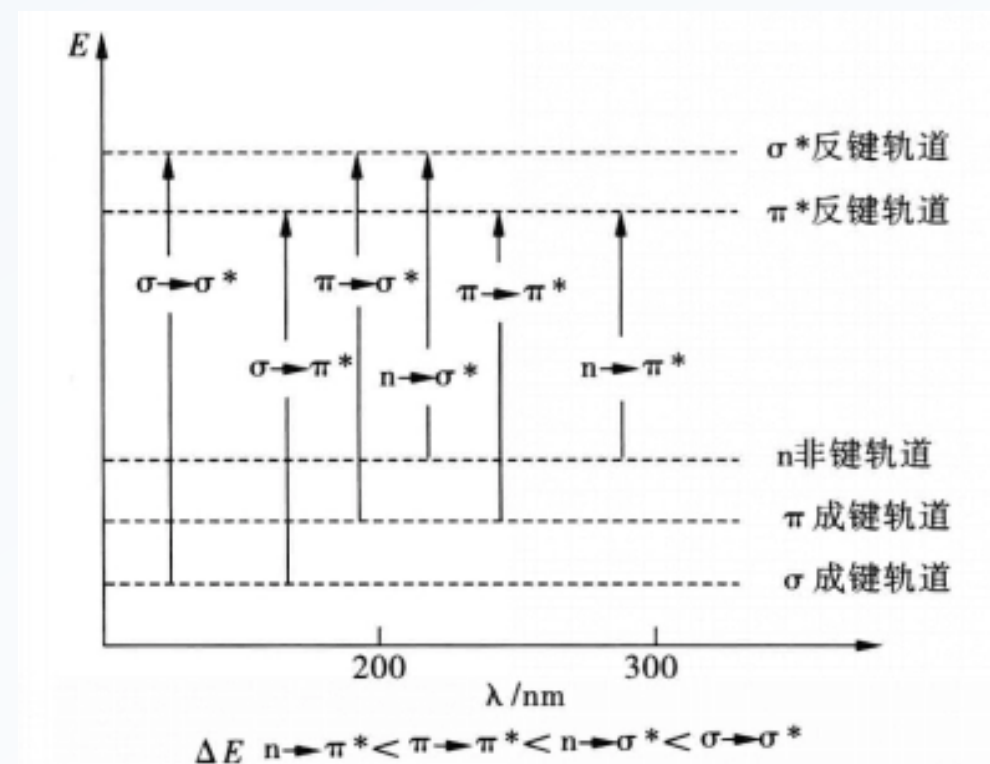


UV, IR, Raman...

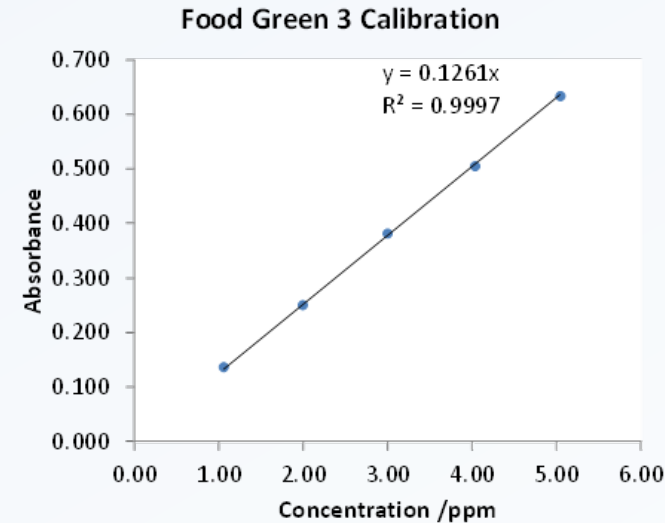
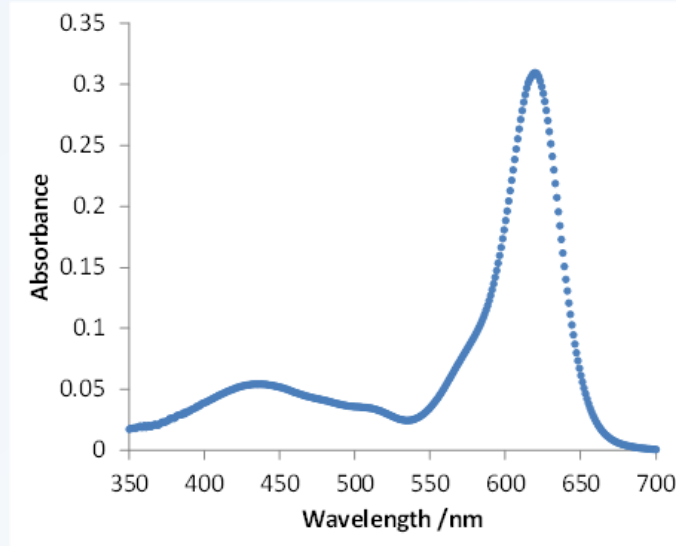
Ultraviolet-Visible Spectroscopy

Its spectral range is 190–800 nm. Because the electron energy level in the molecule is greater than the vibrational and rotational energy levels, when the molecule absorbs light to realize the electron transition, the vibrational and rotational spectra of the molecule must be accompanied, and they overlap each other.

Wavelength used in absorbance analysis in nanometers	What does UV absorbance at this wavelength indicate the presence of?	What causes UV absorbance at this wavelength?
230	Protein	Protein shape
260	DNA and RNA	Adenine, guanine, cytosine, thymine, uracil
280	Protein	Mostly tryptophan and tyrosine



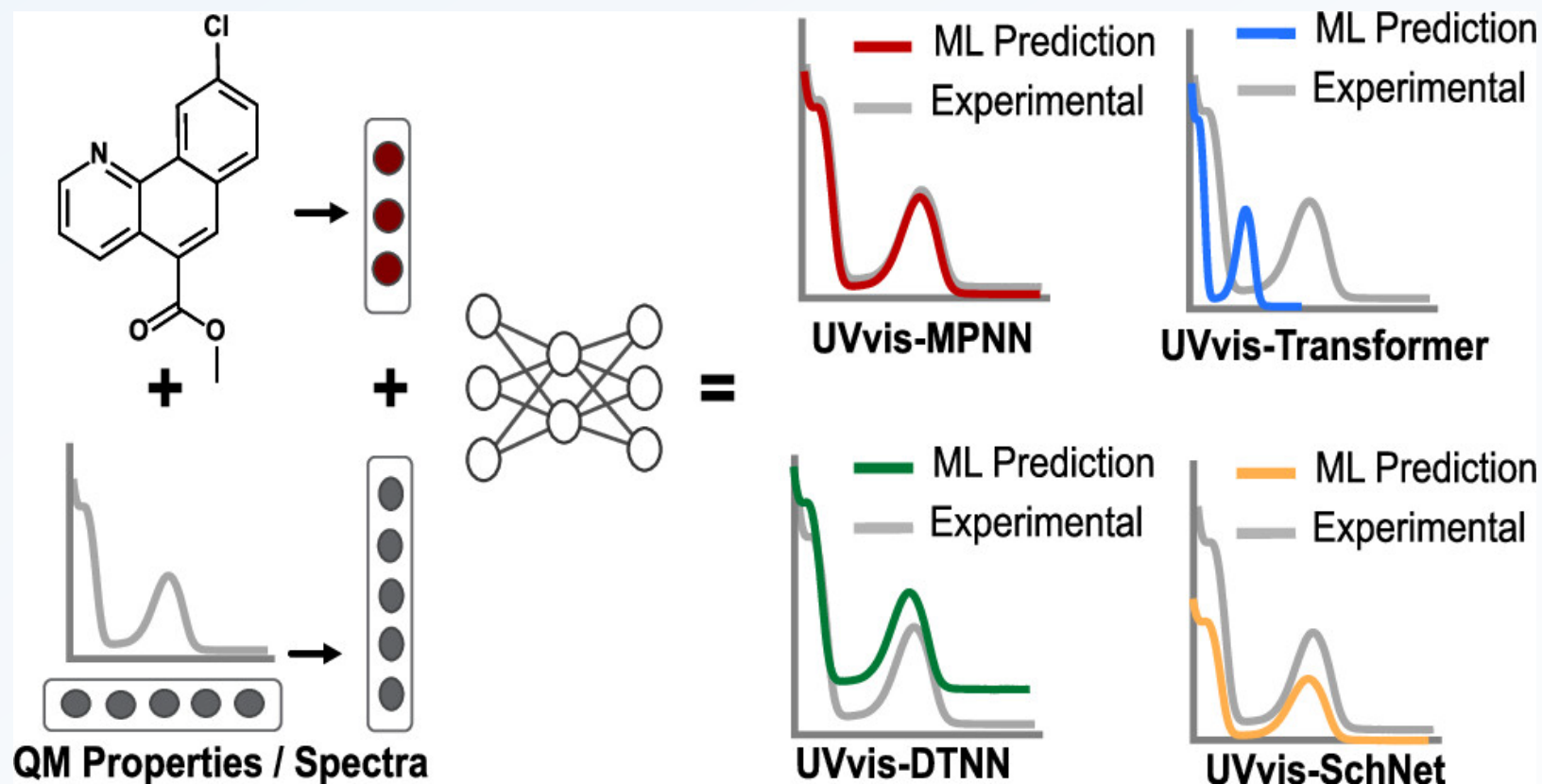
Ultraviolet-Visible Spectroscopy



Due to the diversity and complexity of water components, especially the different components of organic matter in different water systems, their all-band UV absorption spectra are significantly different, and not all the maximum absorption wavelengths appear at 254 nm. Therefore, the applicability of the single-wavelength method is poor, and satisfactory results can not be obtained. Multi-wavelength analysis usually adopts dual wavelength. In addition to 254 nm, another wavelength such as 550 nm or 546 nm is selected for turbidity compensation. At present, some commercial online water quality analyzers adopt dual wavelength measurement. The use of full spectrum (200–750 nm) combined with chemometrics (such as PLS or artificial neural network, ANN) can more comprehensively reflect the internal information of water COD, and establish a multivariate correction model for specific measured water system, which can more accurately estimate the COD value of water.

Ultraviolet-Visible Spectroscopy

Spectrum-structure relationship (prediction and generation).



<https://github.com/PNNL-CompBio/CME-QM>.

https://github.com/PNNL-CompBio/ML_UVvisModels

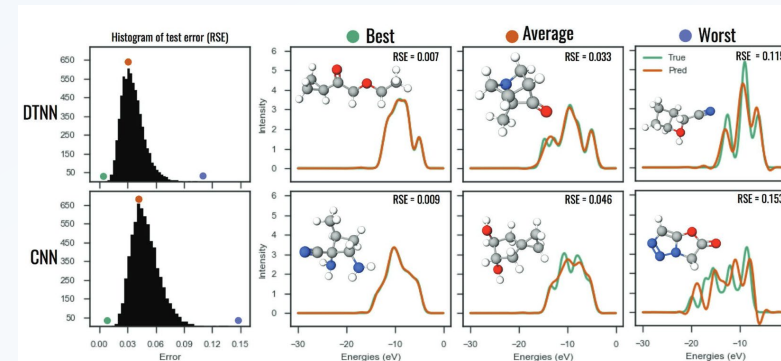
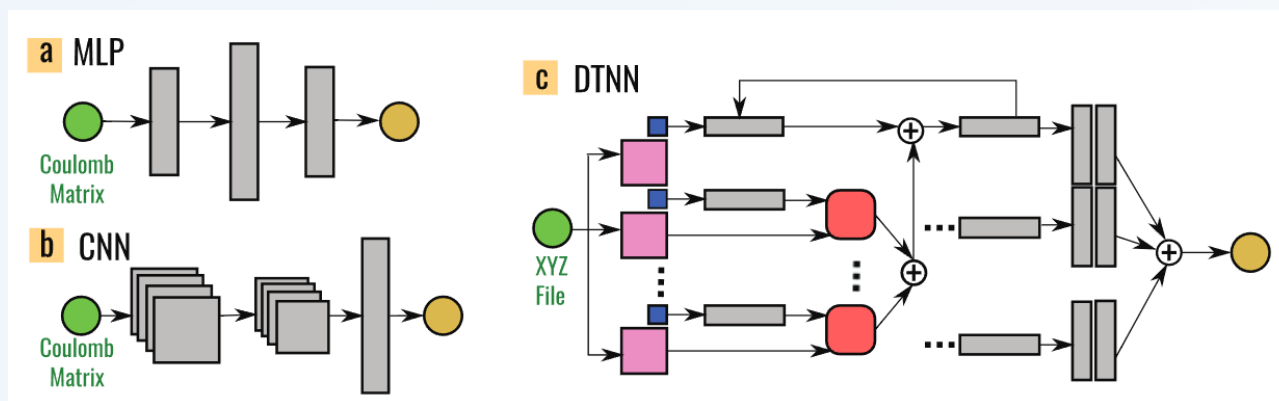
J. Chem. Inf. Model. 2023, 63, 5, 1462–1471

UV, IR, Raman...

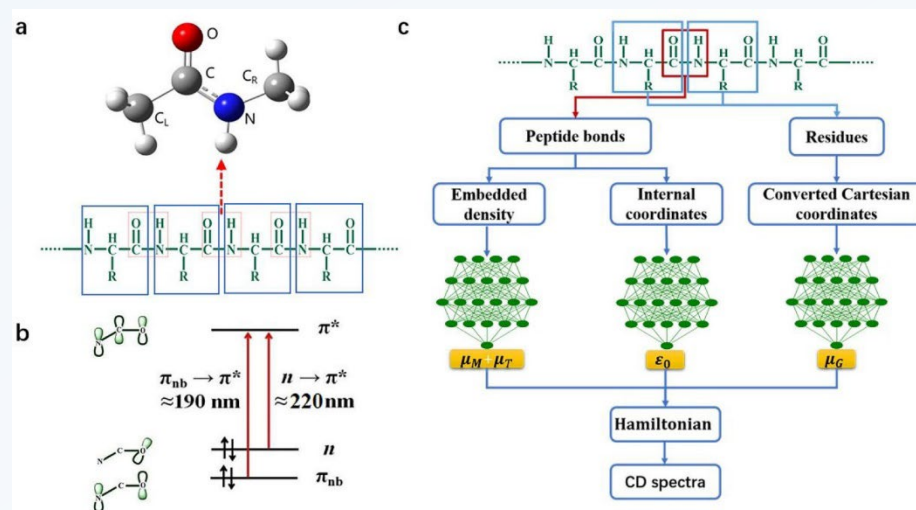
Prof. Pavlo O. Dral

<http://dr-dral.com/>

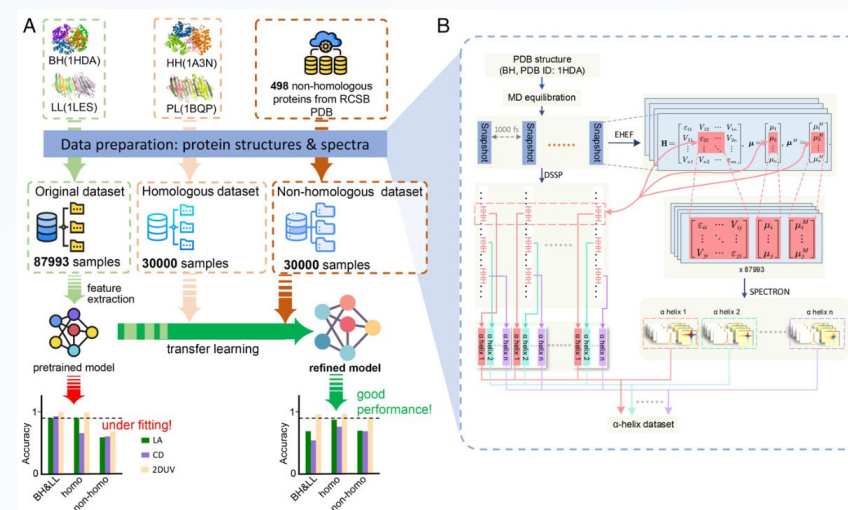
Expand Ultraviolet-Visible Spectroscopy to absorption spectroscopy



Adv. Sci. 2019, 6, 1801367



JACS Au 2021, 1, 12, 2377–2384



Proc Natl Acad Sci USA. 119 (2022) e2202713119

中科大江俊教授课题组<http://staff.ustc.edu.cn/~jiangj1/#link-title>

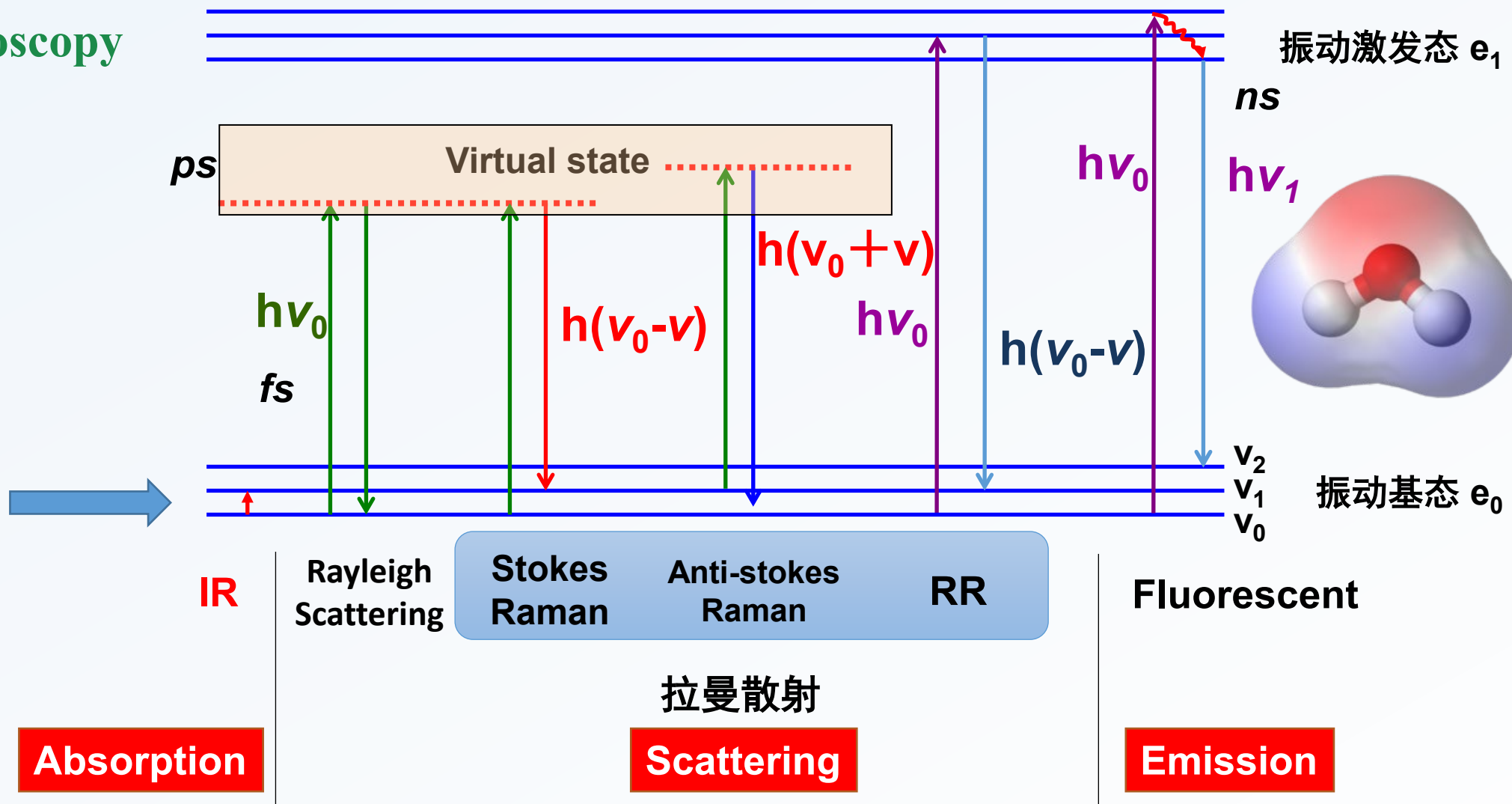
IR Spectroscopy

NIR light is an electromagnetic wave between UV-Vis (UV-Vis) and mid-infrared light (MIR). **Its wavelength range is 700–2500 nm (14,286–4000 cm⁻¹)**, and it could be further divided into two regions: short-wave (700–1100 nm) and long-wave (1100–2500 nm) NIR spectra. The short-wave region is also called the Herschel region in honor of Herschel's discovery of the infrared region (actually the NIR region) in 1800. Instruments extending from the UV-Vis spectrum often take the wavelength (nm or μm) as the horizontal coordinate unit, while instruments extending from the infrared spectrum, especially Fourier-type instruments, take the wavenumber (cm⁻¹) as the horizontal coordinate unit.

The MIR spectroscopy is commonly called IR spectroscopy with **a spectral range of 400–4000 cm⁻¹**, reflecting the spectral information of the vibration and rotation of material molecules, and the fundamental frequency absorption bands of the vast majority of organic compounds and inorganic ions appear in this region. Compared with the NIR spectral region, the MIR spectral region has strong absorption, relatively rich information, and strong group resolution ability, which can distinguish substances with very similar structure. This is also the reason why MIR spectroscopy has been mainly used for the analysis of molecular structure of substances for a long time. In recent years, with the development of instrument manufacturing technology, chemometrics methods, and computers, MIR spectroscopy has been increasingly used in the field of emergency analysis and online process analysis.

UV, IR, Raman...

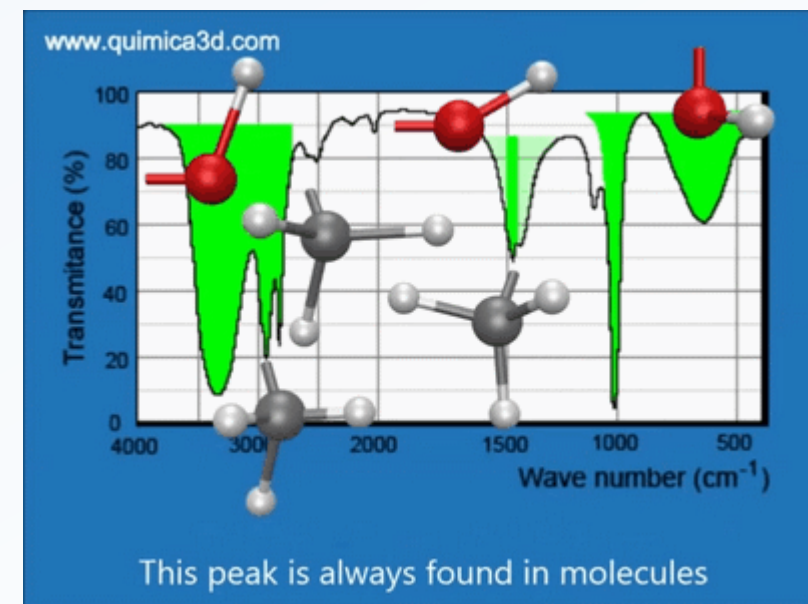
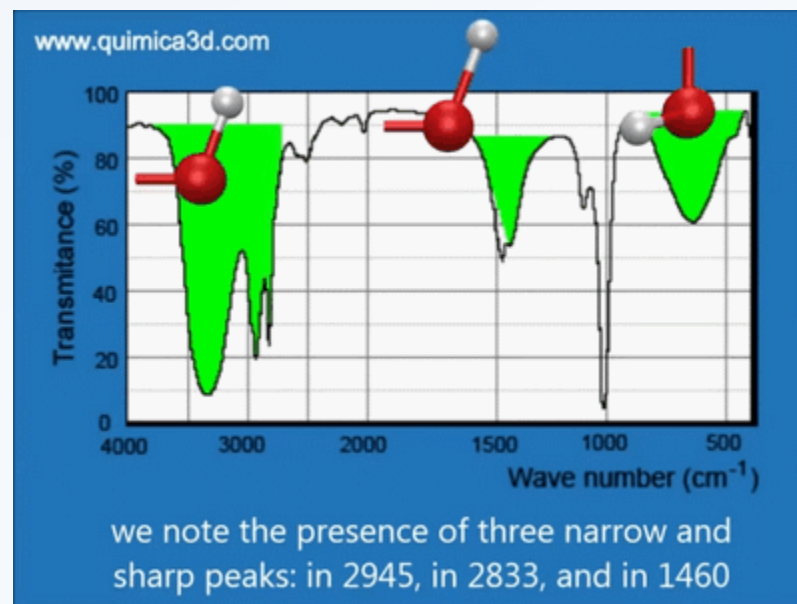
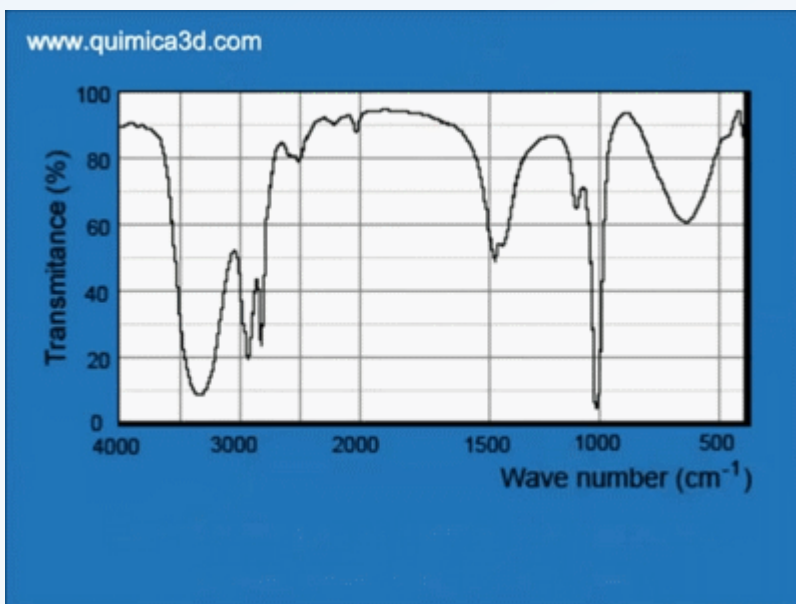
IR Spectroscopy



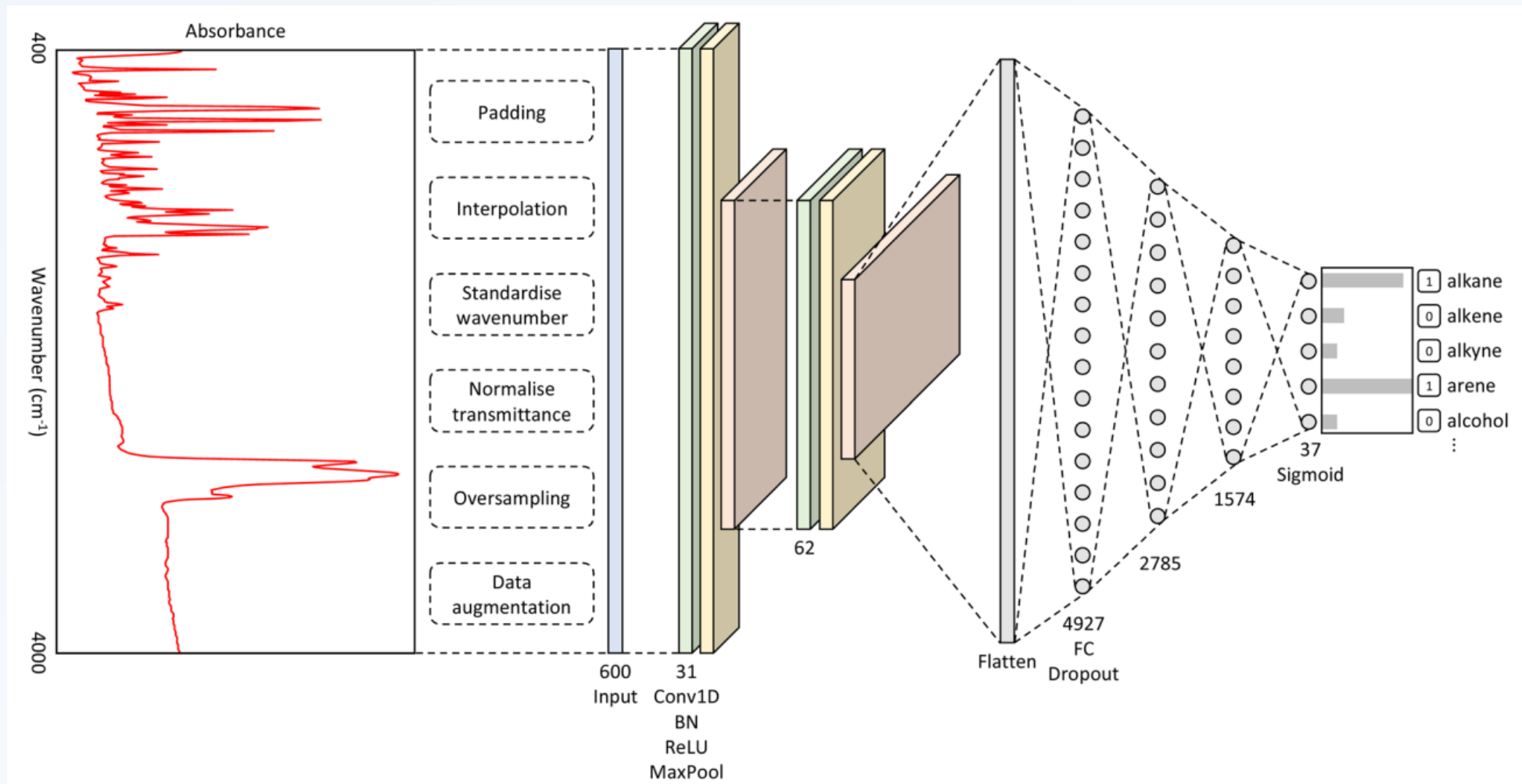
光强比——荧光：入射光：瑞利散射：拉曼散射= $10^2:1:10^{-6}:10^{-10}$

UV, IR, Raman...

IR Spectroscopy



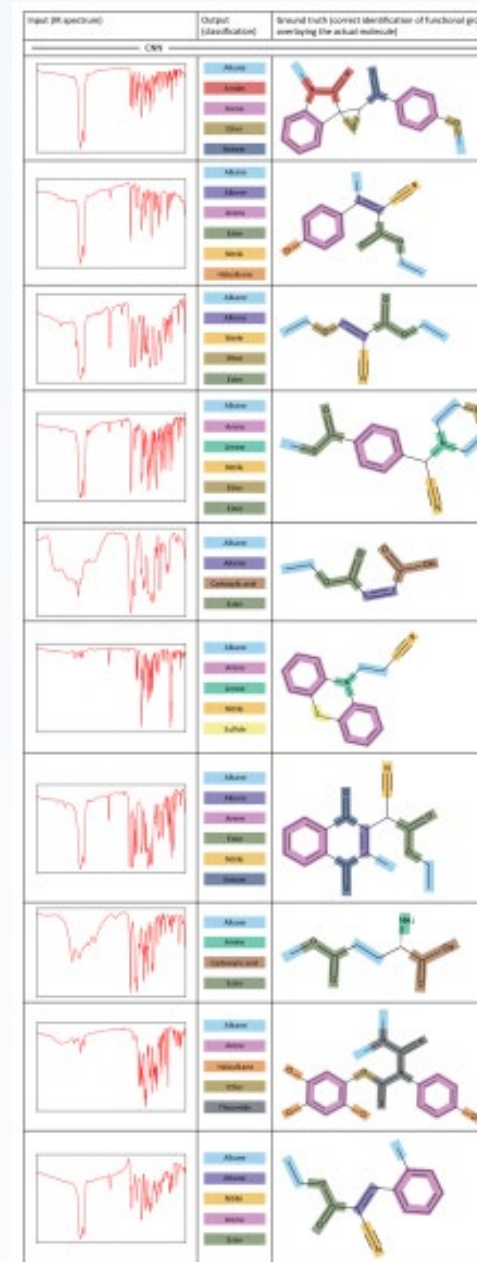
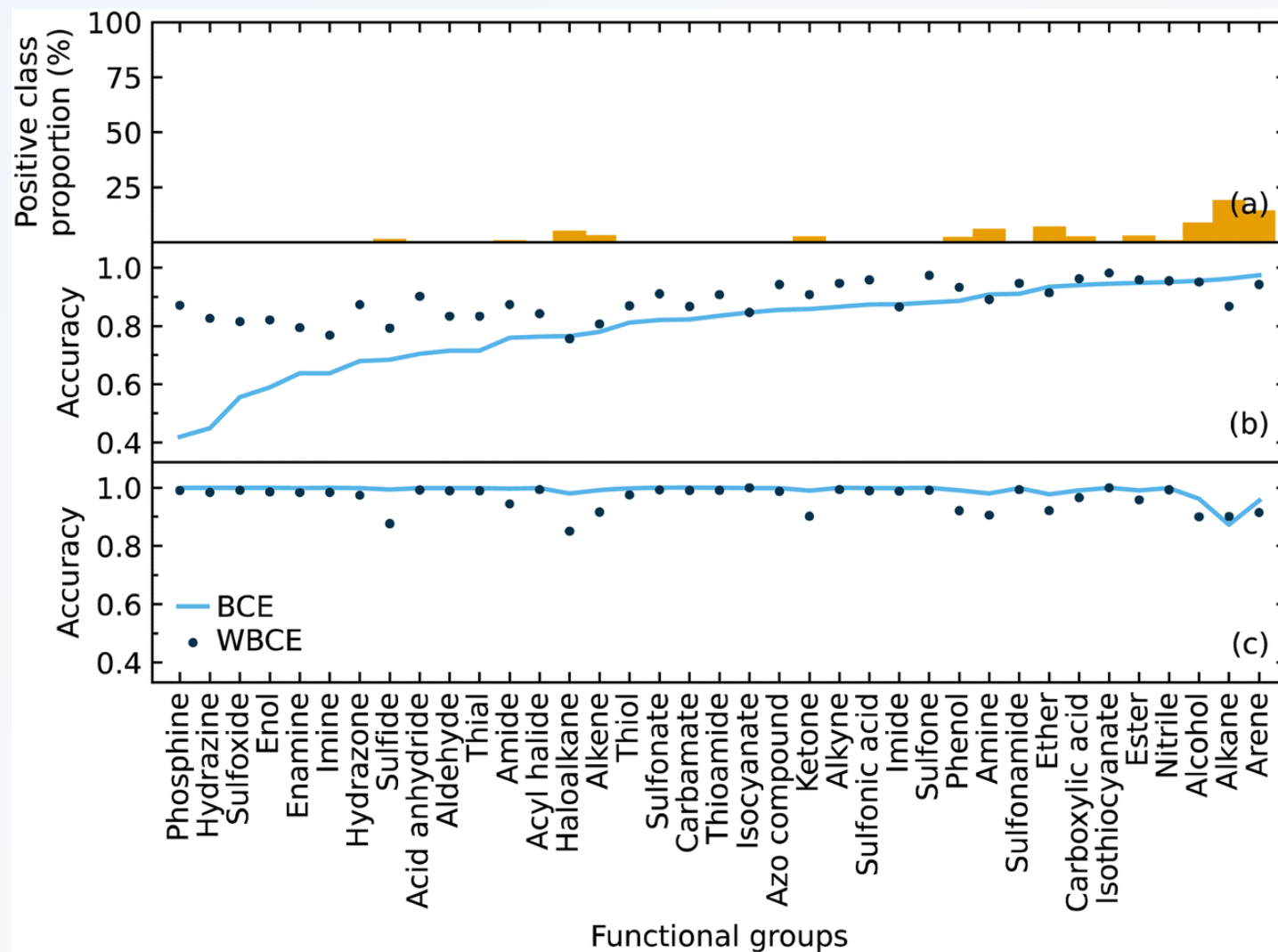
IR Spectroscopy



Chem. Sci., 2023,**14**, 3600-3609

UV, IR, Raman...

IR Spectroscopy



IR Spectroscopy

Code

Spectral Database for Organic Compounds SDBS [Japanese](#) [Introduction](#) [Disclaimer](#) [HELP](#) [Contact](#) [What's New](#) [RIO-DB](#) [FAQ](#) [LINK](#) [AIST](#)

Welcome to Spectral Database for Organic Compounds, SDBS.
This is a free site organized by [National Institute of Advanced Industrial Science and Technology \(AIST\)](#), Japan.

NMR: T.Yamazaki, M.Nabeshima, T.Saito, T.Yamaji, K.Hayamizu, M.Yanagisawa and O.Yamamoto
MS: S.Matsuyama, C.Ono and N.Wasada
ESR: K.Someno
IR: S.Matsuyama, Y.Takizawa, S.Kinugasa, K.Tanabe and T.Tamura
Raman: K.Tanabe and J.Hiraishi

What's New
URL: <https://sdb.sdb.aist.go.jp>
If you can not access to the Search page, check this [FAQ](#).

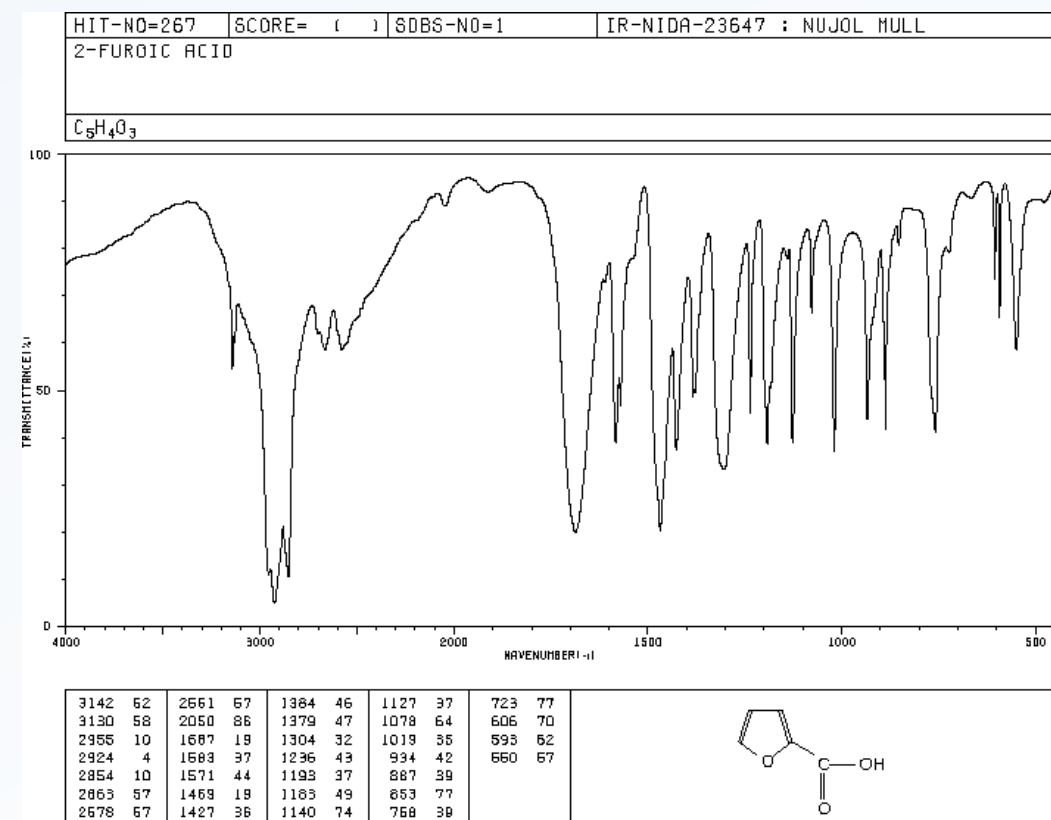
2022.03.31	New data were updated.(360 Spectra)
2021.03.31	New data were updated.(342 Spectra)

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We are doing our best to compile high quality databases. However, there are no such databases without any errors or mistakes. We make no warranties to those effects and shall not be liable for any damage that may result from errors in the database. You can check the page [Known Errors of Mistakes in SDBS](#). When you find new errors or mistakes, please inform us by email (see [this page](#) for contact details).

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SDBSWeb : <https://sdb.sdb.aist.go.jp> (National Institute of Advanced Industrial Science and Technology, date of access)

☐ I agree the disclaimer and use SDBS.



IR Spectroscopy

```
# Define SMARTS
alkane = '[CX4;H3,H2]'
alkene = '[CX3]=[CX3]'
alkyne = '[CX2]#[CX2]'
arene = '[cX3]1[cX3][cX3][cX3][cX3][cX3]1'
haloalkane = '[#6][F,Cl,Br,I]'
alcohol = '[#6][OX2H]'
aldehyde = '[CX3H1](=O)[#6,H]'
ketone = '[#6][CX3](=O)[#6]'
carboxylic_acid = '[CX3](=O)[OX2H]'
acid_anhydride = '[CX3](=[OX1])[OX2][CX3](=[OX1])'
acyl_halide = '[CX3](=[OX1])[F,Cl,Br,I]'
ester = '[#6][CX3](=O)[OX2H0][#6]'
ether = '[OD2]([#6])[#6]'
amine = '[NX3;H2,H1,H0;!$(NC=O)]'
amide = '[NX3][CX3](=[OX1])[#6]'
nitrile = '[NX1]#[CX2]'
imide = '[CX3](=[OX1])[NX3][CX3](=[OX1])'
```

```
if mol is not None:
    for fg in fg_list:
        pattern = Chem.MolFromSmarts(fg)
        match = mol.HasSubstructMatch(pattern)
        if match == True:
            label_temp.append(1)
        elif match == False:
            label_temp.append(0)
```

UV, IR, Raman...

IR Spectroscopy

#	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM
1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	InChI=1S/nist	
2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	InChI=1S/nist	
3	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	InChI=1S/nist	
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	InChI=1S/nist	
5	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	InChI=1S/nist	
6	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	InChI=1S/nist	
7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	InChI=1S/nist	
8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	InChI=1S/nist	

VM	VN	VO	VP	VQ	VR	VS	VT	VU	VV	VW	VX	VY	VZ	WA	WB	WC
0. 810666	0. 820548	0. 829278	0. 834218	0. 839159	0. 8441	0. 846875	0. 85	0. 852673	0. 853125	0. 85307	0. 846258	0. 833251	0. 818614	0. 804113	0. 8	InChI=1S/
0. 746333	0. 766097	0. 790138	0. 814842	0. 833103	0. 8375	0. 8375	0. 842339	0. 854894	0. 869717	0. 878125	0. 876254	0. 875562	0. 880503	0. 886512	0. 89351	InChI=1S/
0. 916167	0. 901327	0. 896875	0. 896875	0. 896875	0. 893049	0. 8875	0. 8875	0. 892846	0. 91307	0. 93454	0. 947492	0. 948876	0. 938994	0. 929113	0. 922115	InChI=1S/
0. 863249	0. 88551	0. 889805	0. 899687	0. 904784	0. 9059	0. 903125	0. 903982	0. 908923	0. 915228	0. 915515	0. 903762	0. 889501	0. 877242	0. 869738	0. 865625	InChI=1S/
0. 904416	0. 909375	0. 908222	0. 90625	0. 90625	0. 910076	0. 919958	0. 932964	0. 940173	0. 937897	0. 92796	0. 915008	0. 9125	0. 917256	0. 934456	0. 949279	InChI=1S/
0. 932541	0. 9375	0. 936347	0. 934375	0. 936034	0. 93785	0. 940625	0. 935786	0. 925904	0. 912897	0. 903015	0. 893133	0. 877001	0. 862364	0. 847863	0. 84375	InChI=1S/
0. 925	0. 925	0. 932403	0. 937343	0. 945409	0. 95	0. 941335	0. 91275	0. 870144	0. 817609	0. 77456	0. 733162	0. 692942	0. 658356	0. 633225	0. 619231	InChI=1S/
0. 922521	0. 925	0. 916445	0. 888752	0. 790843	0. 699593	0. 644709	0. 689518	0. 721837	0. 77438	0. 822095	0. 832504	0. 798068	0. 744089	0. 688325	0. 660337	InChI=1S/
0. 883084	0. 873202	0. 859042	0. 838283	0. 815237	0. 79965	0. 796875	0. 805697	0. 820519	0. 835739	0. 83421	0. 817516	0. 795189	0. 777989	0. 763488	0. 759375	InChI=1S/
0. 670687	0. 68551	0. 696055	0. 705937	0. 712694	0. 719451	0. 729333	0. 745768	0. 770068	0. 798809	0. 818805	0. 814391	0. 790941	0. 763858	0. 73635	0. 72524	InChI=1S/
0. 617666	0. 647347	0. 665958	0. 674843	0. 689353	0. 714952	0. 756038	0. 809429	0. 853087	0. 867783	0. 846435	0. 808779	0. 783251	0. 778125	0. 779194	0. 784135	InChI=1S/

IR Spectroscopy

```
In [5]: from iterstrat.ml_stratifiers import MultilabelStratifiedShuffleSplit

shuffle_splits = 1
# 定义分层抽样器1, 使测试集占全集20%。test_size=1/(1+1+3)=0.2
str_shuffle_1 = MultilabelStratifiedShuffleSplit(n_splits=shuffle_splits, test_size=0.2, random_state=SEED)
# 定义分层抽样器2, 使验证集占剩训练+验证数据的25%。test_size=1/(1+3)=0.25
str_shuffle_2 = MultilabelStratifiedShuffleSplit(n_splits=shuffle_splits, test_size=0.25, random_state=SEED)

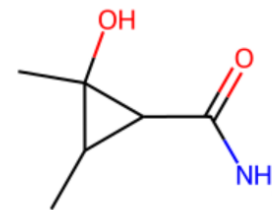
for train_val_index, test_index in str_shuffle_1.split(data, label):
    X_train_val = data[train_val_index]
    X_test = data[test_index]
    y_train_val = label[train_val_index]
    y_test = label[test_index]
    smiles_train_val = smiles[train_val_index]
    smiles_test = smiles[test_index]
for train_index, val_index in str_shuffle_2.split(X_train_val, y_train_val):
    X_train = X_train_val[train_index]
    X_val = X_train_val[val_index]
    y_train = y_train_val[train_index]
    y_val = y_train_val[val_index]
    smiles_train = smiles_train_val[train_index]
    smiles_val = smiles_train_val[val_index]
X_train.shape, y_train.shape, X_val.shape, y_val.shape, X_test.shape, y_test.shape
```

```
Out[5]: ((6507, 500), (6507, 10), (2169, 500), (2169, 10), (2170, 500), (2170, 10))
```

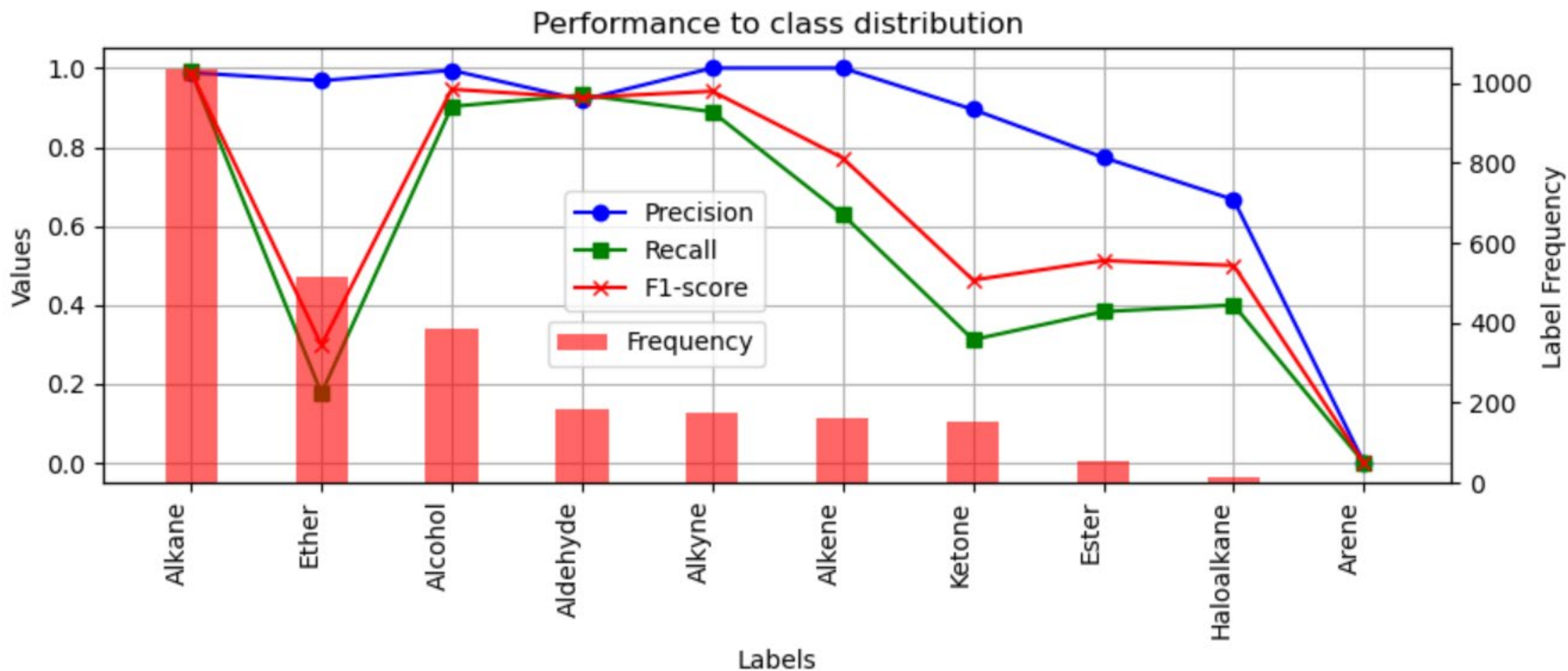
```
label_names = [
    'Alkane',
    'Alkene',
    'Alkyne',
    'Arene',
    'Haloalkane',
    'Alcohol',
    'Aldehyde',
    'Ketone',
    'Ester',
    'Ether',
]
```

```
from rdkit import Chem
index = 201
mol = Chem.MolFromSmiles(smiles[index])
print([label_names[i] for i in range(10) if label[index][i]])
mol
```

['Alkane', 'Alcohol']



IR Spectroscopy



IR Spectroscopy

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JOURNAL OF THE AMERICAN CHEMICAL SOCIETY

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Article

A Machine Learning Protocol for Predicting Protein Infrared Spectra

Sheng Ye,[⊥] Kai Zhong,[⊥] Jinxiao Zhang,[⊥] Wei Hu, Jonathan D. Hirst, Guozhen Zhang, Shaul Mukamel, and Jun Jiang*



Cite This: *J. Am. Chem. Soc.* 2020, 142, 19071–19077



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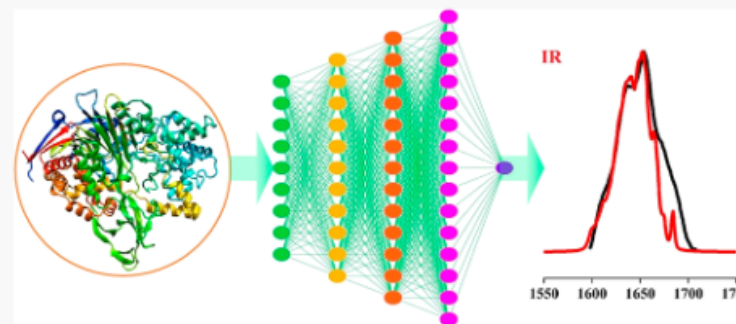


Article Recommendations

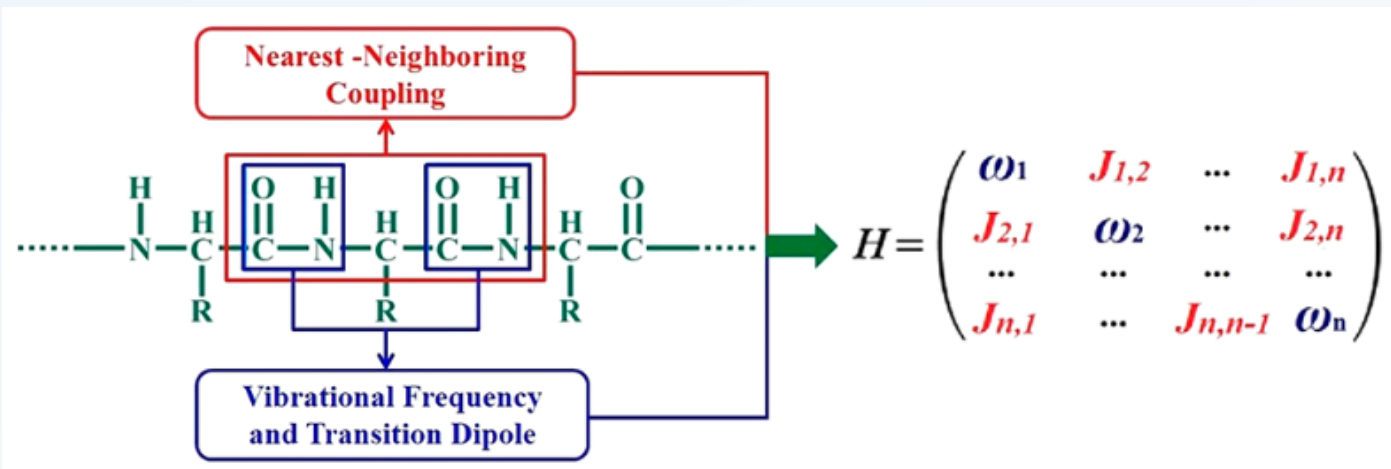


Supporting Information

ABSTRACT: Infrared (IR) absorption provides important chemical fingerprints of biomolecules. Protein secondary structure determination from IR spectra is tedious since its theoretical interpretation requires repeated expensive quantum-mechanical calculations in a fluctuating environment. Herein we present a novel machine learning protocol that uses a few key structural descriptors to rapidly predict amide I IR spectra of various proteins and agrees well with experiment. Its transferability enabled us to distinguish protein secondary structures, probe atomic structure variations with temperature, and monitor protein folding. This approach offers a cost-effective tool to model the relationship between protein spectra and their biological/chemical properties.



IR Spectroscopy

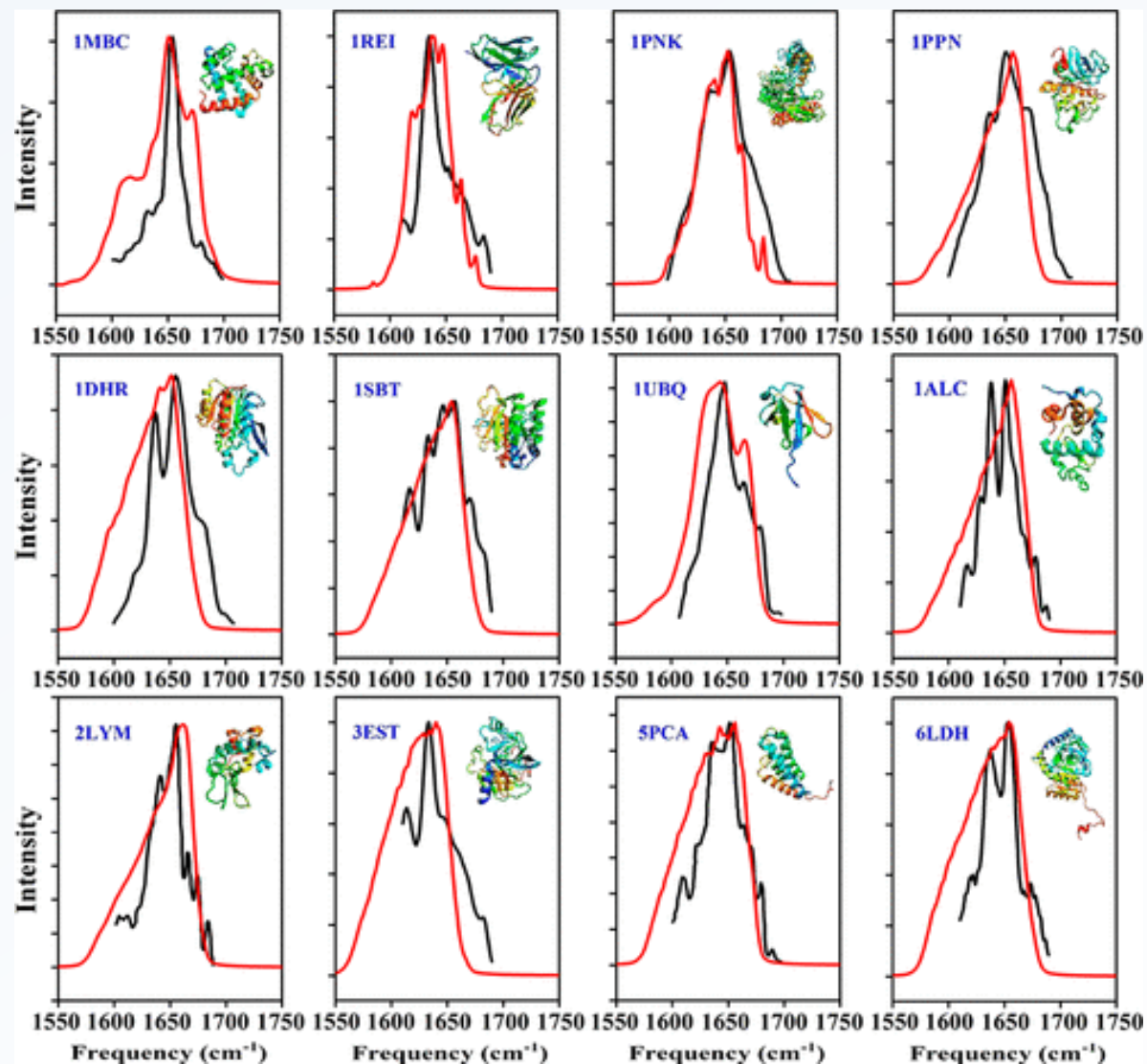
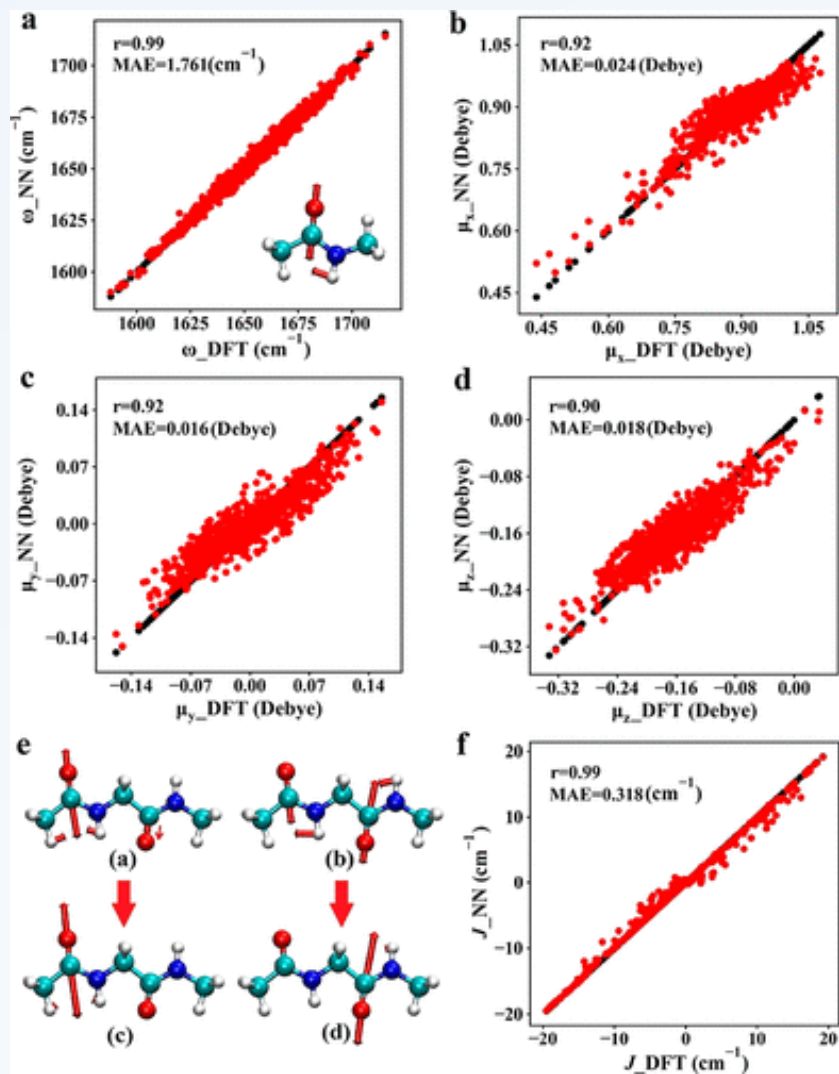


$$J_{ij} = \frac{1}{4\pi\epsilon_0} \left(\frac{\vec{\mu}_i \cdot \vec{\mu}_j}{r_{ij}^3} - 3 \frac{(\vec{\mu}_i \cdot \vec{r}_{ij})(\vec{\mu}_j \cdot \vec{r}_{ij})}{r_{ij}^5} \right), \text{ where } \epsilon_0 \text{ is the dielectric}$$

constant, $\vec{\mu}_i$ ($\vec{\mu}_j$) is the transition dipole of peptide bond i (j), and r_{ij} is the vector connecting dipole i and j . For two neighboring oscillators, the couplings are computed directly using a dipeptide model.^{10,25}

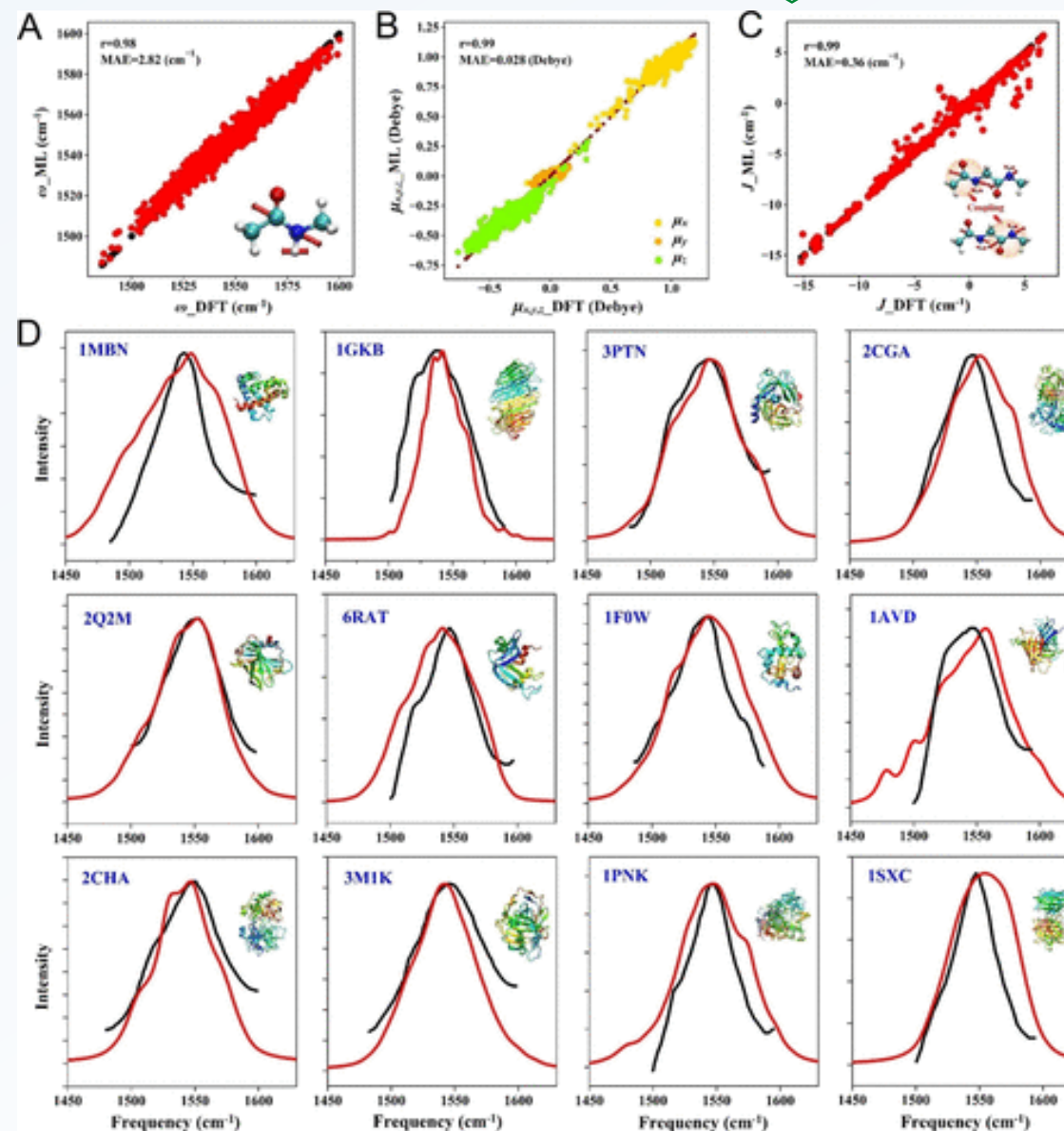
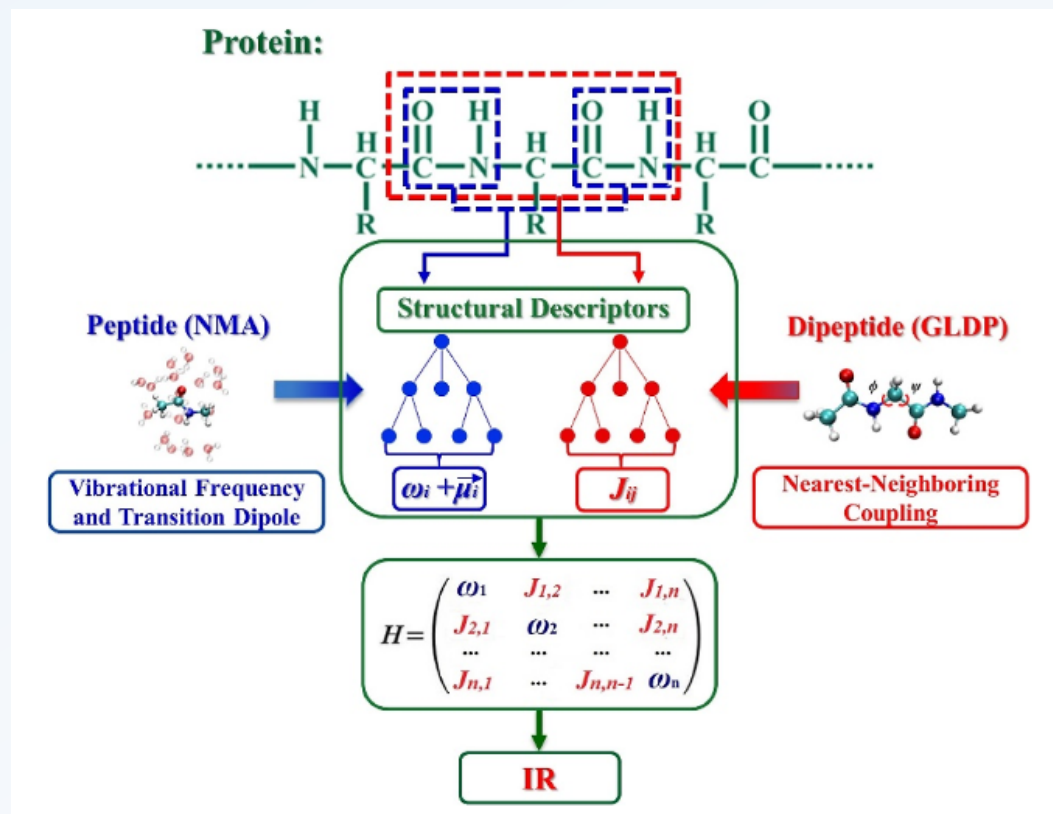
Neural Network Architecture and Descriptors. Multi-layer perceptron (MLP) with a supervised training scheme using a back-propagation algorithm implemented in TensorFlow³² was used to predict the properties $(\omega_i, \vec{\mu}_i, J_{ij})$ from the geometric descriptors. We have chosen MLP for two reasons: (1) it handles regression problems well, which this work belongs to; (2) it is simple and easy to implement.^{13,33,34} The NN consists of one input layer, three hidden layers, and one output layer. For each hidden NN layer we used the rectified linear unit activation function.³⁵ The number of hidden layer neurons are 32, 64, and 128, respectively. We adopted different learning rates of the Adam optimizer³⁶ in TensorFlow for the training process to avoid being trapped into local minima. The learning rate is set to be halved every 500 steps, and the initial learning rate is set to 0.0004. For the training, we added L2 regularization³⁷ to the architecture of the neural network to prevent overfitting. Hyperparameters were optimized by using a random search algorithm³⁸ in TensorFlow (including neurons for hidden layers, learning rate, and L2 regularization parameters) to create a reasonable ML protocol in this work.

IR Spectroscopy



UV, IR, Raman...

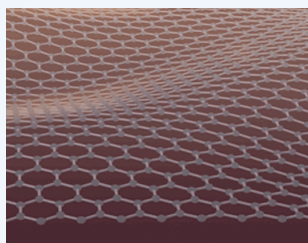
IR Spectroscopy



J. Am. Chem. Soc. 2024, 146, 4, 2663-2672

UV, IR, Raman...

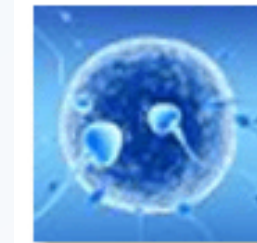
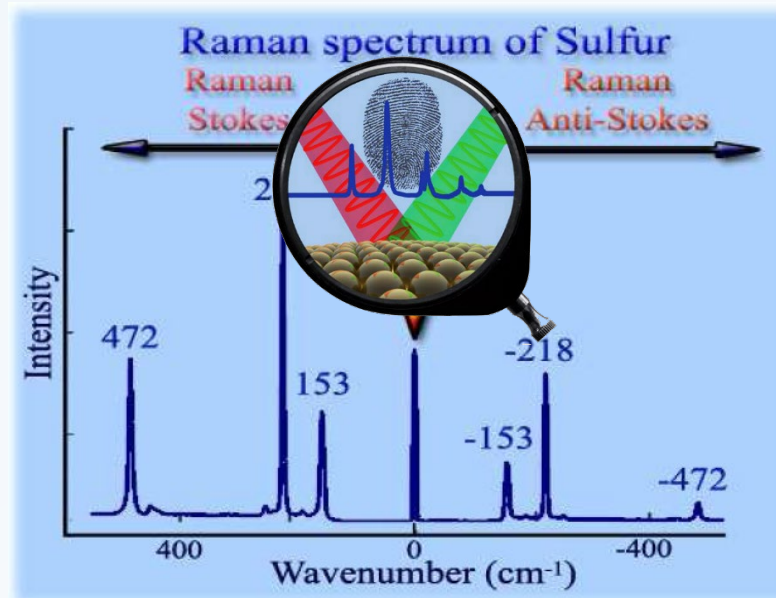
Raman Spectroscopy



Physics



Security



Biology



Space



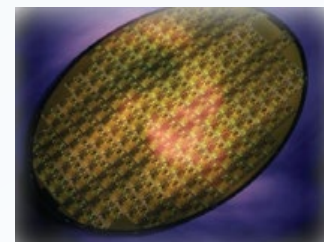
Environment



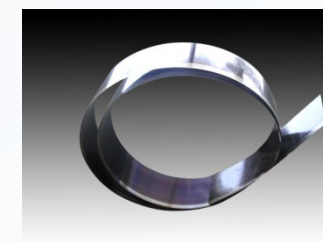
Medicine



Minerals



Semiconductor



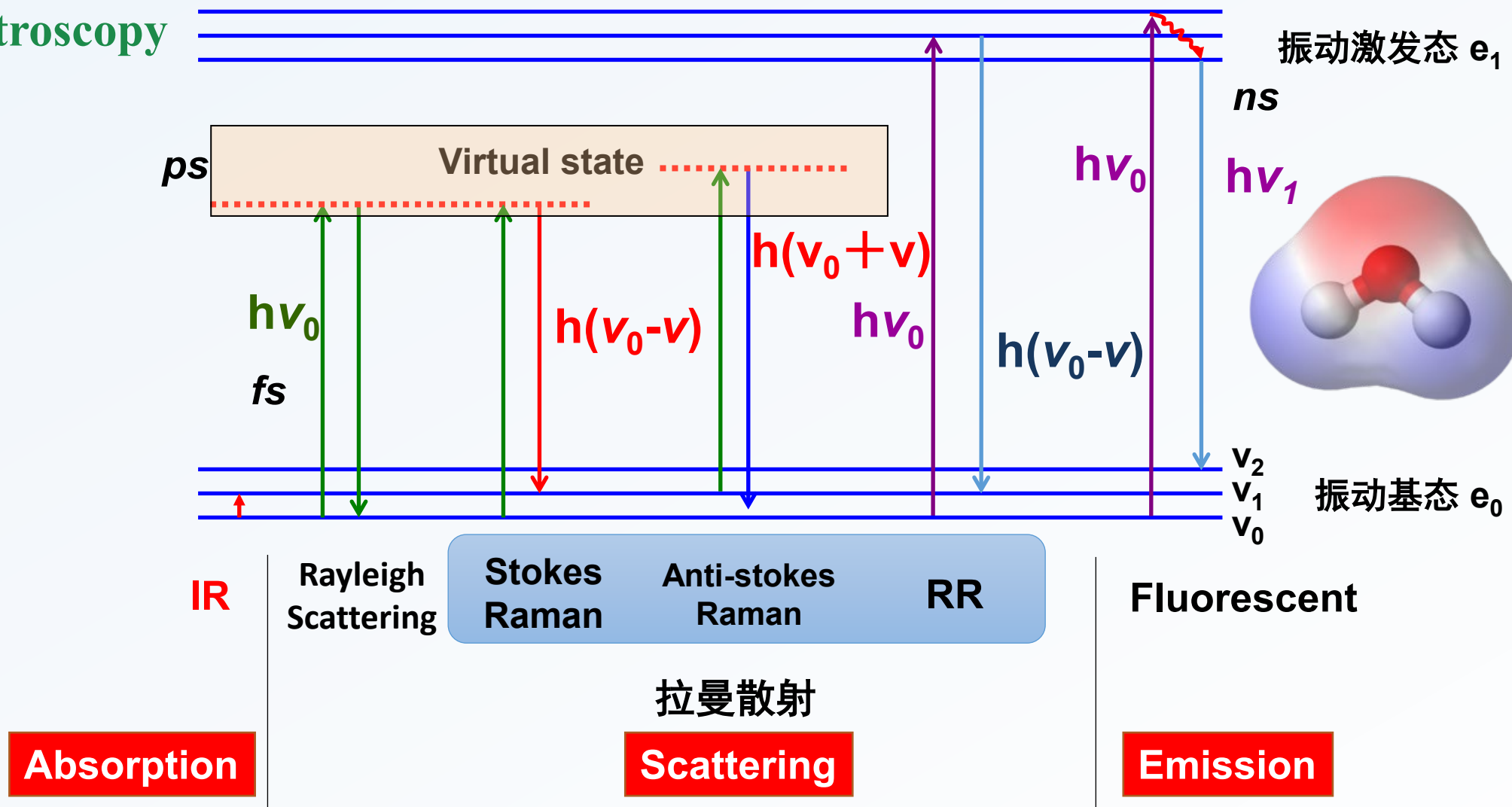
Chemistry



Food

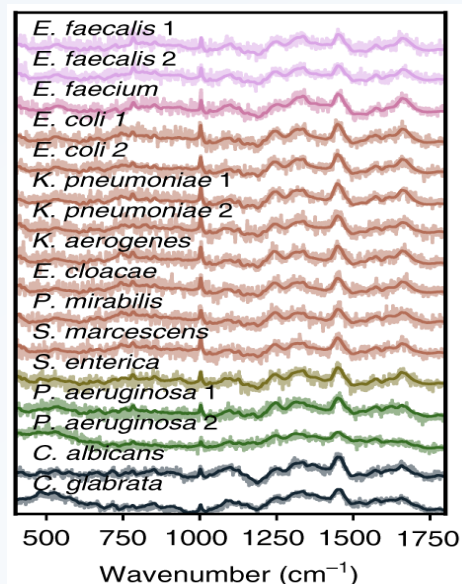
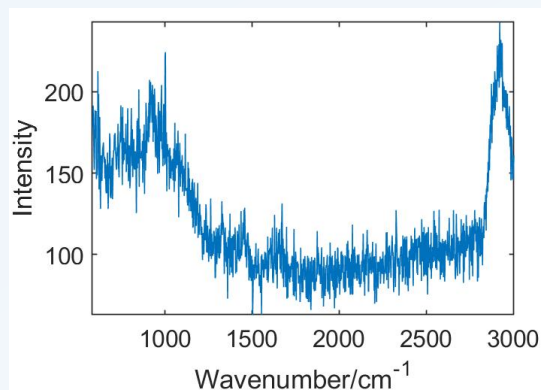
UV, IR, Raman...

Raman Spectroscopy



光强比——荧光：入射光：瑞利散射：拉曼散射= $10^2:1:10^{-6}:10^{-10}$

Raman Spectroscopy



谱图整形

降噪、弱信号提取

峰识别

物种识别、分类

解释性

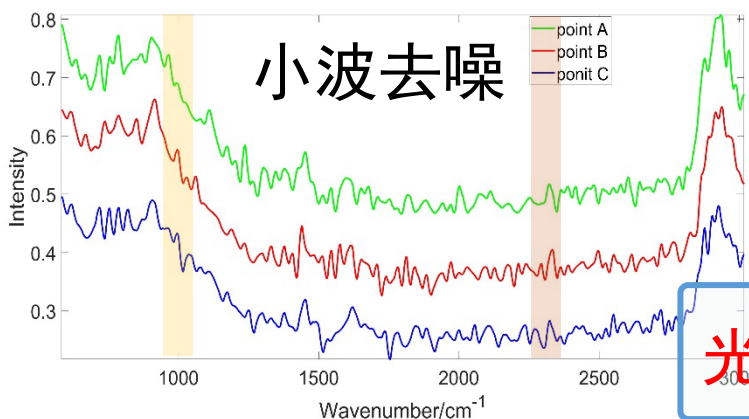
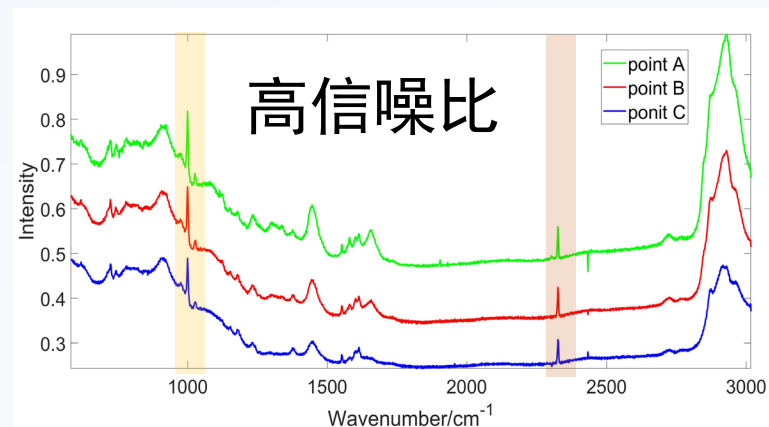
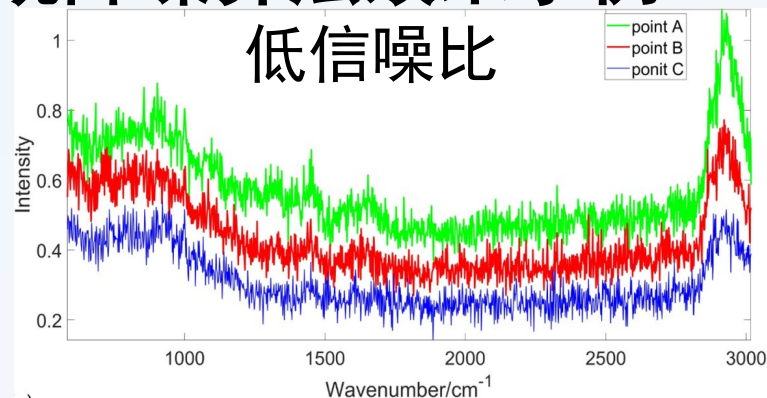
Chemometrics



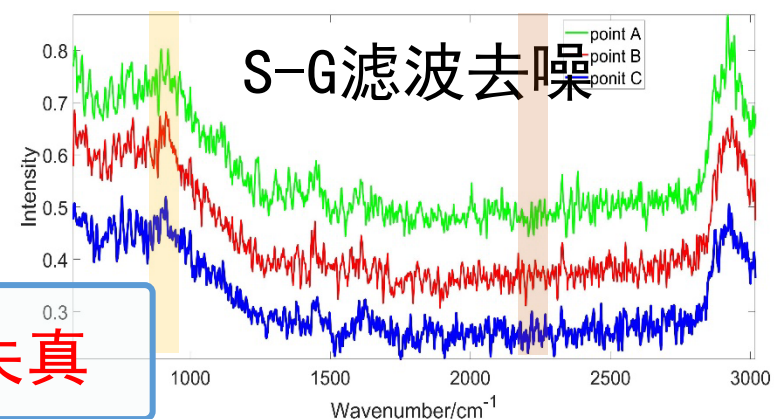
AI

Raman Spectroscopy

传统降噪算法效果示例：



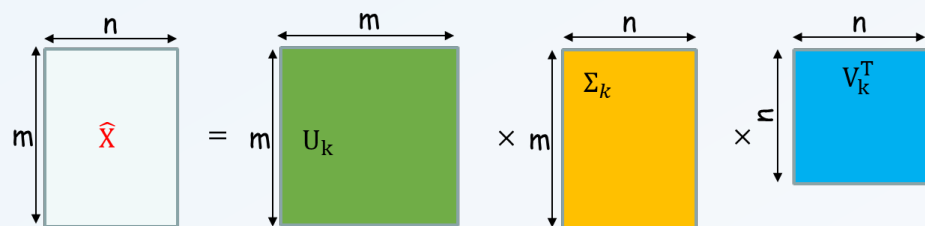
光谱失真



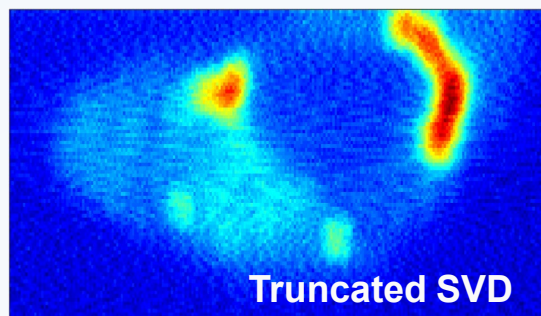
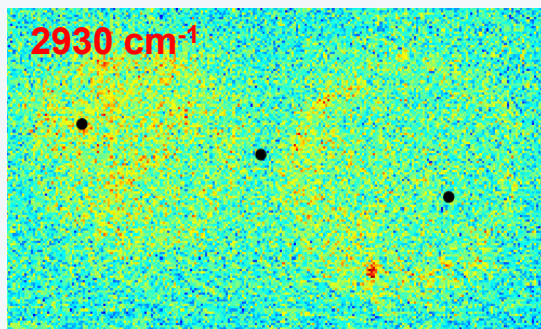
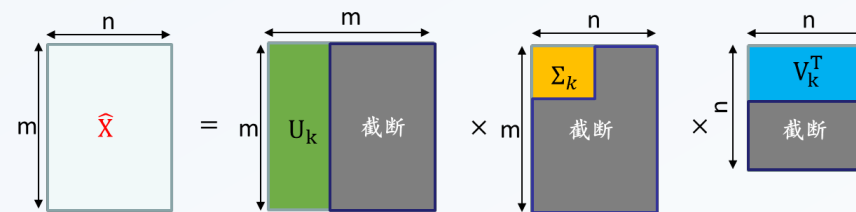
传统降噪算法可信度低；参数选择困难；对样品有选择性等。

Raman Spectroscopy

SVD



截断SVD低秩矩阵近似



SVD利用成像数据之间的之间的相关性，实现信号和噪音的分离，在拉曼成像中得到广泛应用。

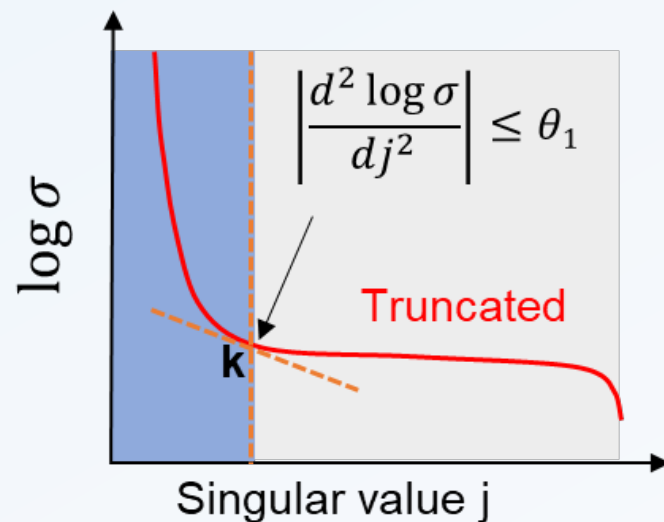
截断奇异值的低秩矩阵逼近：大的奇异值分量是信号分量，小的奇异值分量是噪音分量，通过保留前面少数奇异值分量达到信号提取的目的；

存在问题：可能把弱的信号分量丢掉，保留强的噪音分量，缺少合理的奇异值分量选择标准，去噪效果不佳

自适应低秩矩阵逼近(ALRMA)信号提取算法

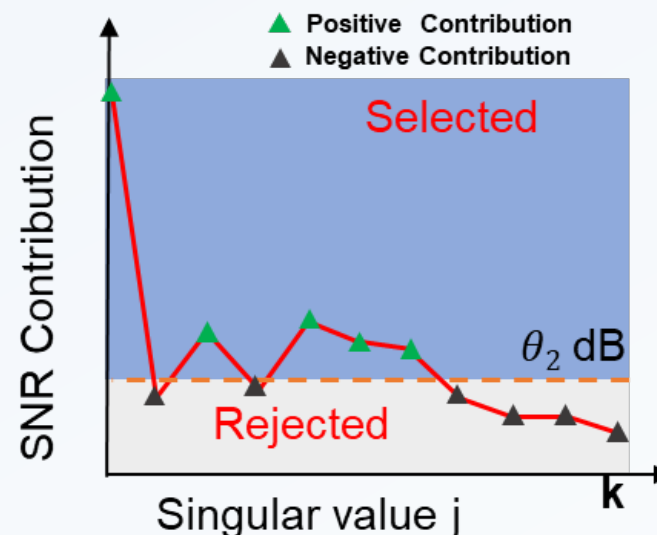
利用SNR贡献指标来筛选信号成分

Phase 1 Truncation

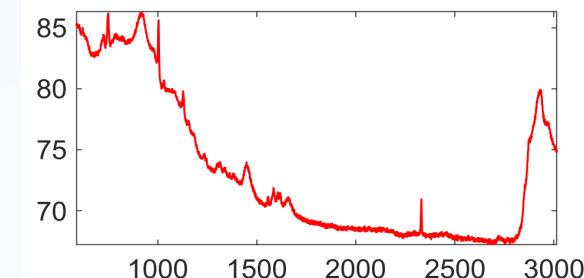
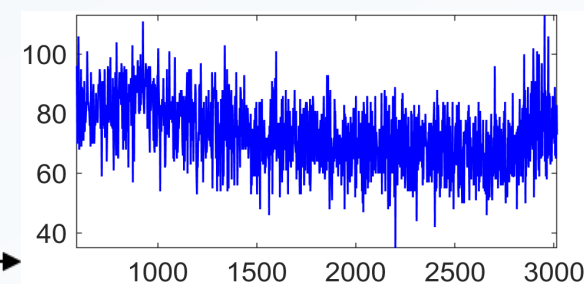
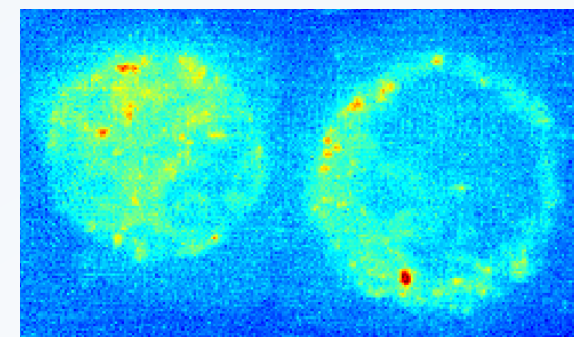


根据奇异值曲线进行截断，来缩小奇异值搜索空间，加快算法的运行速度

Phase 2 Selection



对保留下来的奇异值分量进行SNR贡献评估，
+SNR分量--信号重构
-SNR分量--剔除



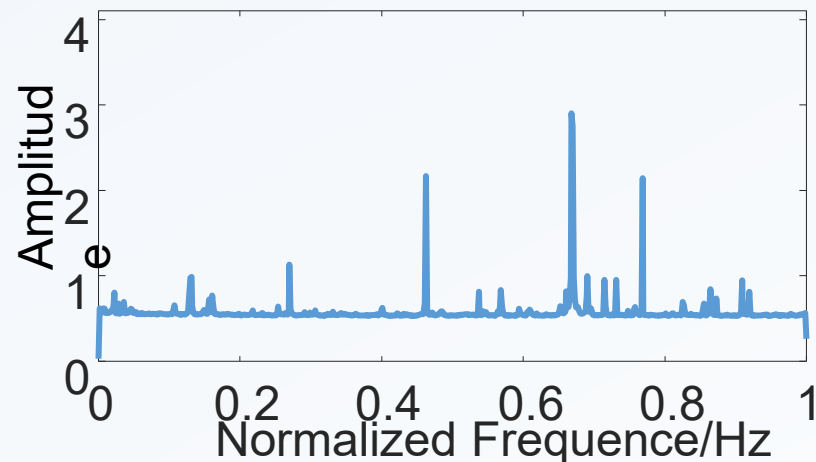
ALRMA信号提取性能强，适用于成像有相关的数据，提升成像速度5~10倍

Raman Spectroscopy

仪器噪音是否有特定模式？



Raman-11, Nanophoton

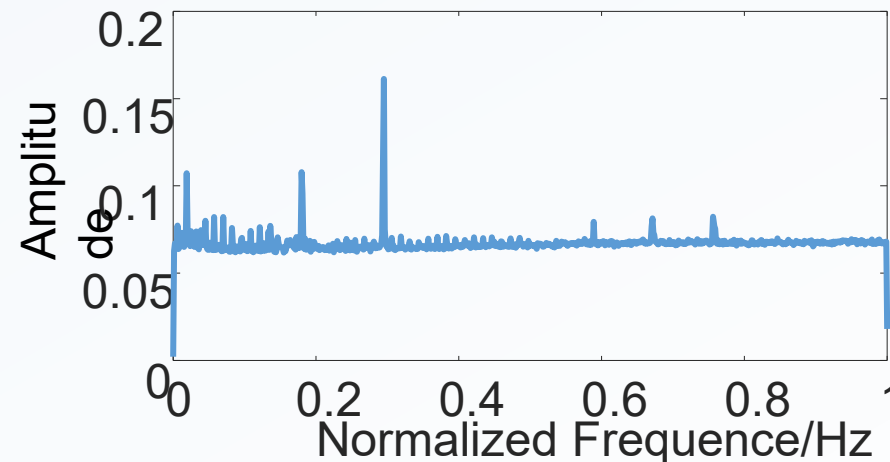


Raman-11: 1万条噪音谱功率谱

拉曼光谱仪器噪音存在很强的规律性！



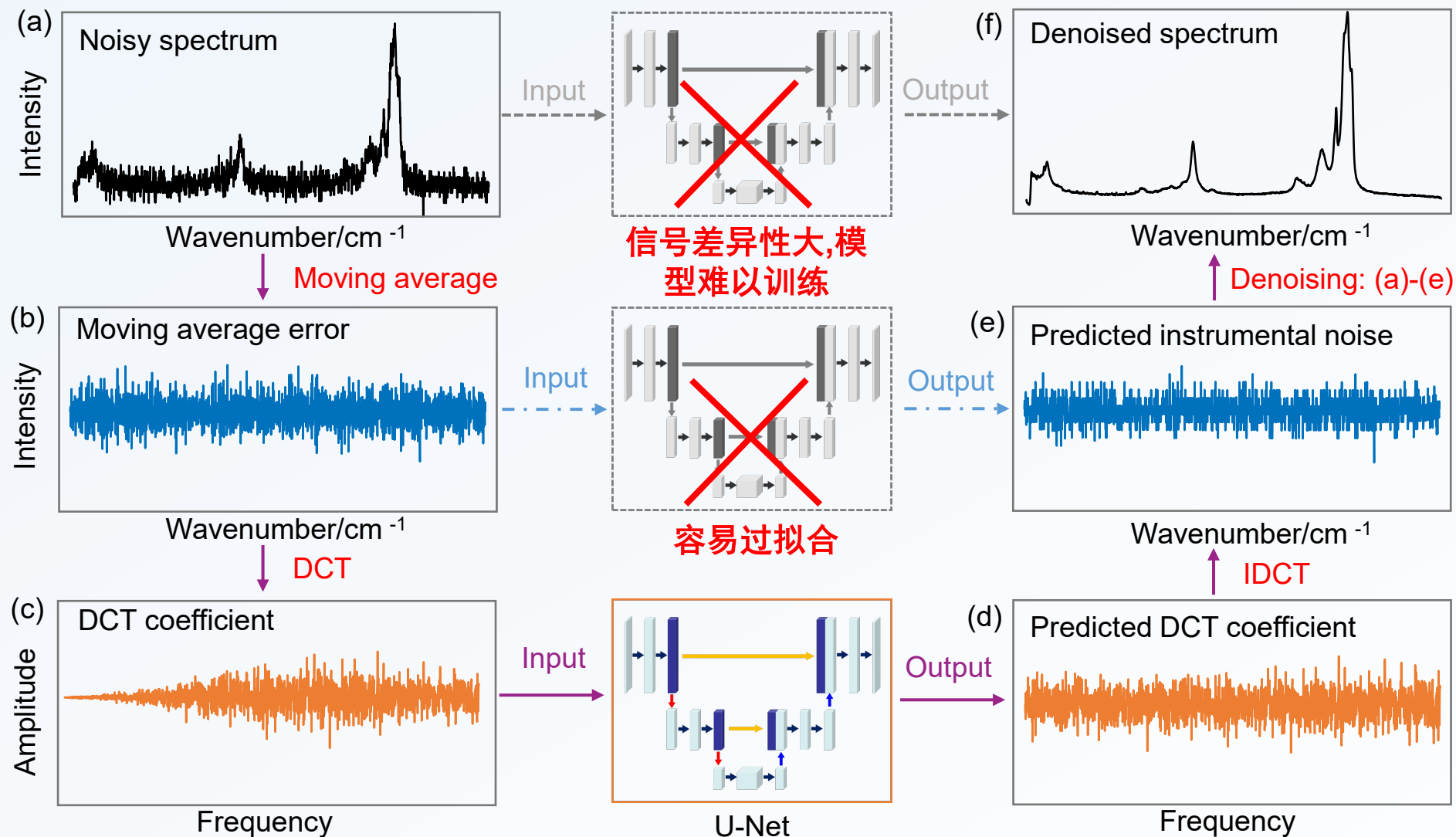
LabRAM HR 800, Horiba



LabRAM HR: 1万条噪音谱功率谱

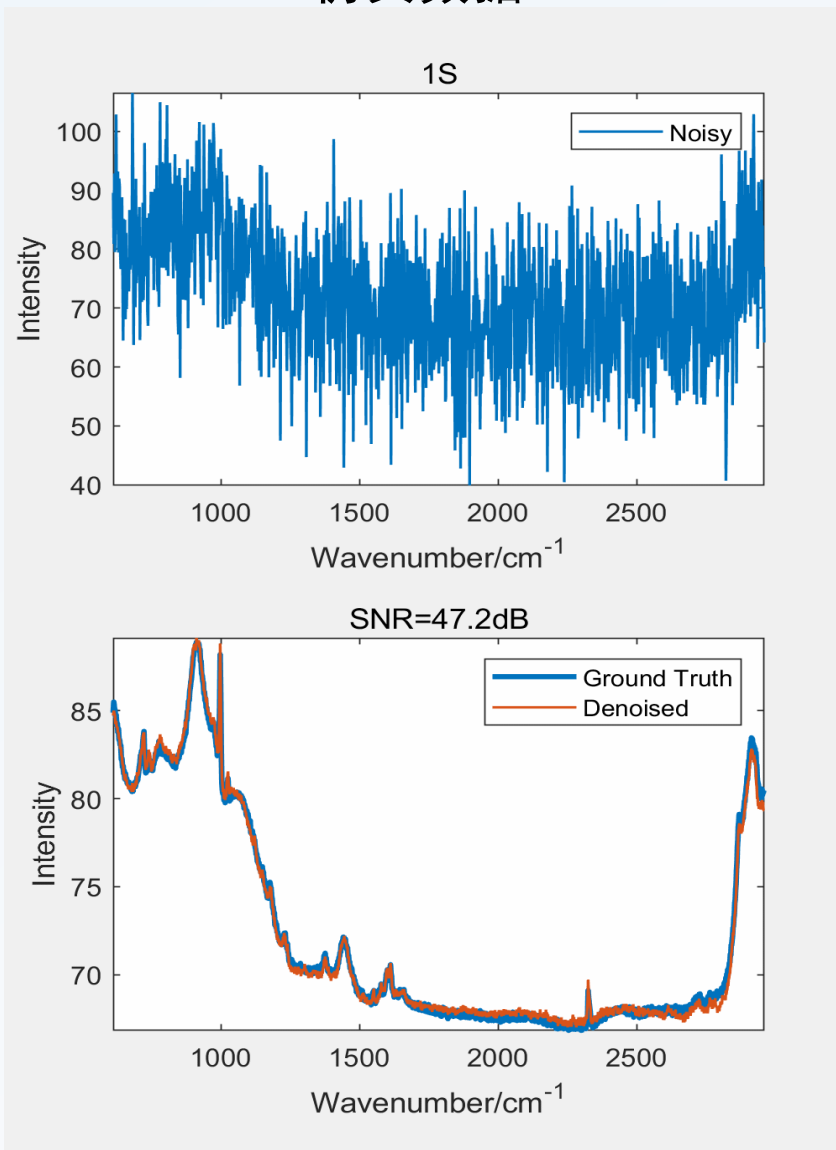
UV, IR, Raman...

使用U-Net学习仪器噪音模式：三种技术路线

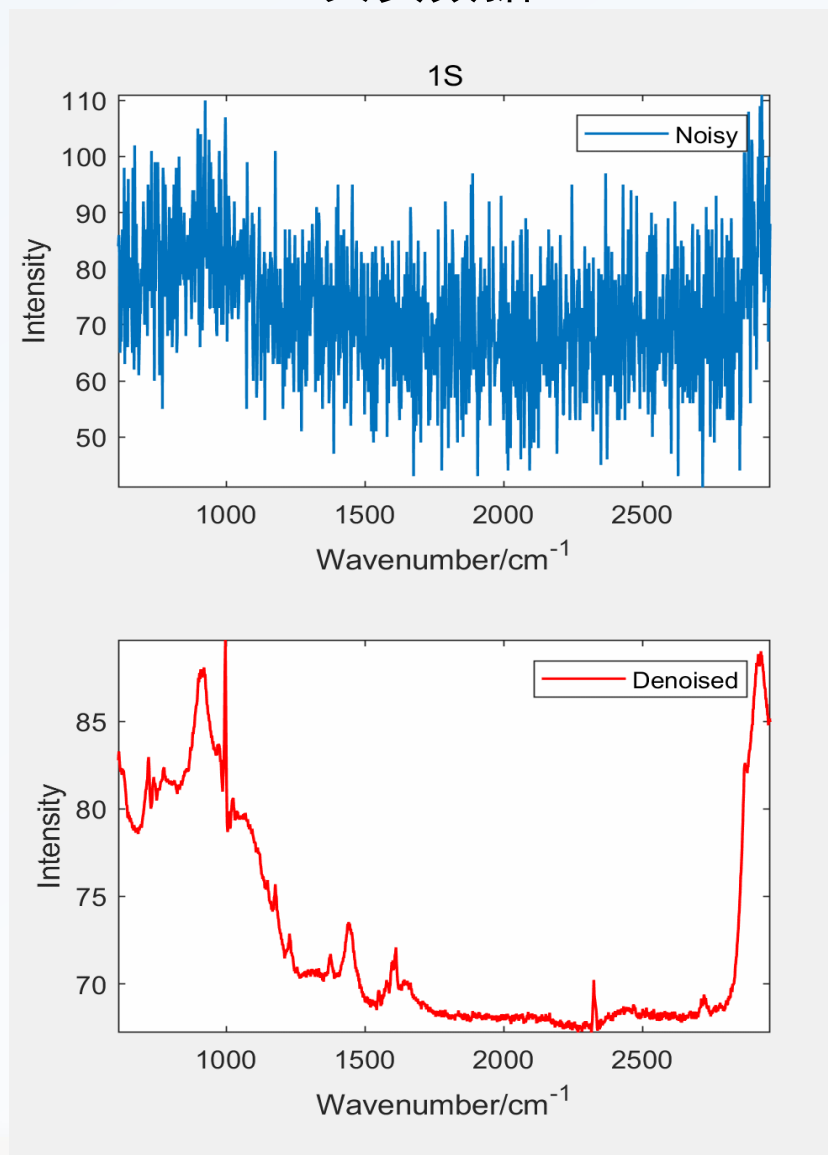


----- Roadmap 1 - . - . - Roadmap 2 ——— Roadmap 3

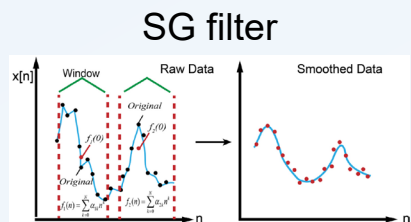
仿真数据



真实数据

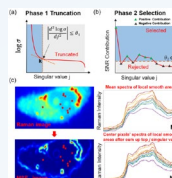


UV, IR, Raman...



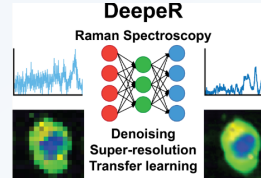
Savitzky Abraham et al. Anal. Chem. 1964, 36, 8, 1627

ALRMA



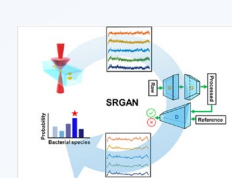
H. He et al. Journal of Raman Spectroscopy. 2020, 51, 2552

ResUNet



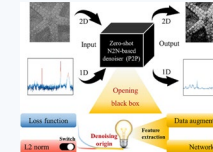
Horgan et. al. Anal.Chem.2021, 93, 15850

SRGAN



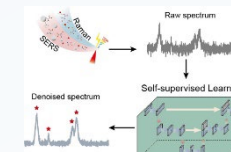
X.Y Ma et. al. Anal.Chem. 2022, 94, 577

P2P



S.H. Luo et. al. Anal.Chem. 2024, 96, 4086

GRSD



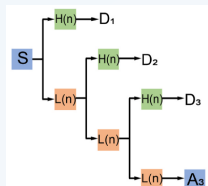
S.Y. Wu et. al. Anal.Chem. 2024, 96, 17476

Empirical method

Supervised Learning

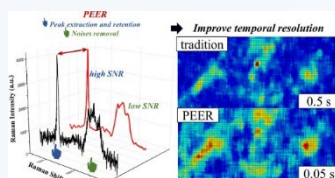
Self-Supervised Learning

WTD



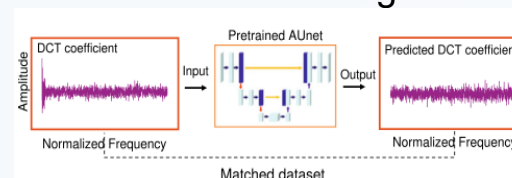
S.Y. Fang et. al. TrAC Trends in Anal. Chem. 2024, 172, 117578 (Review)

PEER



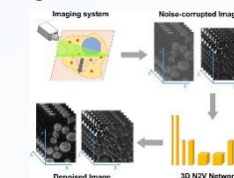
S.H. Luo et. al. Anal.Chem. 2021, 93, 8408

Noise Learning



H. He et al. Nat. Commun. 2024, 15, 754

3D Noise2Void



Z.L. Jiang et. al. Anal. Chem. 2025, 97, 32, 17380

Poor performance
Parameter tuning

Data cost
Poor generality

Time cost
Parameter tuning

Technical route is determined as the self-supervised learning (SSL) method

UV, IR, Raman...

Faster

Training of a
single spectrum



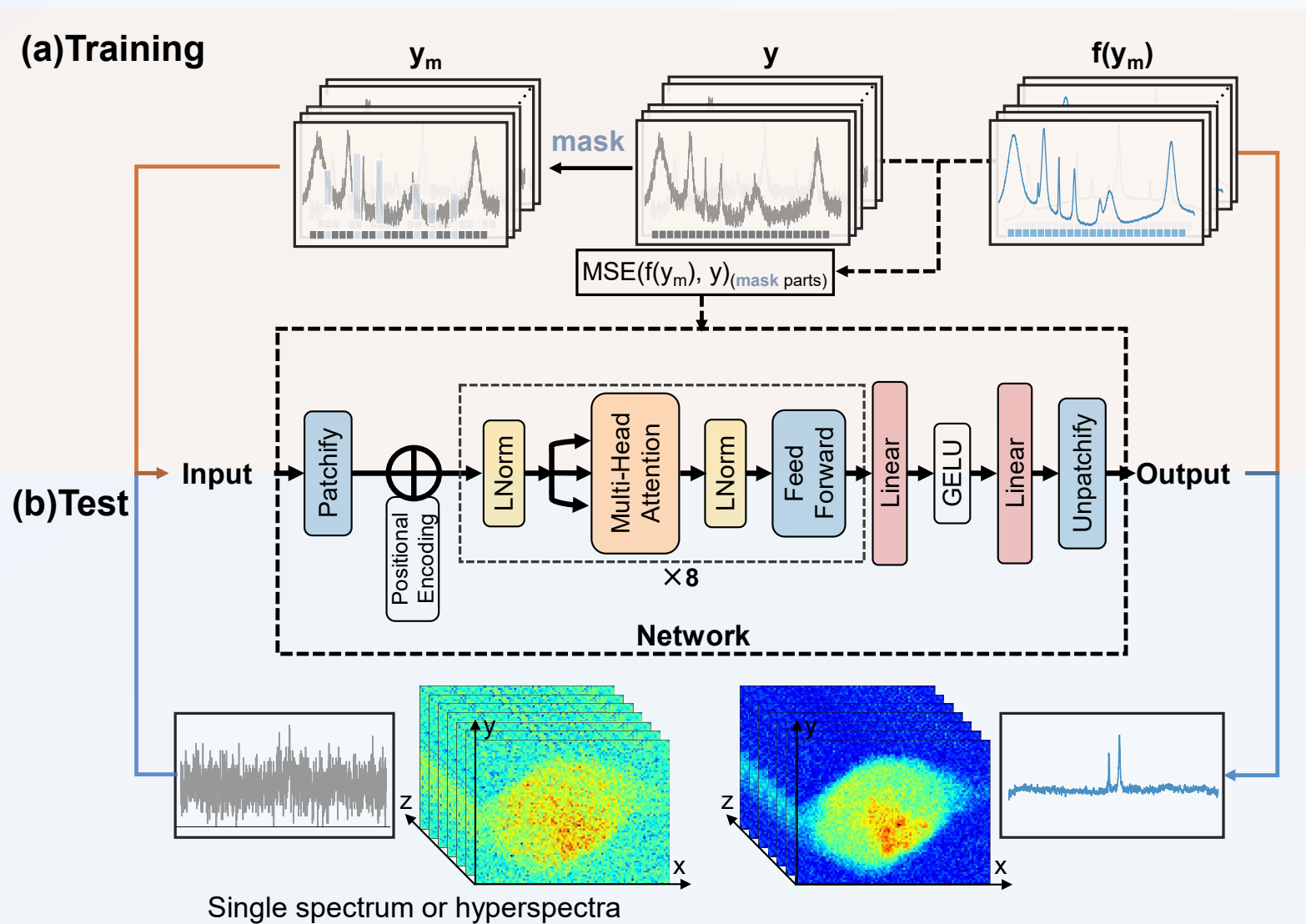
Direct inference

Automatic

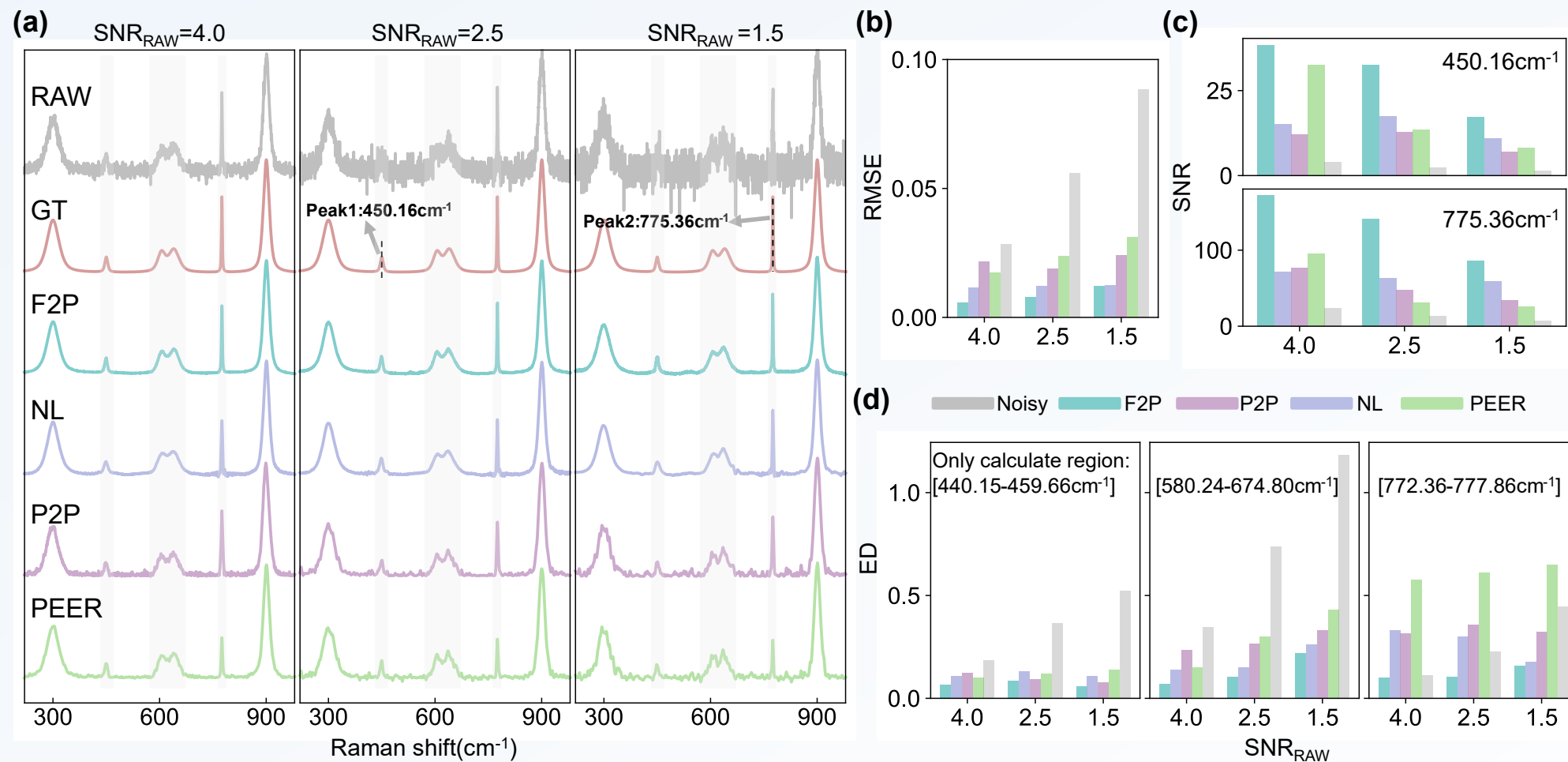
Convolution-based
network



Attention-based
network

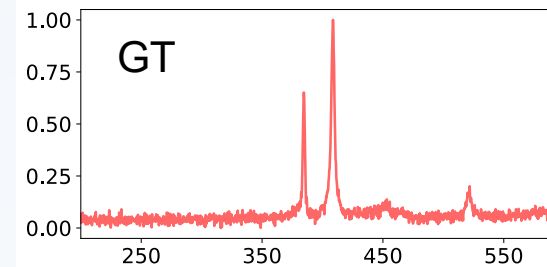
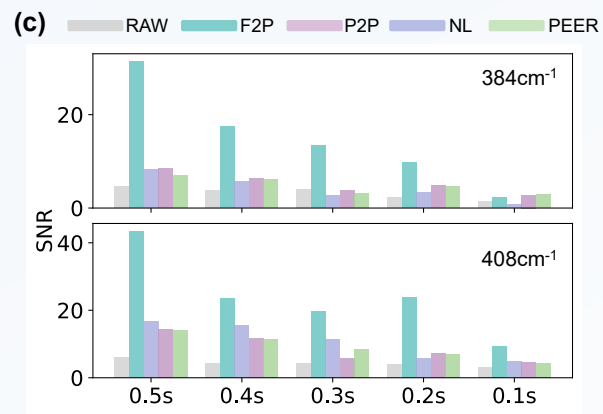
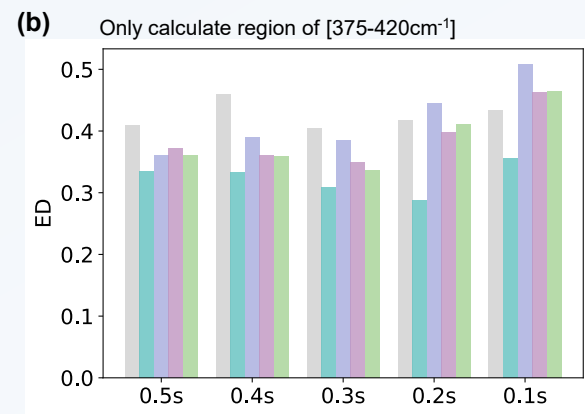
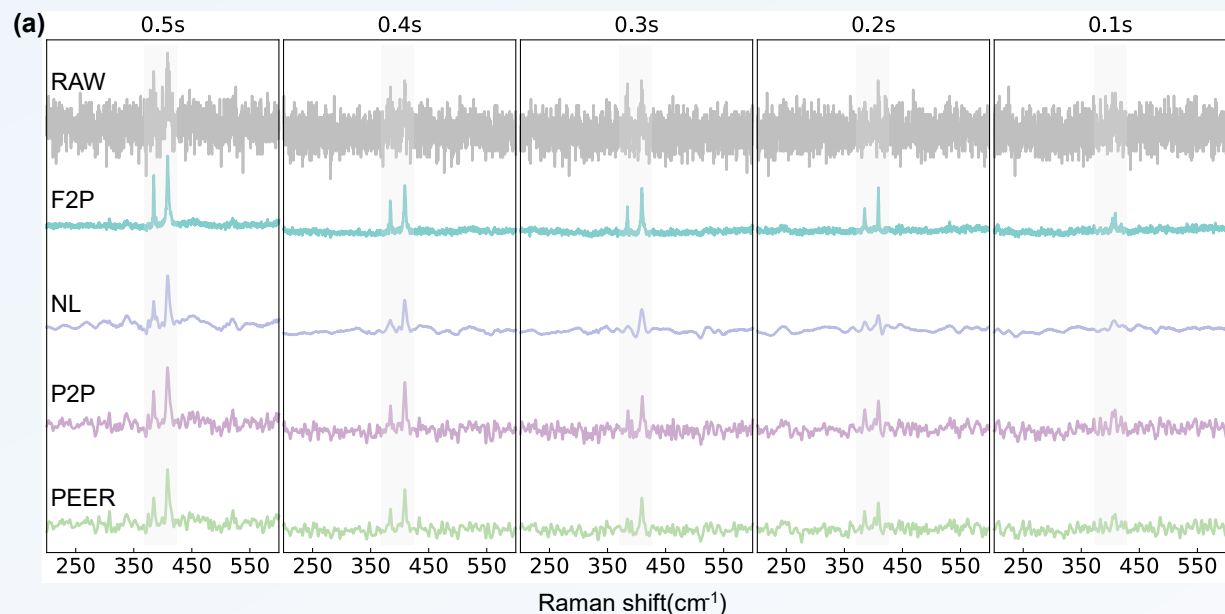


UV, IR, Raman...



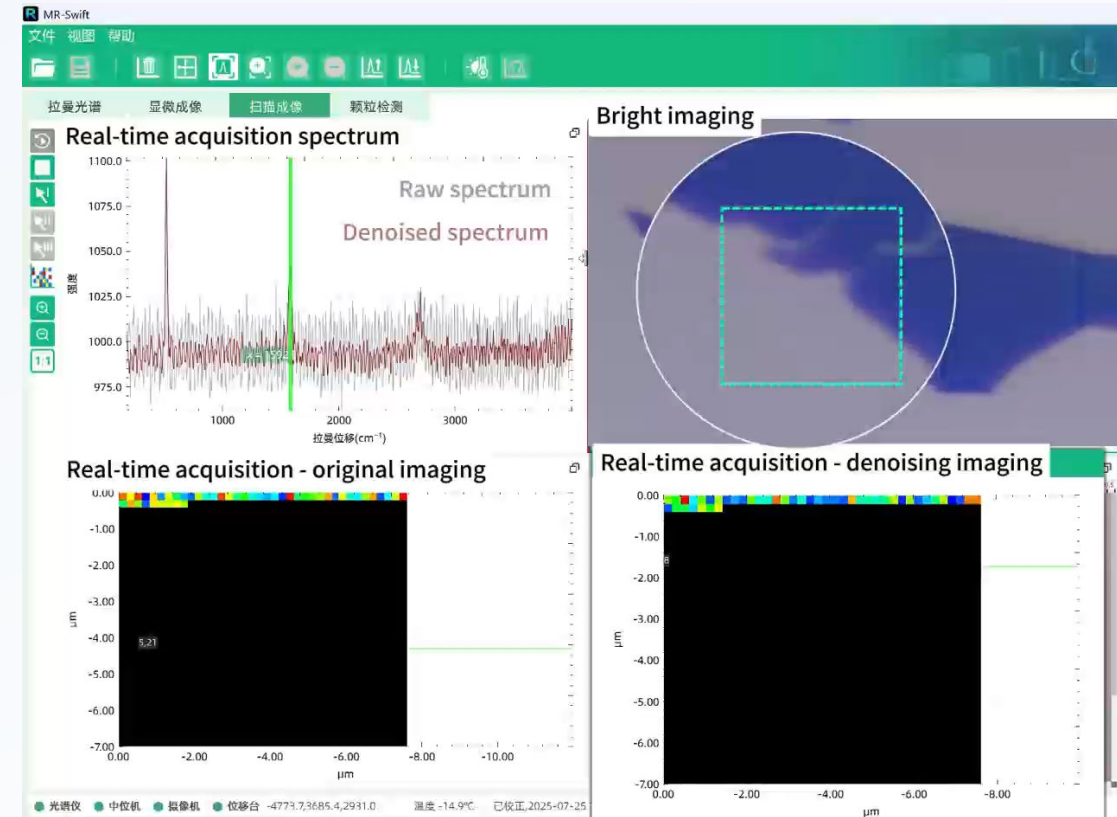
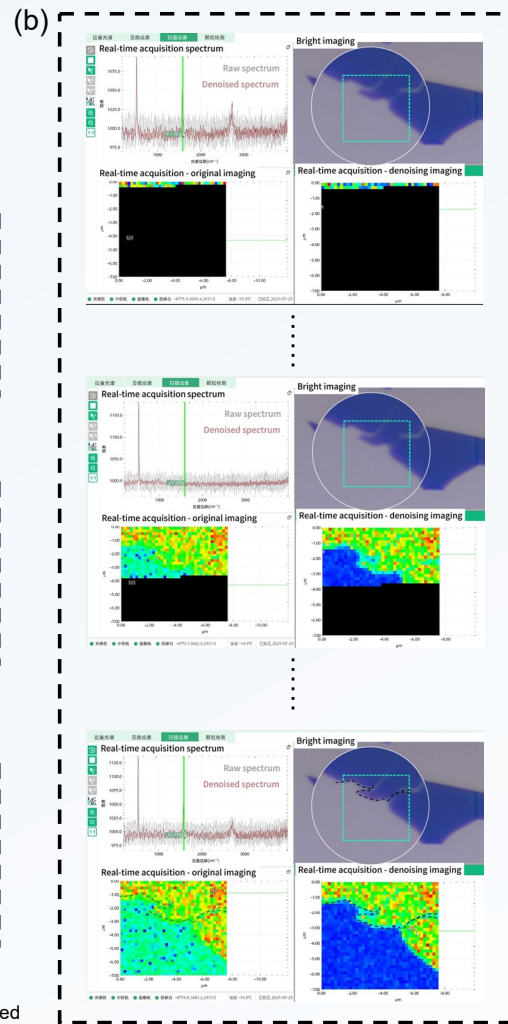
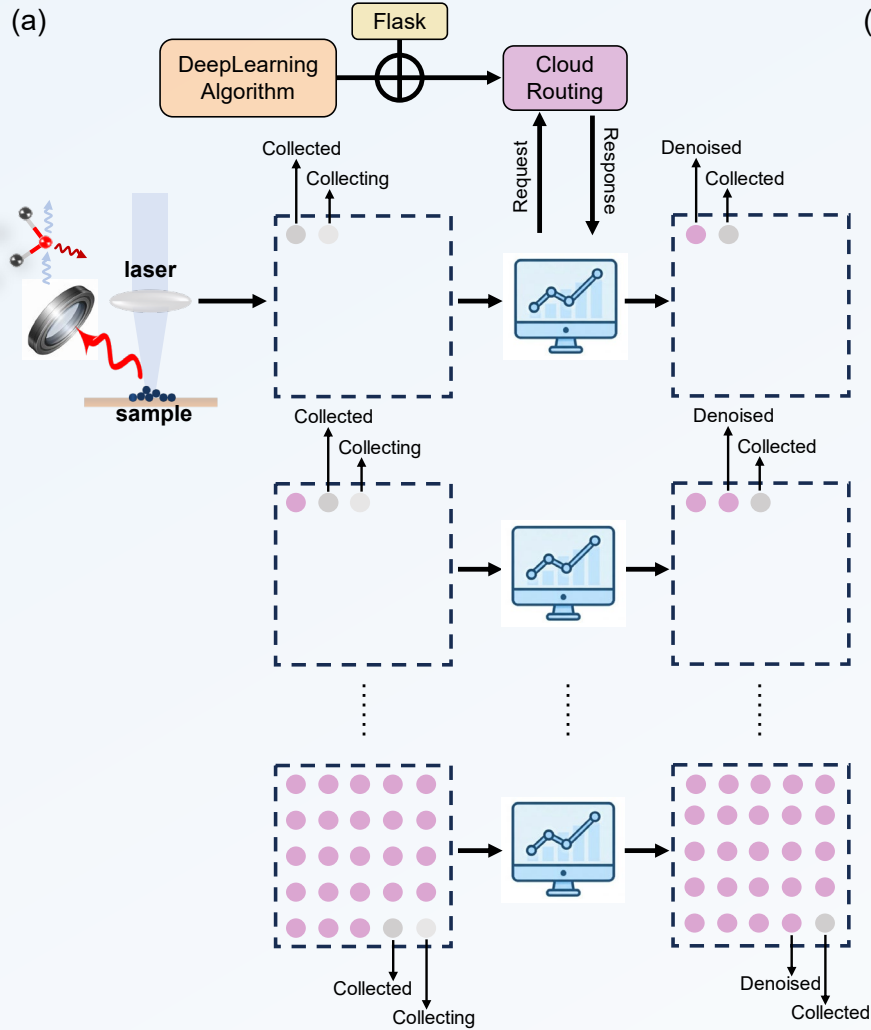
The F2P fidelity (ED) is the highest, and the noise reduction effect (SNR) is also excellent.

UV, IR, Raman...



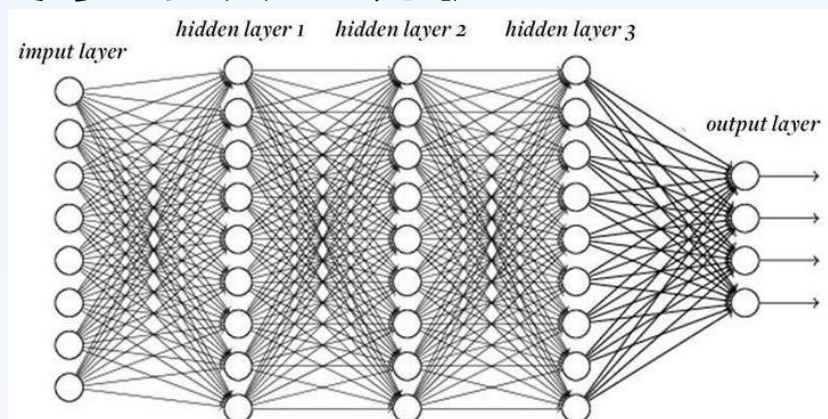
1. In complex experimental scenarios, the effects of P2P and PEER are very poor, and parameter adjustment is required
2. NL needs to collect the noise data of the current experimental scenario and retrain the model
3. Extreme scenarios(0.1s), F2P can't help either.

UV, IR, Raman...



Real-time denoising on home-made Raman instruments

深度学习模型现状：



过度参数化

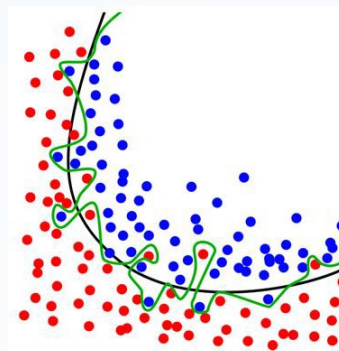


Su, et al., IEEE T EVOLUT COMPUT **2019**, 23, 828–841

模型鲁棒性差



难解释的黑箱



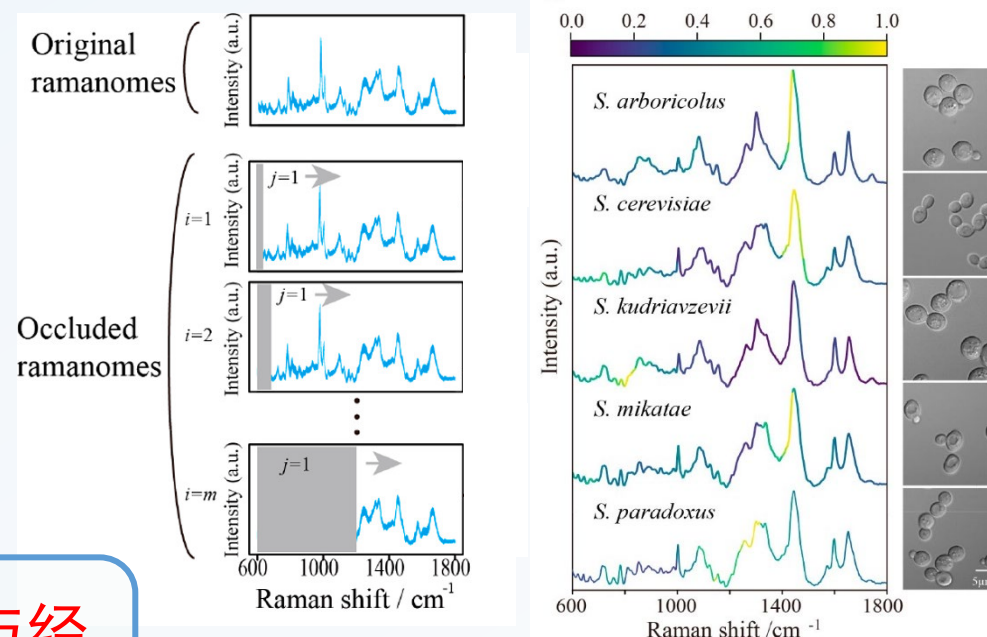
模型过拟合

高性能模型 ≠ 可靠的模型，对深度模型的合理解释十分必要

敏感性分析:

关注待解释样本特征值变化
对模型最终决策结果的影响

存在问题: 解释结果粗糙;
无法体现特征之间的相关性
对最终决策结果的影响; 看似差异小的地方反而被高亮



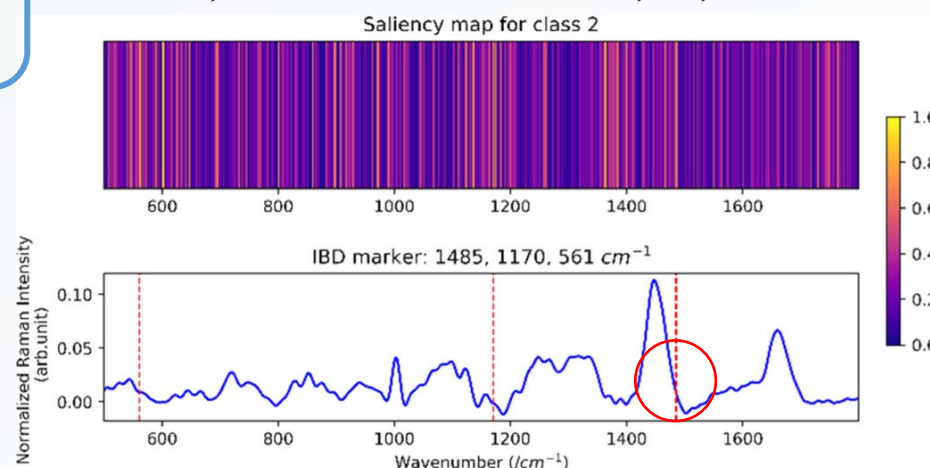
Y. V. Fu, et al. Anal. Chem. **2020**, 92, 6288-6296

反向传播:

解释结果与经
验判断不符

将有关决策重要性的信号从
模型的输出层逐层反向传播
到模型的输入特征

存在问题: 提供的解释结果
难以理解; 被高亮的地方没
有有效信息



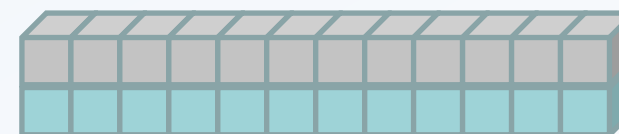
Stallmach, et al. Anal. Chem. **2020**, 92, 13776-13784

问题根源：光谱局部背景的变化误导模型的决策

解决思路：弱化背景影响，强化峰形特征

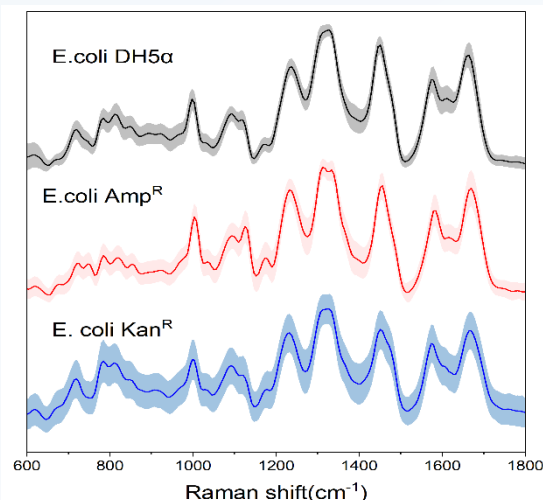


 Raw spectrum



 Derivative spectrum

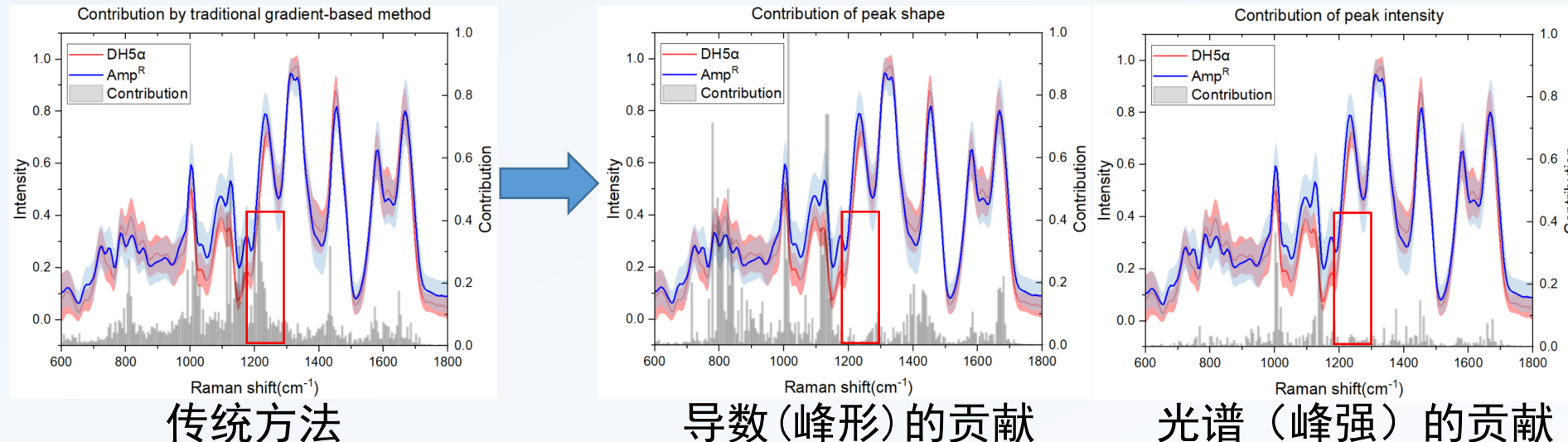
拉曼光谱的一阶导数可以滤除局部背景，将其拼接在原有的一维向量上，可以强化谱峰的峰形特征，弱化局部背景的干扰



细菌数据

	Accuracy(%)
1D spectra	85.33
Derivation	84.67
2D spectra	88.00

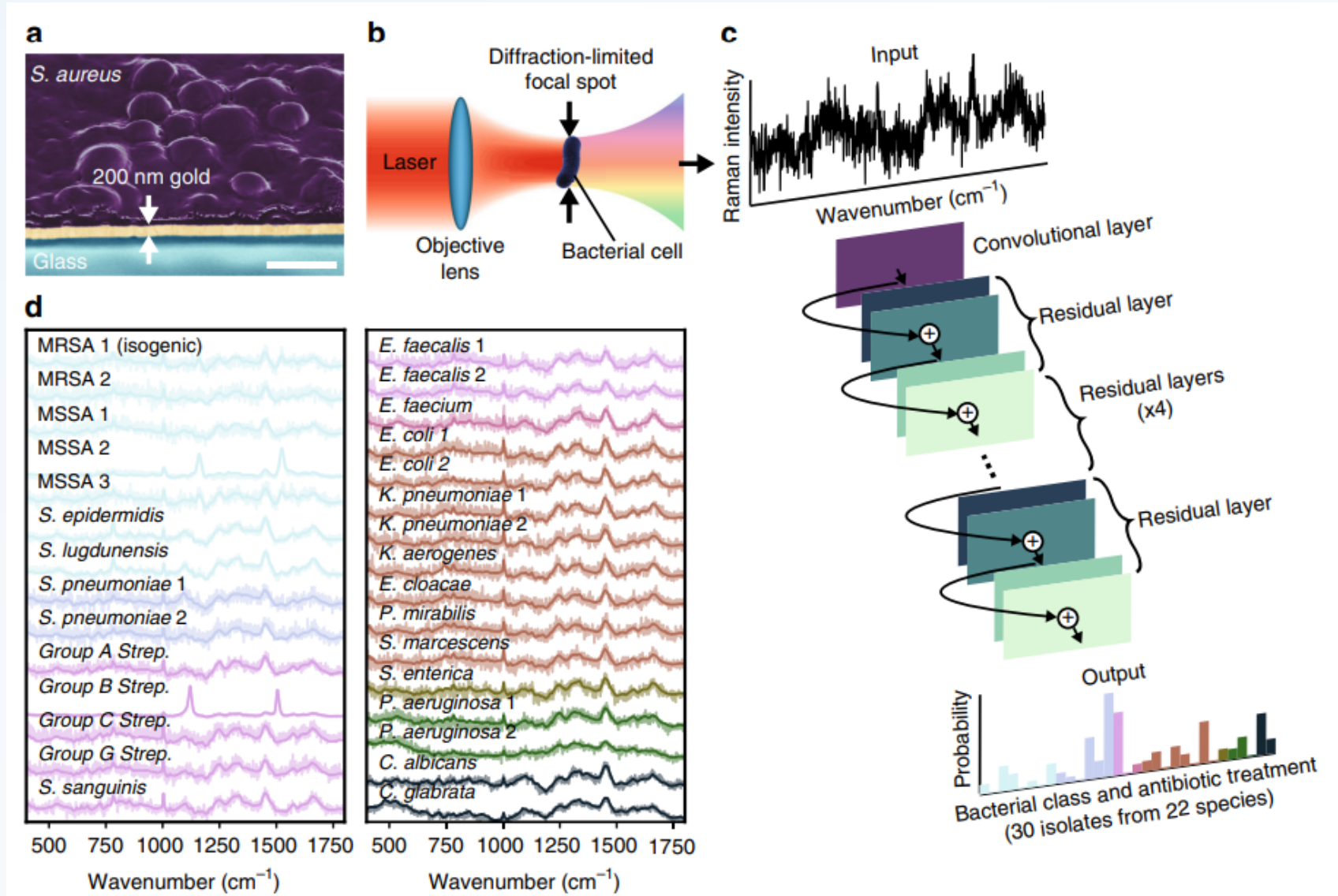
不同输入的准确率



无效贡献被滤除，实现对差异峰准确指认

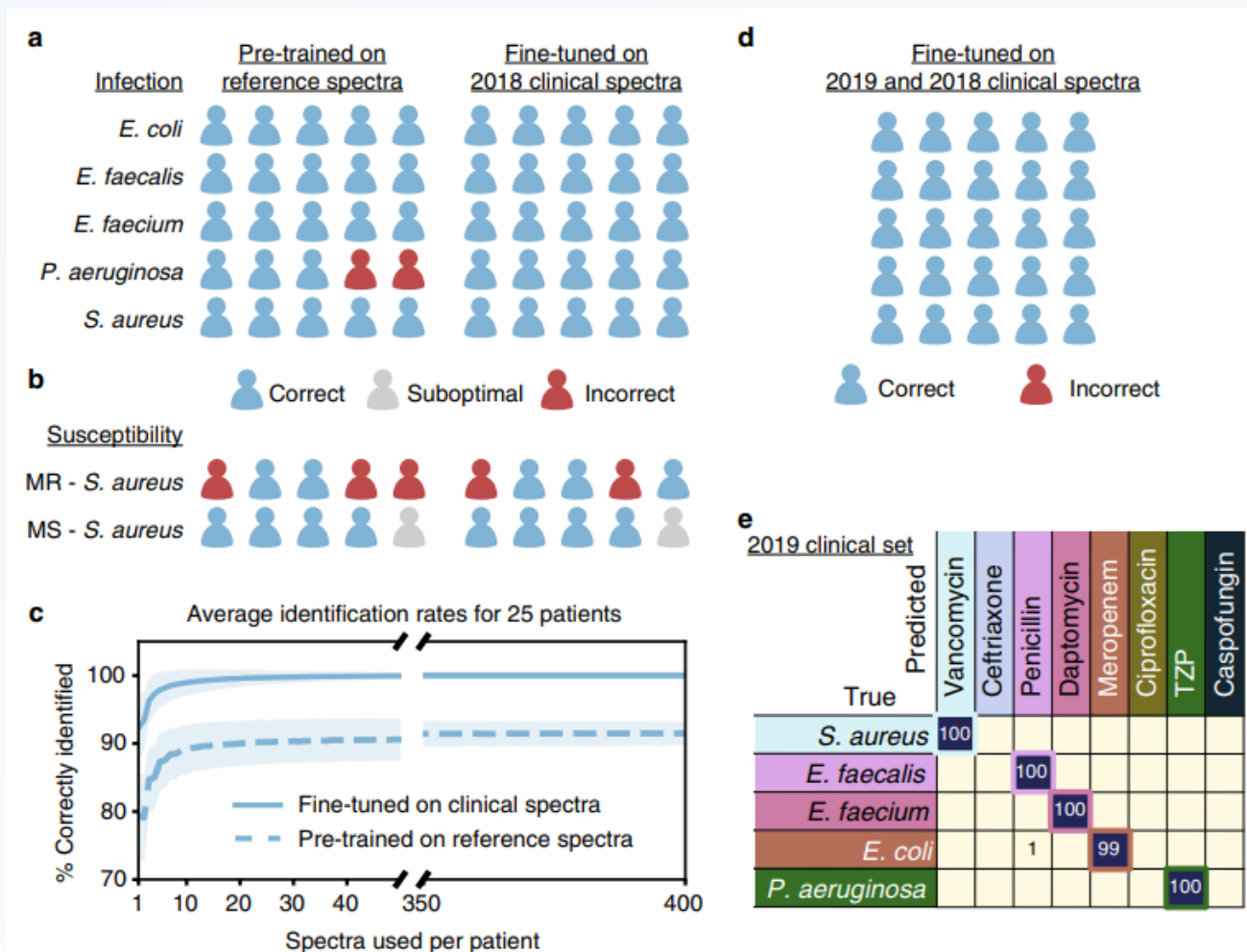
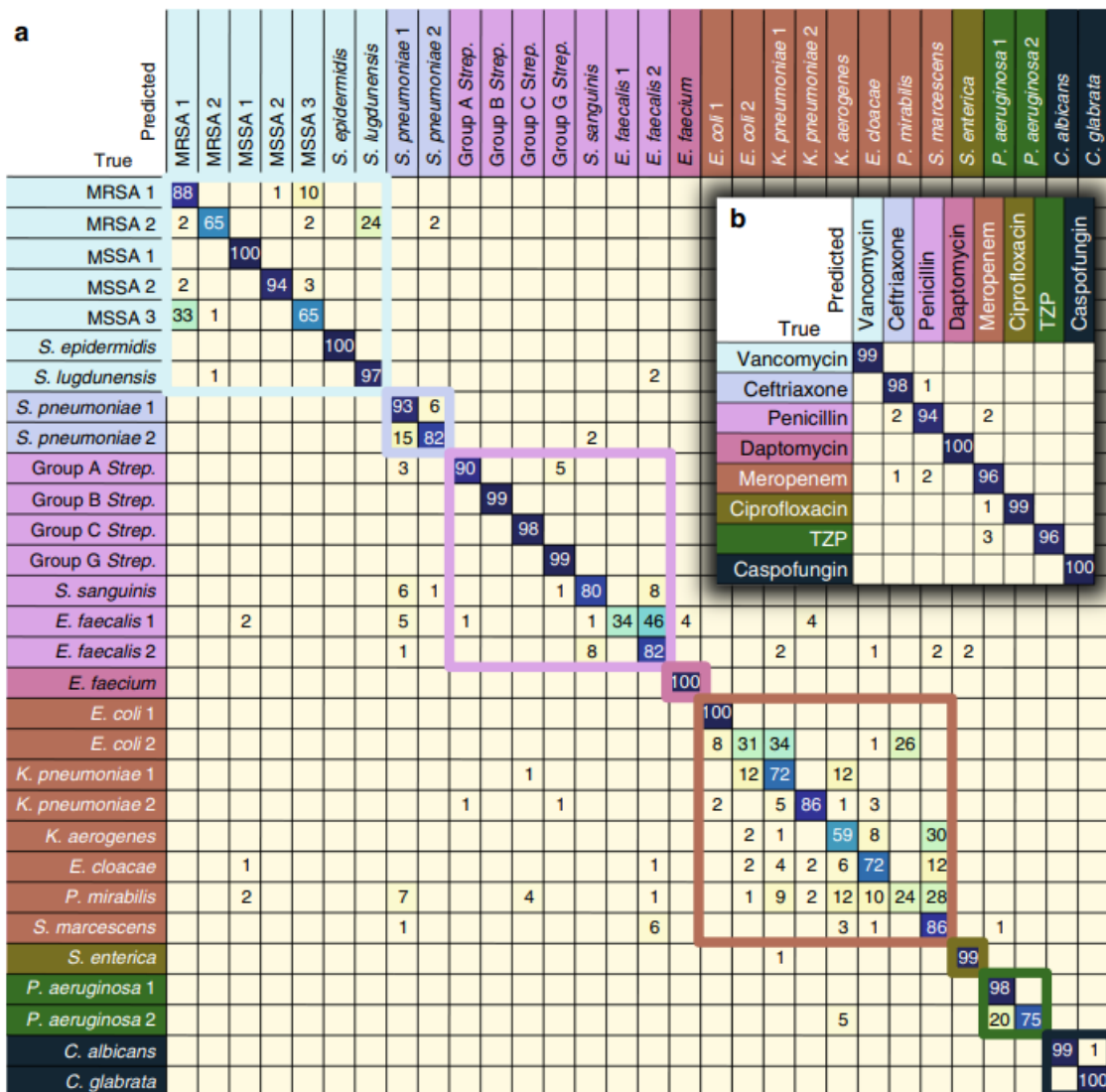
Raman shift(cm ⁻¹)	810~816cm ⁻¹	1002cm ⁻¹	1125~1130cm ⁻¹	1453cm ⁻¹
Assignment	C' ₅ -O-P-O-C' ₃ phosphodiester bands in RNA	Phenylalanine (苯丙氨酸)	n(C-C) skeletal of acyl backbone in lipid (transconformation)	Phospholipids (磷脂质)
Sources of difference	Shape	Shape & Intensity	Shape & Intensity	Shape

UV, IR, Raman...



Dionne, et al., Nat. Commun. **2019**, 10, 4927

UV, IR, Raman...



Dionne, et al., Nat. Commun. **2019**, 10, 4927

UV, IR, Raman...

<https://github.com/csho33/bacteria-ID/>

The screenshot displays the GitHub interface for the repository `csho33/bacteria-ID`. The repository is public and contains 5 commits. The file list includes:

File	Commit Message	Time
data	completed finetuning demo	5 years ago
.gitignore	completed finetuning demo	5 years ago
1_reference_finertuning.ipynb	adding prediction demo and clinical demo	5 years ago
2_reference_prediction.ipynb	adding prediction demo and clinical demo	5 years ago
3_clinical_finertuning.ipynb	correcting message output in notebook 3	5 years ago
LICENSE	Initial commit	5 years ago
README.md	updating link to data	5 years ago
clinical_pretrained_model.ckpt	adding prediction demo and clinical demo	5 years ago
config.py	adding prediction demo and clinical demo	5 years ago
datasets.py	completed finetuning demo	5 years ago
finetuned_model.ckpt	adding prediction demo and clinical demo	5 years ago
pretrained_model.ckpt	adding prediction demo and clinical demo	5 years ago
resnet.py	completed finetuning demo	5 years ago
training.py	completed finetuning demo	5 years ago

Repository statistics:

- 77 stars
- 34 forks
- 2 watching
- Activity
- MIT license
- Readme

Releases: No releases published

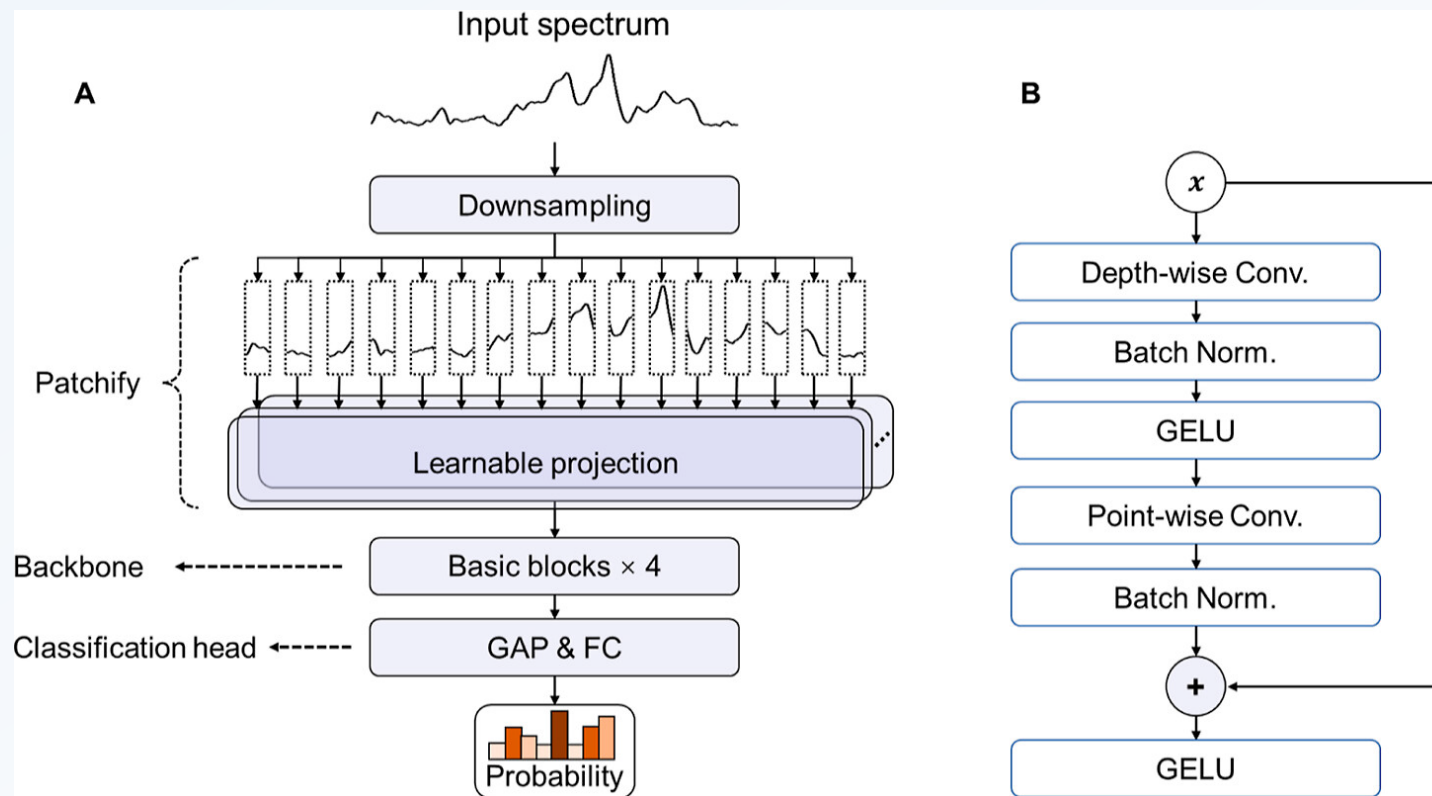
Packages: No packages published

Languages: Jupyter Notebook 96.2%, Python 3.8%

Dionne,et al., Nat. Commun. **2019**, 10, 4927

Code

Raman Spectroscopy

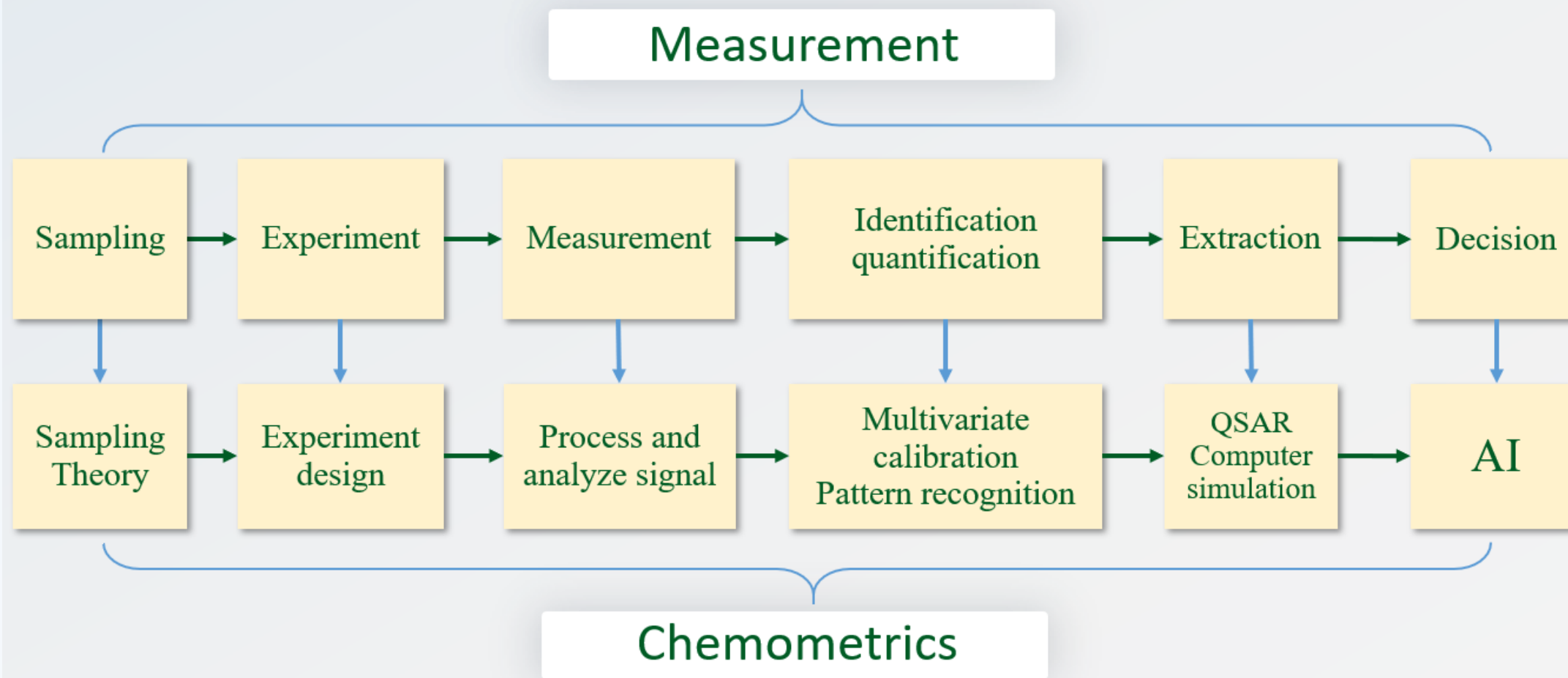


Anal. Chem. 2024, 96, 7, 2866–2873

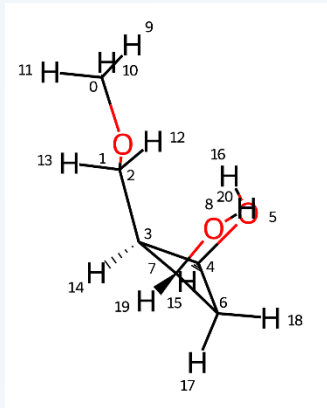
Thinking: The spectral range depends on the interests of researchers, how to handle spectra with different length with only one model?

Content of Chemometrics

Its content almost covers the entire process of chemical measurement.



UV, IR, Raman...

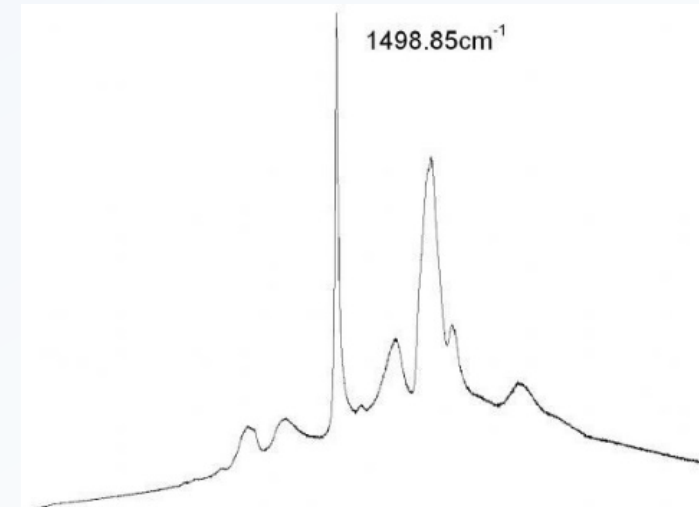


Input Molecule

DFT

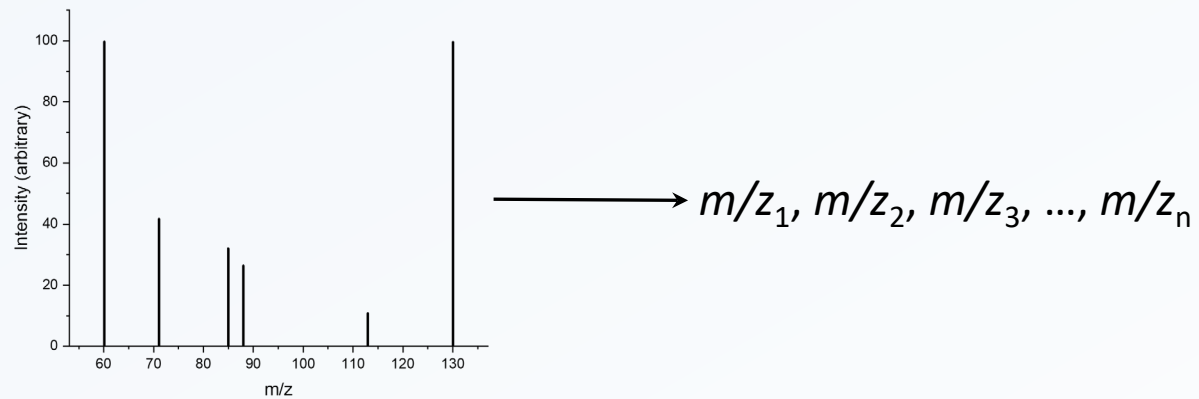
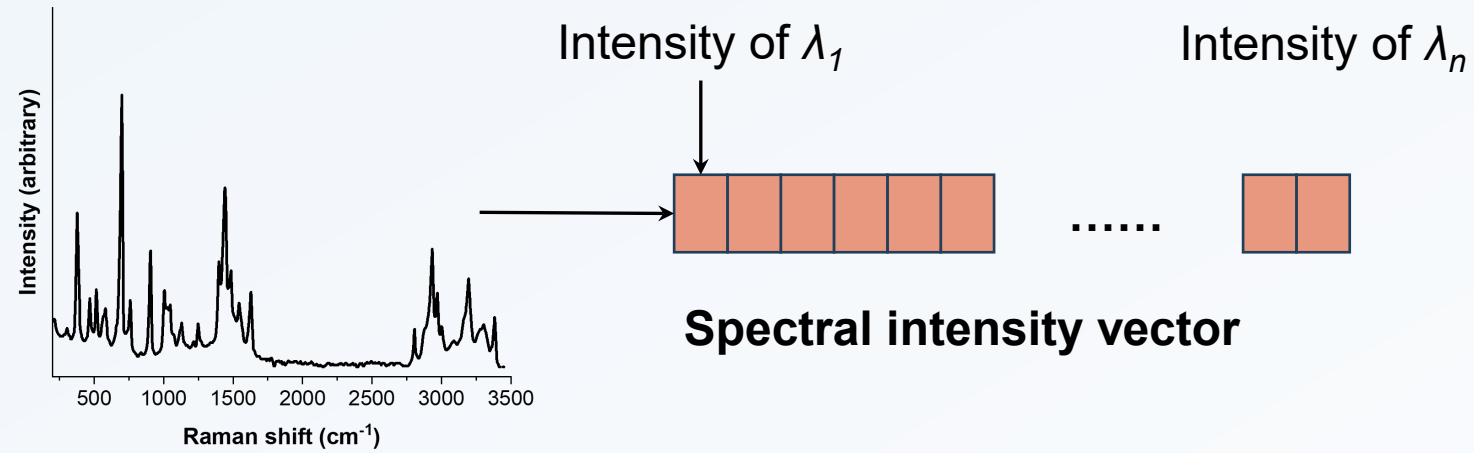


takes too much time

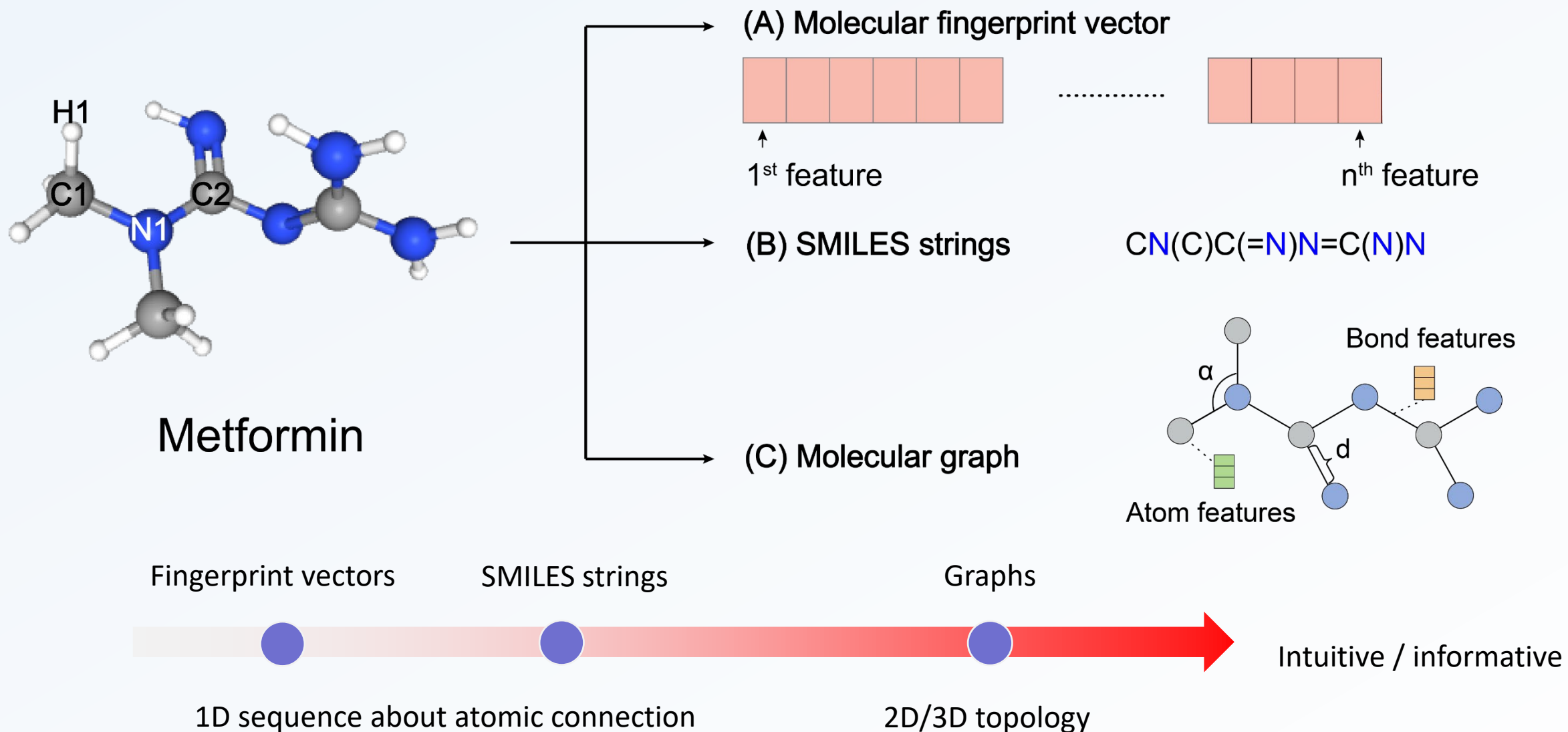


Molecular Raman spectrum

UV, IR, Raman...



UV, IR, Raman...



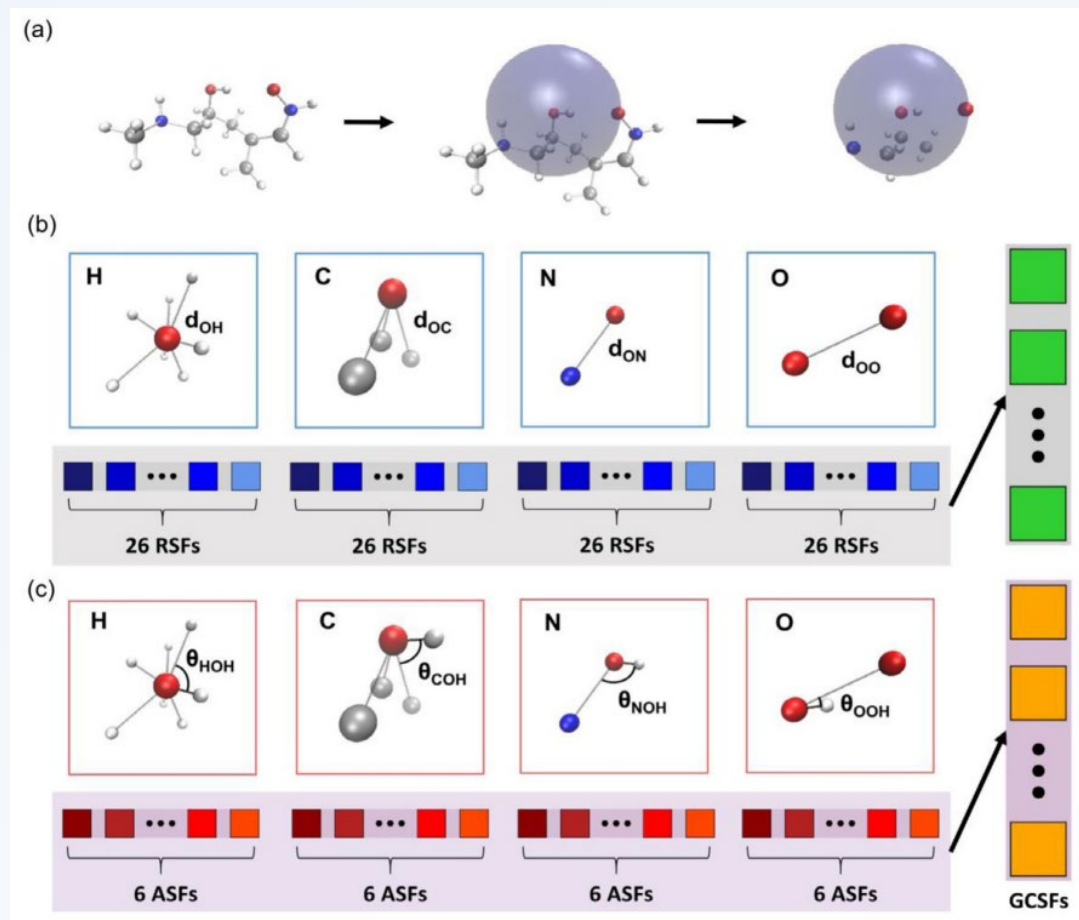
There are two schools (流派) of AI for spectral prediction.

Hand-crafted features

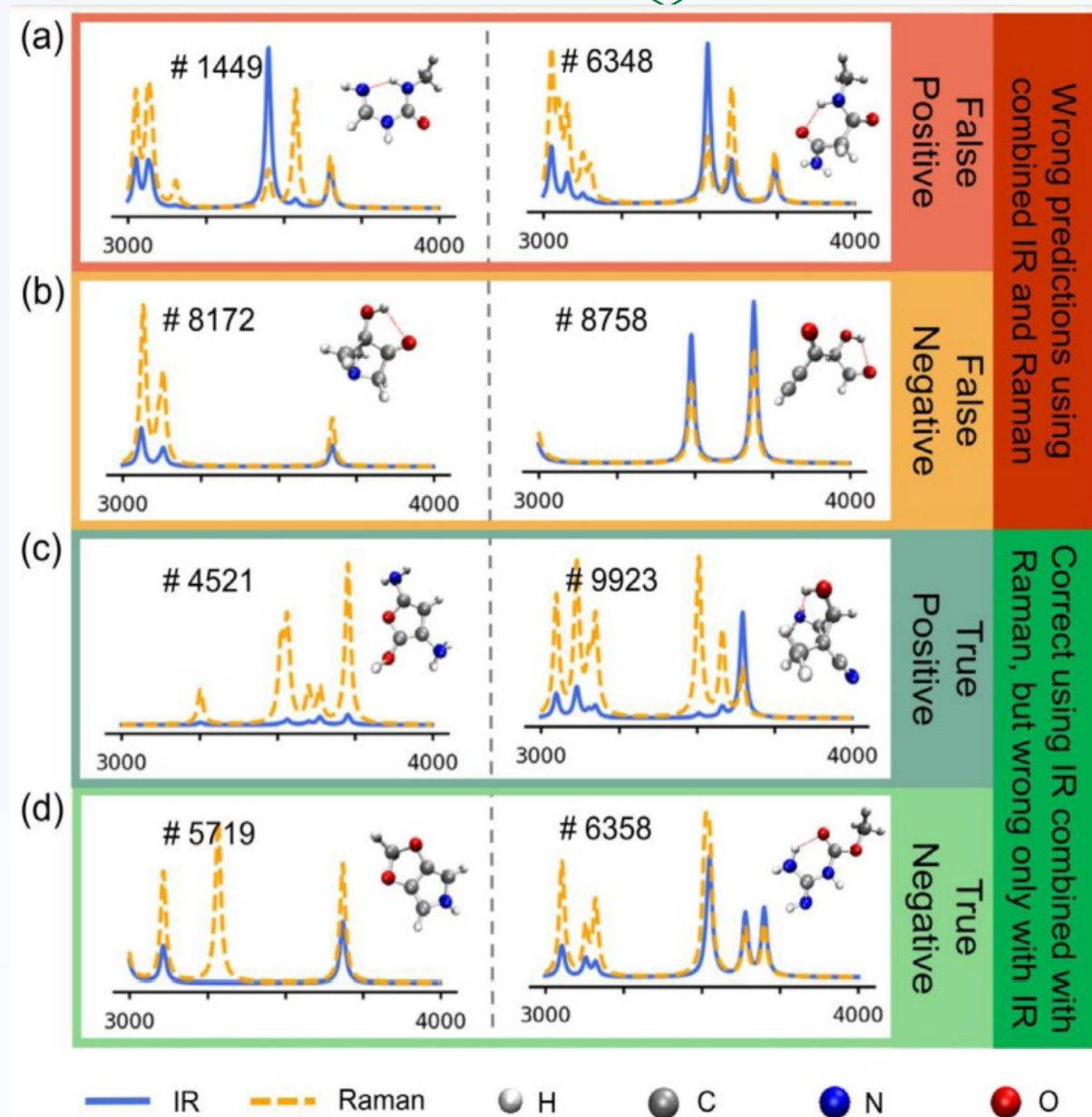
Extract features from bigdata

UV, IR, Raman...

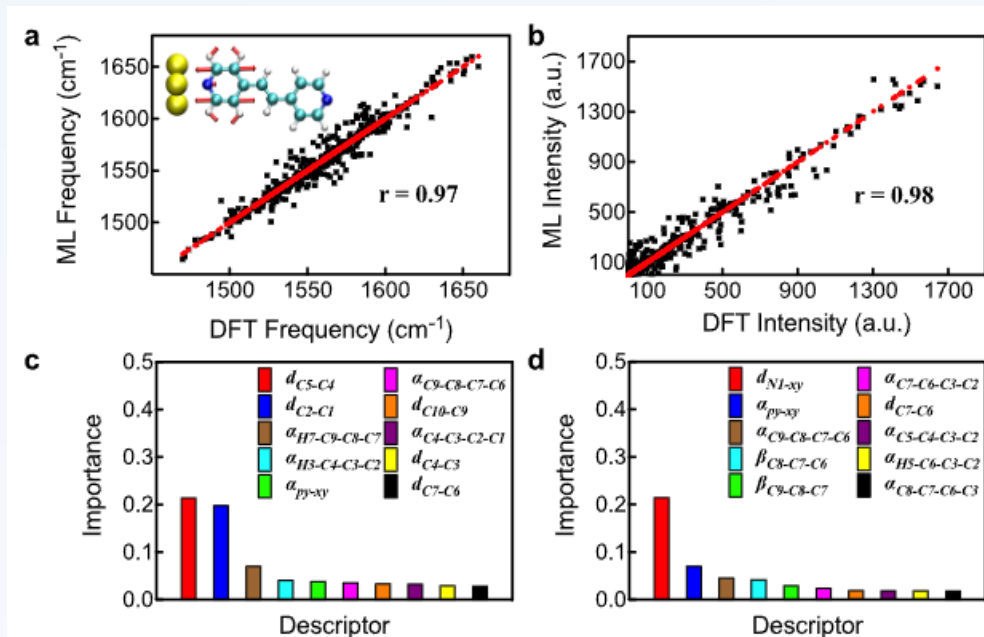
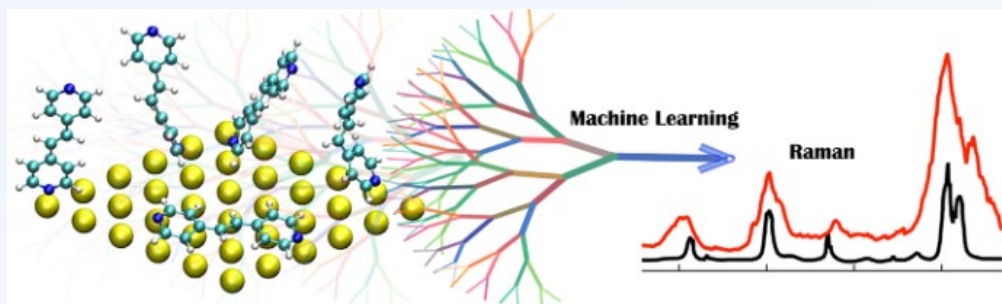
Hand-crafted features



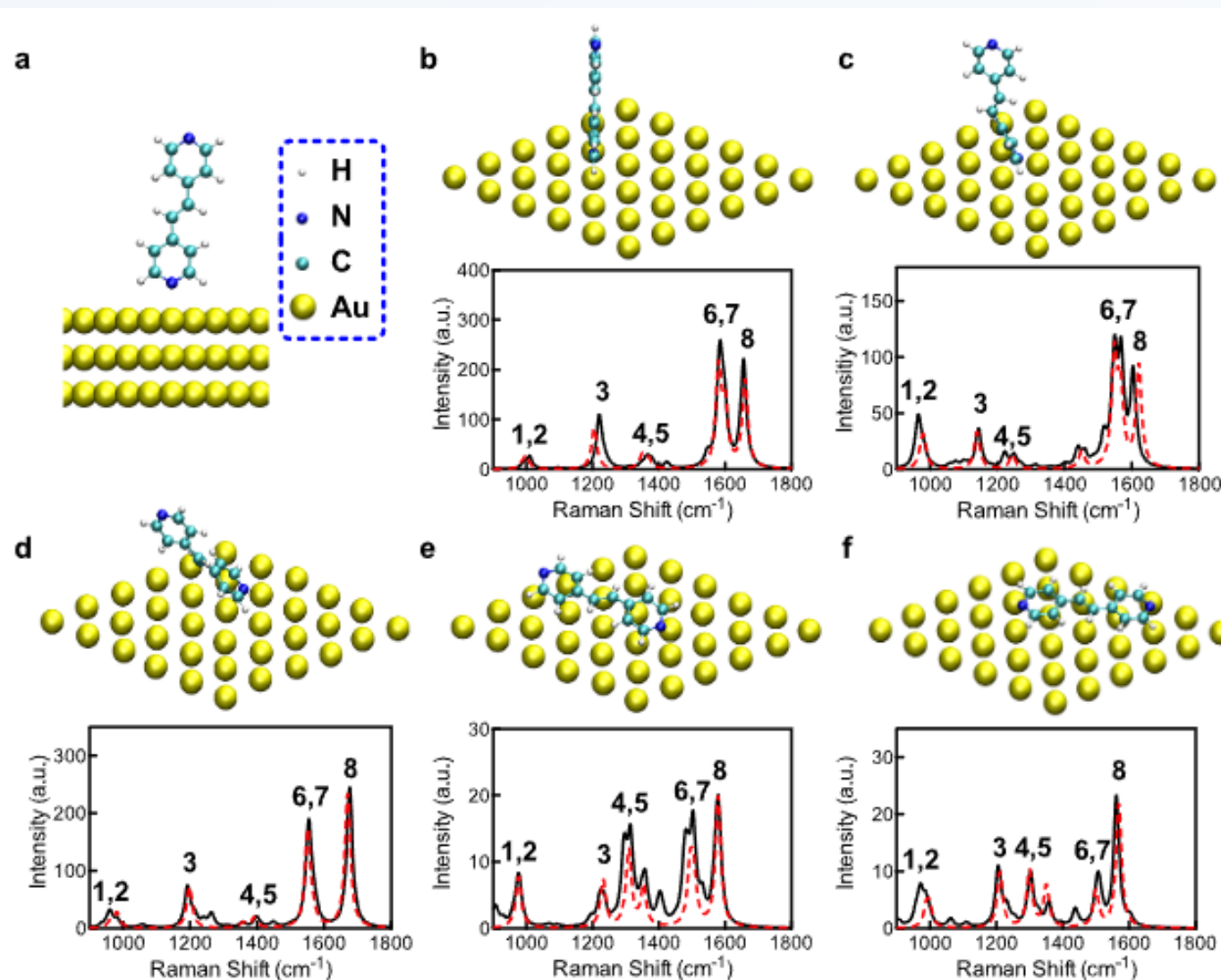
Fundamental Research 2021, 1, 488–494



UV, IR, Raman...



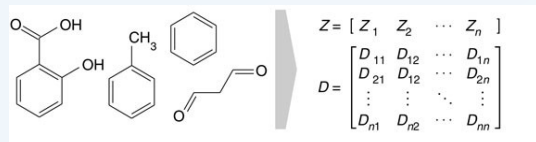
J.Phys.Chem.Lett. 2019, 10, 6026–6031



UV, IR, Raman...

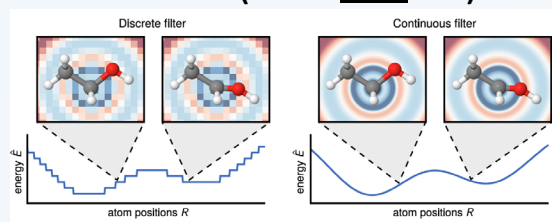
Historical map of GNNs in modeling 3D interaction

Deep tensor neural network (DTNN)



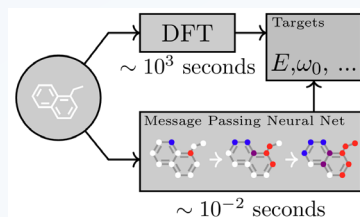
K.-R. Muller et al. Nat. Commun., 2017, 8, 13890

SchNet (K. T. Schütt)



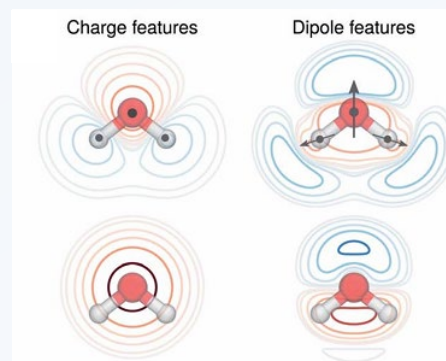
K. T. Schütt, K.-R. Muller et al., NeurIPS 2017, 30

Message Passing Neural Networks (MPNNs)



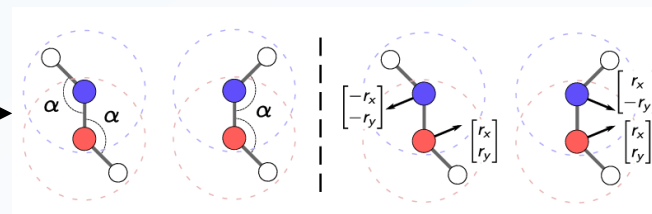
G. E. Dahl et al. ICML 2017, 1263

FieldSchNet



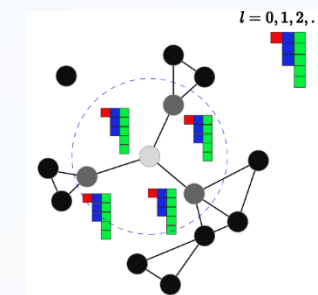
M. Gastegger, K. T. Schütt, K.-R. Muller, Chem. Sci., 2021, 12, 11473

PaiNN



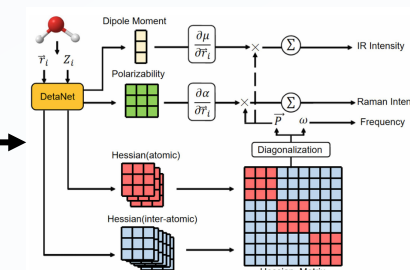
K. T. Schütt, M. Gastegger et al. ICML 2021, 9377

Neural Equivariant Interatomic Potentials (NequIP)



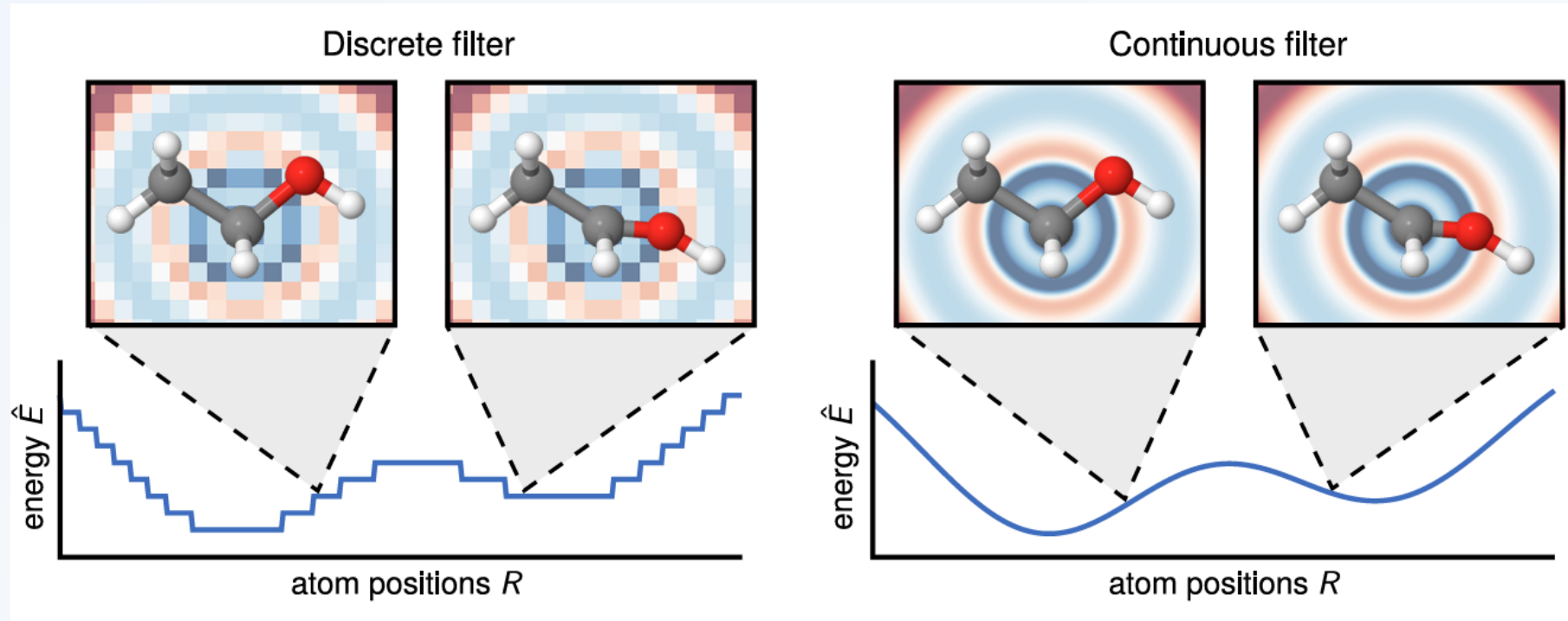
B. Kozinsky et al., Nat. Commun., 2022, 13, 2453

DL for predicting molecular spectra



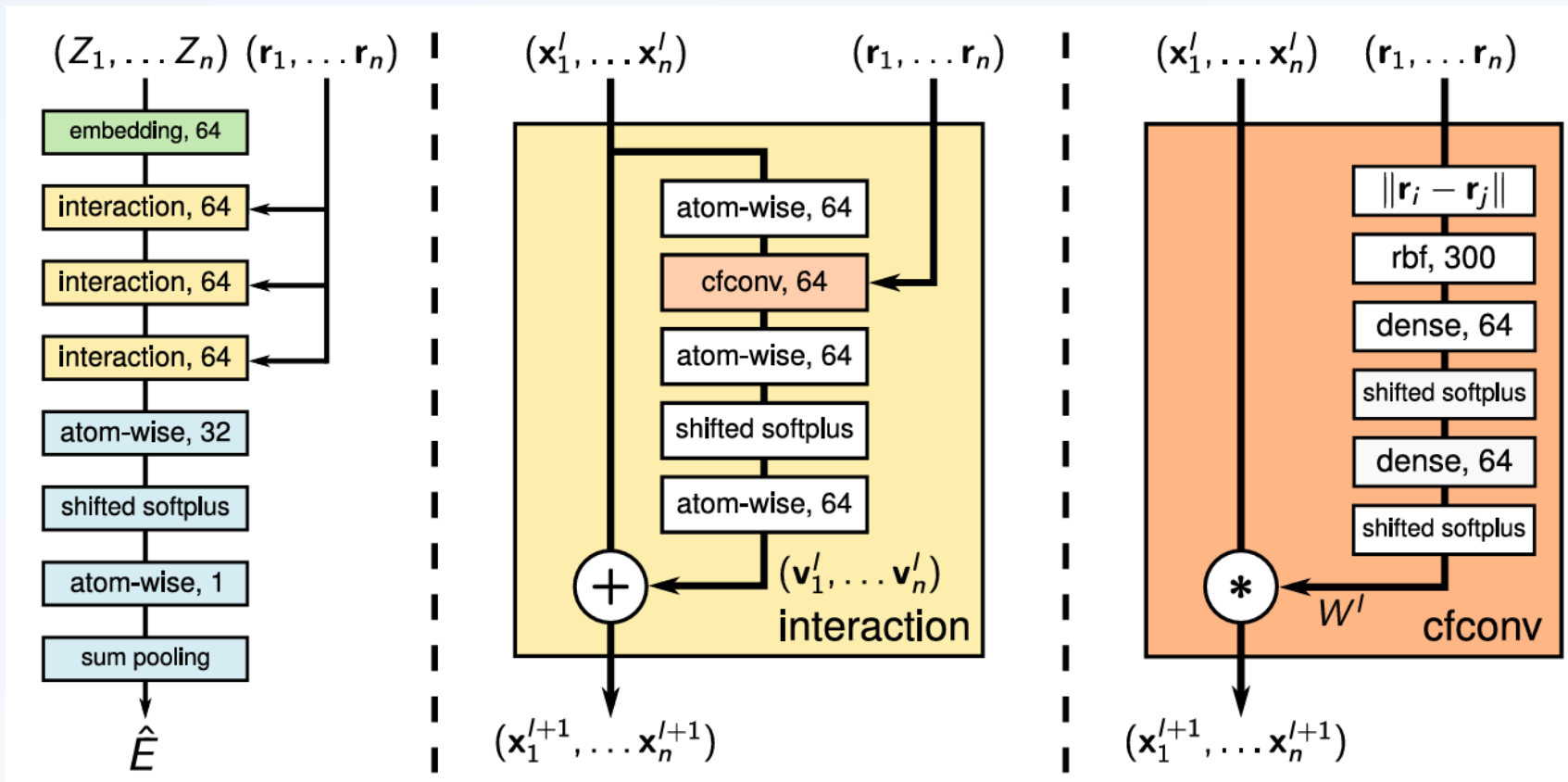
W. Hu et al., Nat. Comput. Sci., 2023, 3, 957

UV, IR, Raman...

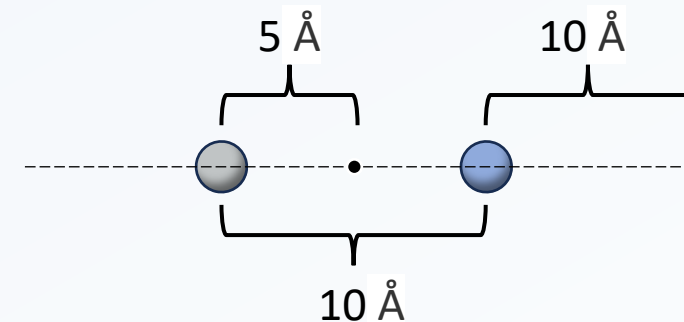


How to predict **continuous** coordinate-energy curves by **discrete** data (coordinates)?

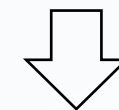
UV, IR, Raman...



$$x_i^{l+1} = \sum_j x_i^l \odot W^l(r_i - r_j)$$



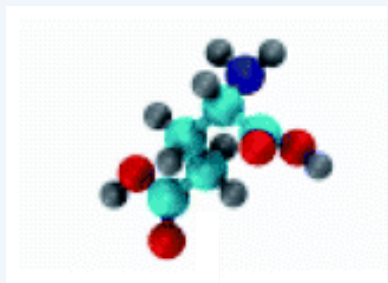
Resolution = 0.1 Å
Length = 30 Å



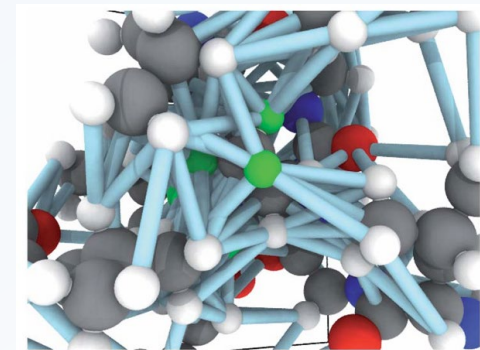
Fixed position → energy
Variable position → energy

Different ways to construct 3D graphs

A. D. White et al. Chem. Sci., 2021, 12, 10802



Create edges with the 16 closest atoms



atoms with
coordinates

Add edges
(bonded and
non-bonded)

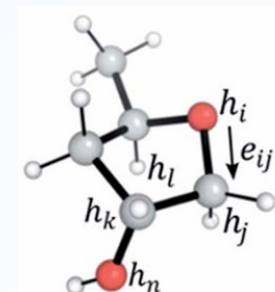
3D graph

GNN

Readout &
predict

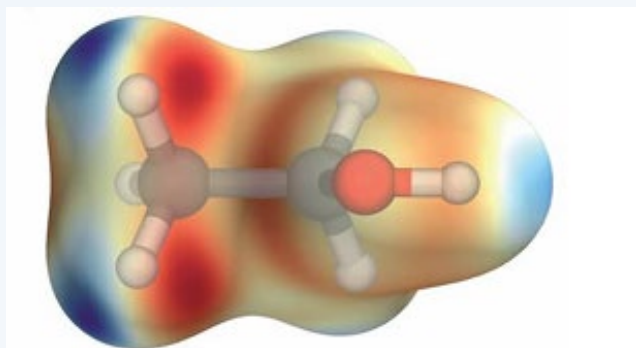


if interatomic distances less than
5 Å, an edge is created

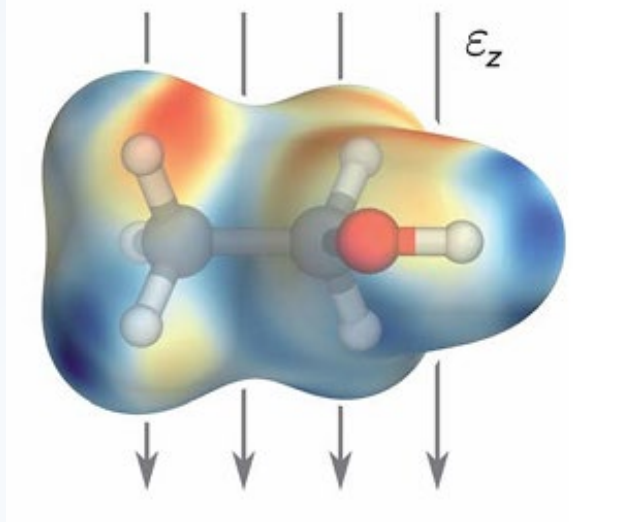


Prediction of spectra considering field effects

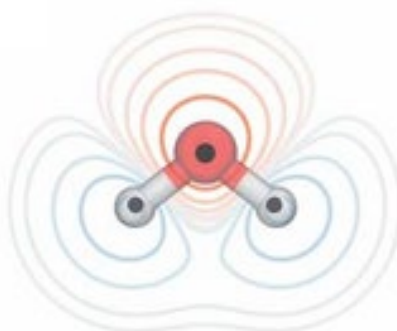
Without field



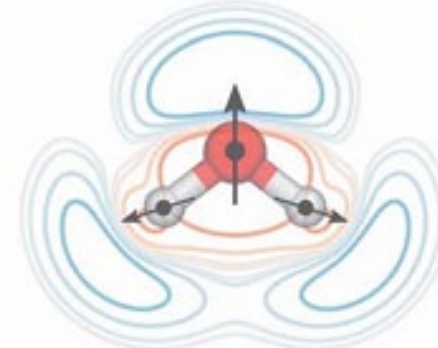
With field



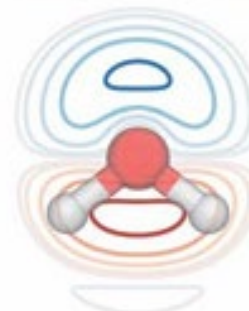
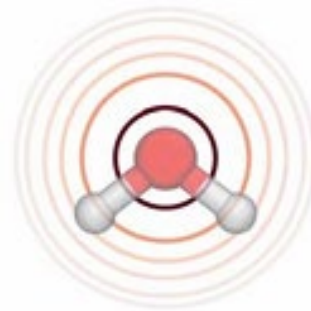
Charge features



Dipole features



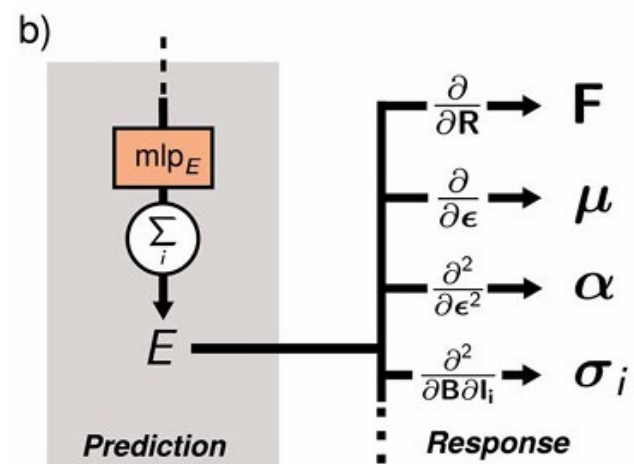
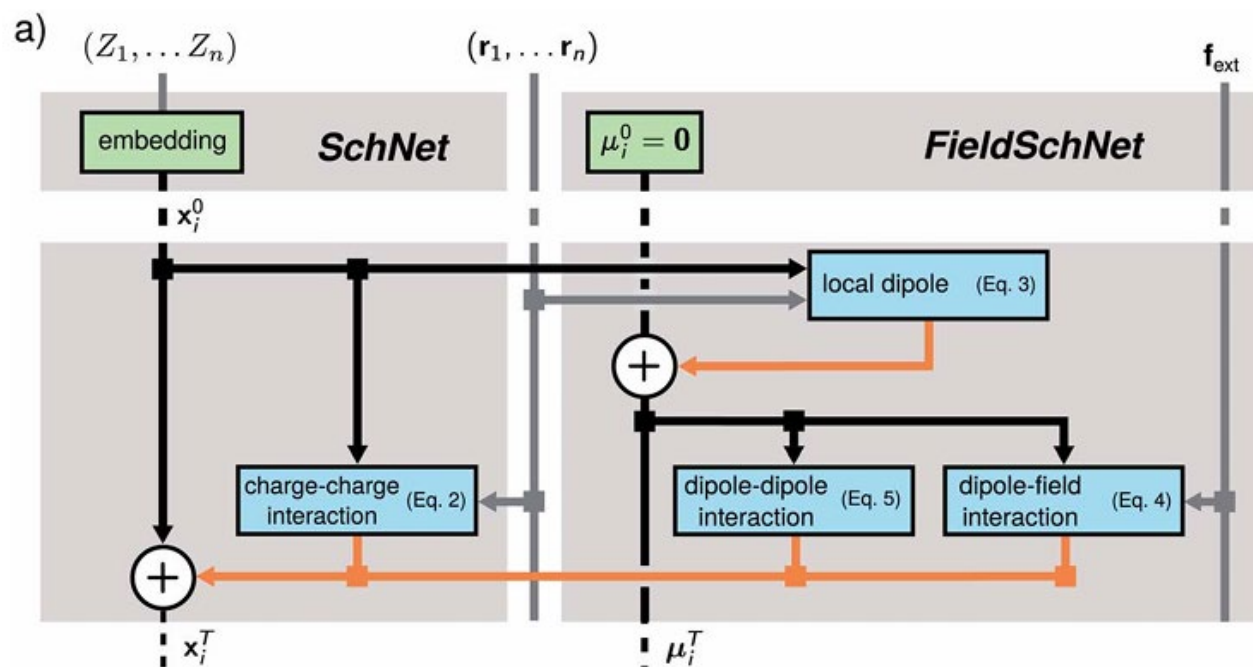
Whole molecule



Only oxygen atom

UV, IR, Raman...

FieldSchNet

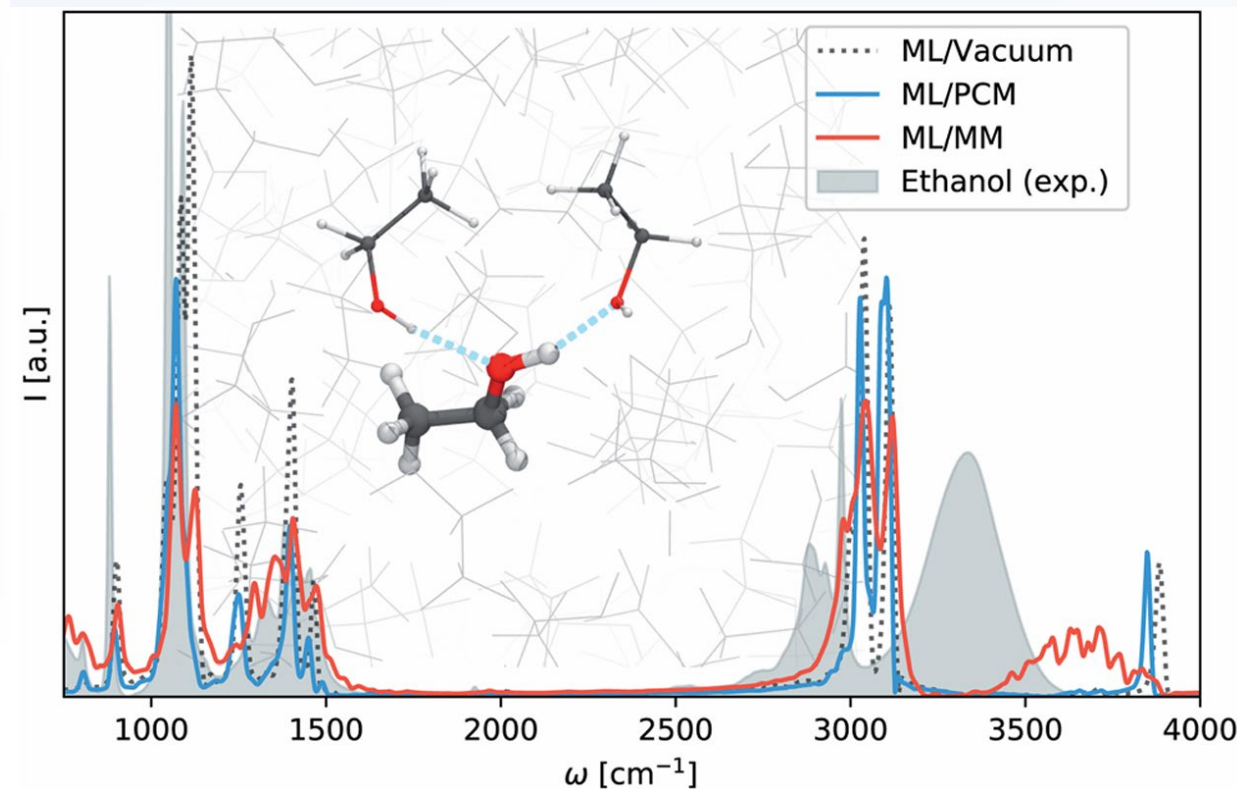


Input:

1. Atom type
2. Coordinates
3. Field vector

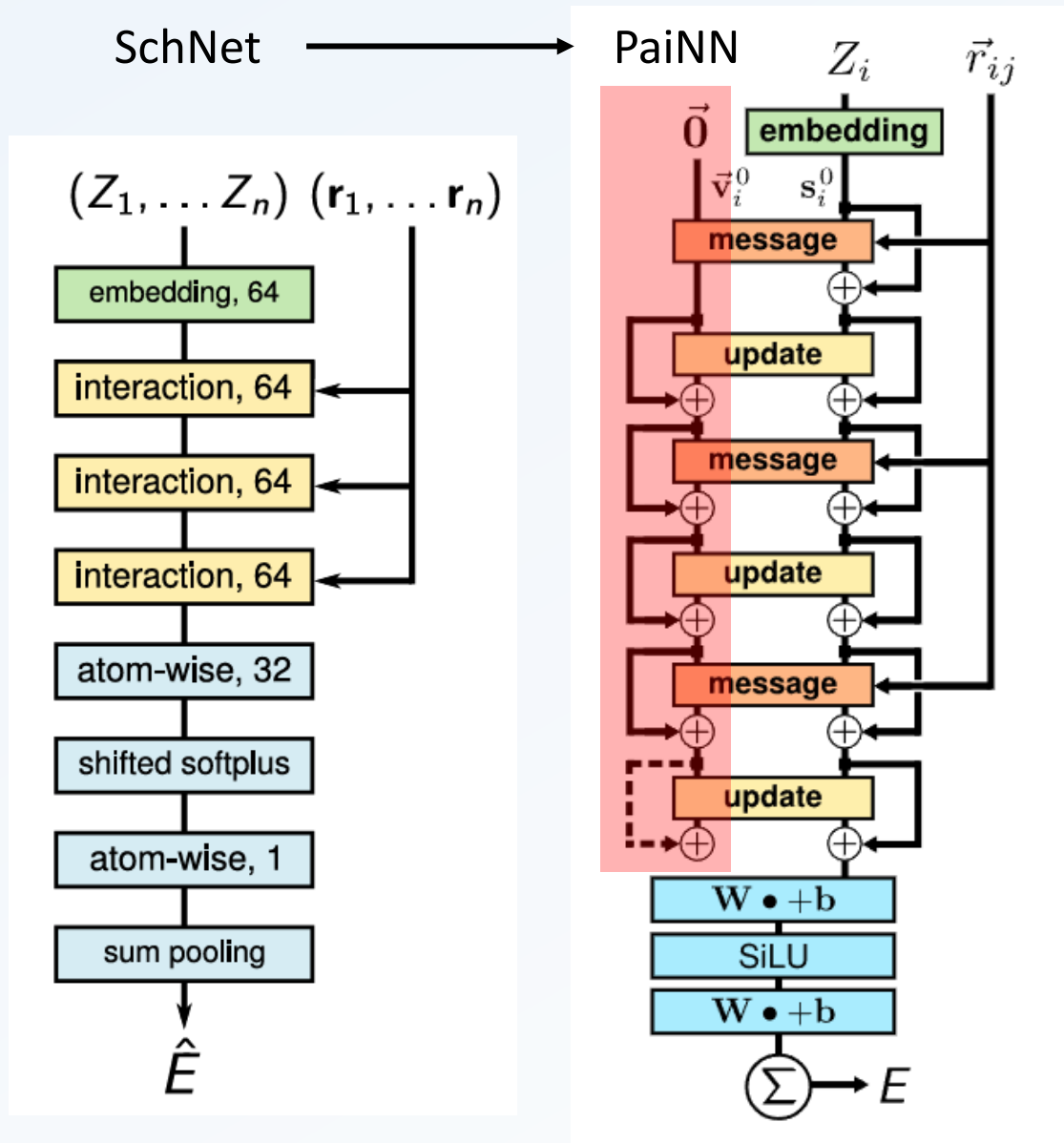
Output:

Molecular energy



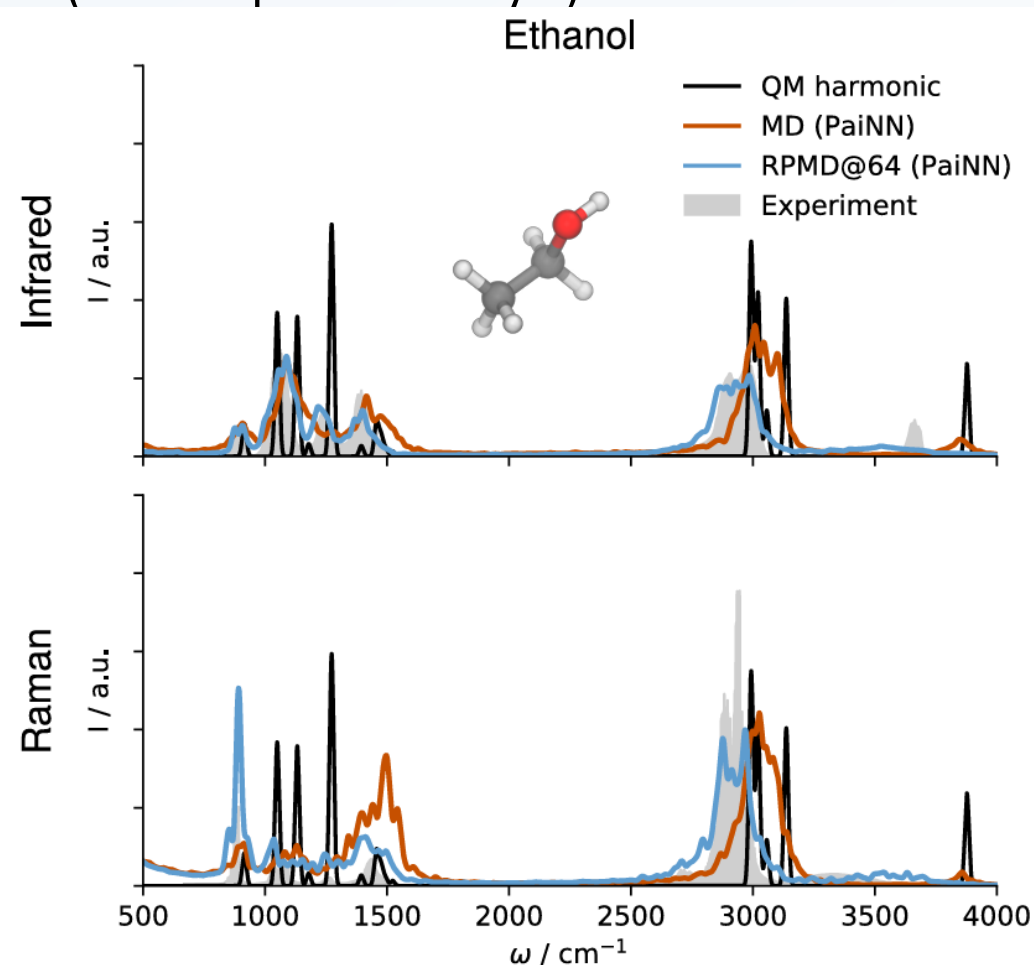
How to further improve the interaction?

UV, IR, Raman...



Passing tensorial molecular properties

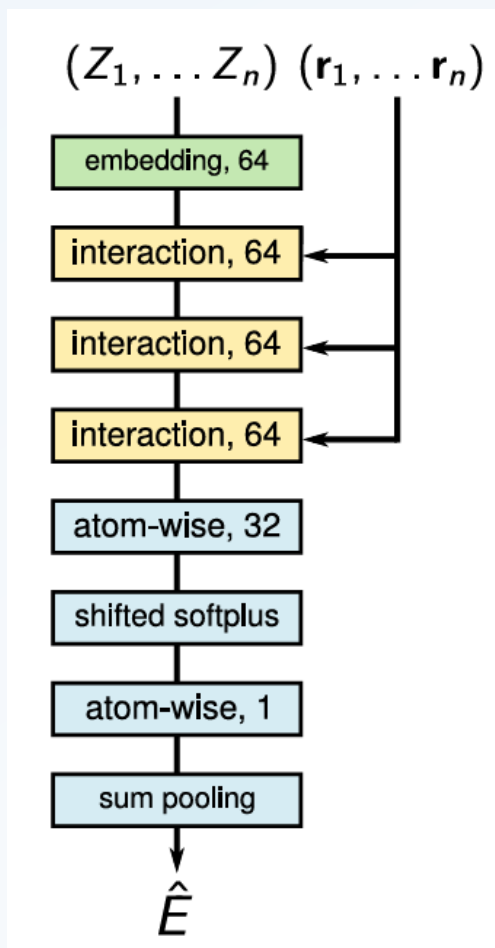
(such as polarizability...)



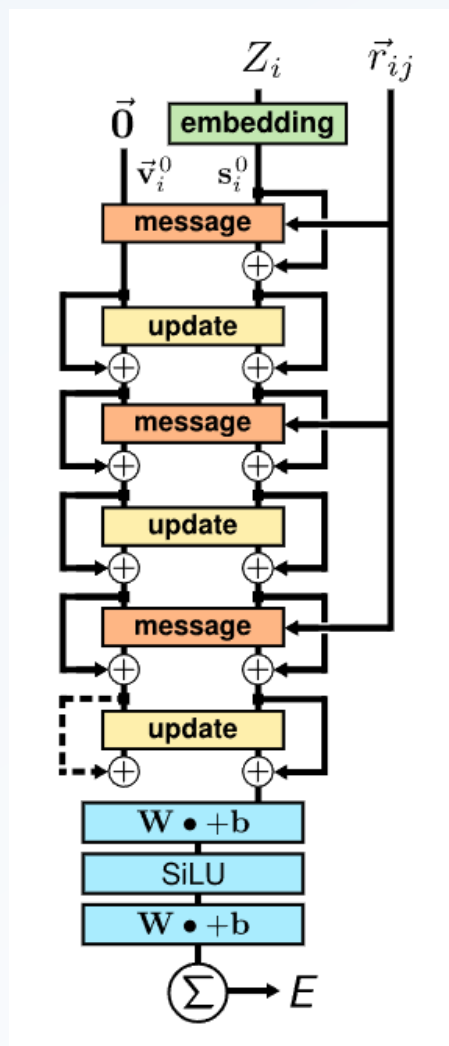
K. T. Schütt, M. Gastegger et al. ICML 2021, 9377

UV, IR, Raman...

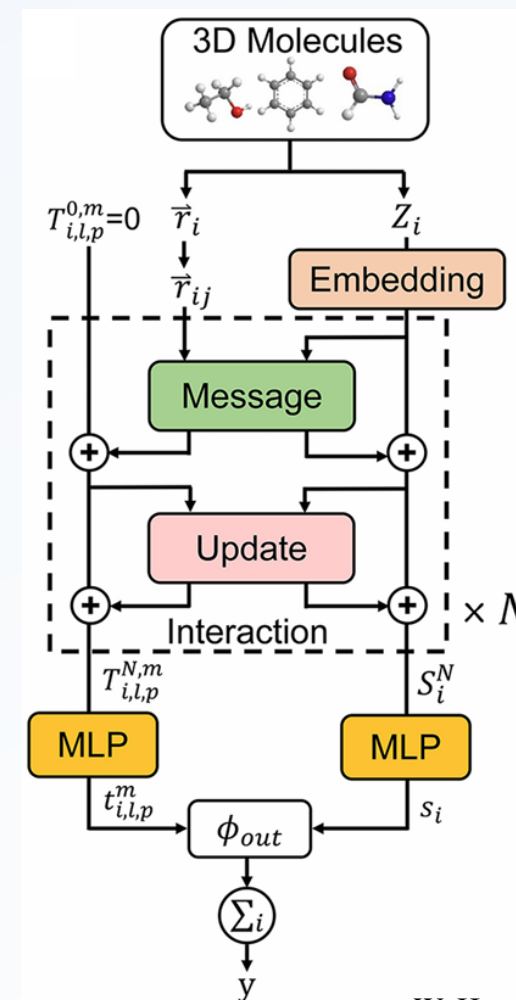
2017
SchNet



2021
PaiNN

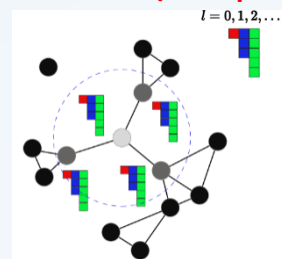


2023
DetaNet



W. Hu et al., Nat. Comput. Sci., 2023, 3, 957

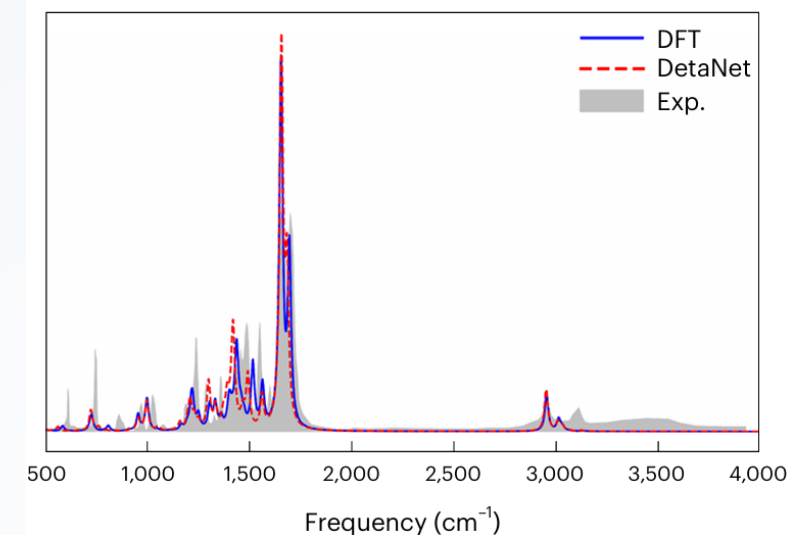
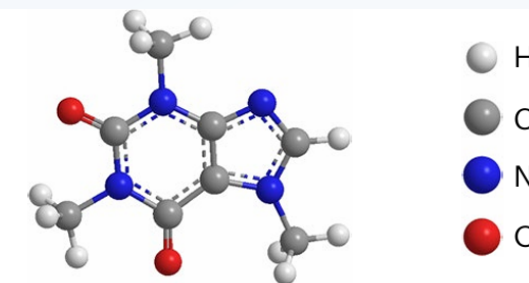
Neural Equivariant Interatomic Potentials (NequIP)



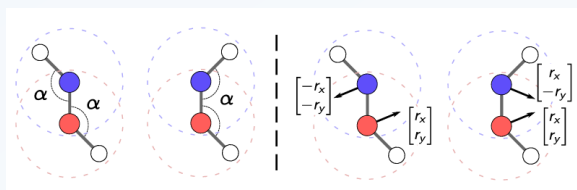
B. Kozinsky et al., Nat. Commun., 2022, 13, 2453

DetaNet

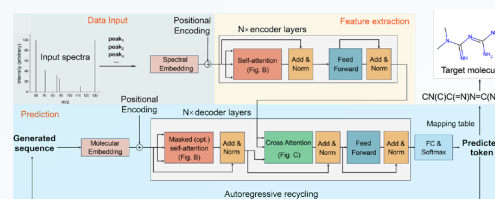
hampers efficiency. Here we develop DetaNet, a deep-learning model combining **E(3)-equivariance group** and **self-attention mechanism** to predict molecular spectra with improved efficiency and accuracy. By passing **high-order geometric tensorial messages**, DetaNet is able to generate a wide variety of molecular properties, including scalars, vectors, and second- and third-order tensors—all at the accuracy of quantum chemistry calculations.



PaiNN



Transformer



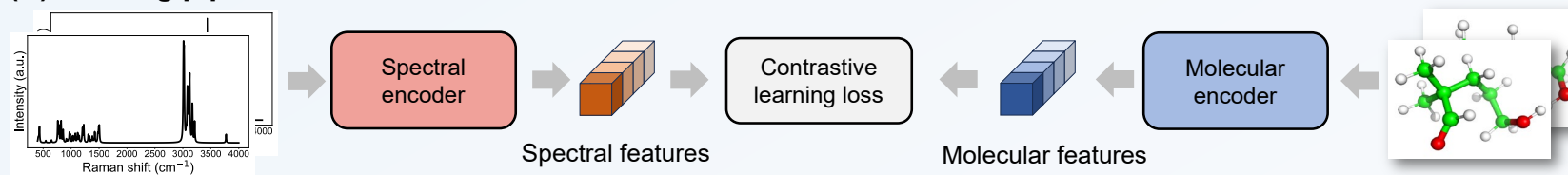
More tricks lead to higher accuracy

K. T. Schütt, M. Gastegger et al. ICML 2021, 9377

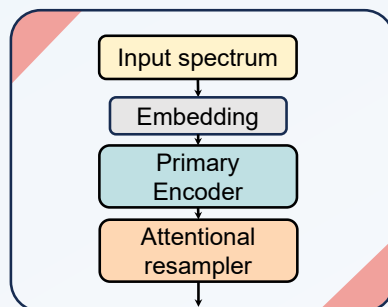
A. Vaswani et al. NeurIPS 2017, 30

W. Hu et al., Nat. Comput. Sci., 2023, 3, 957

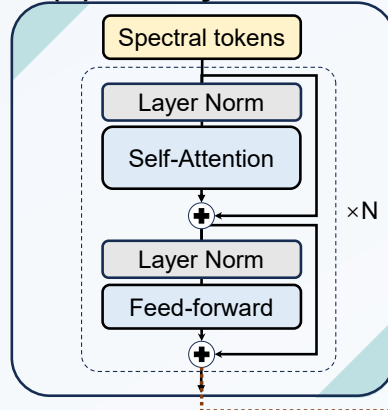
(A) Training pipeline of Vib2Conf



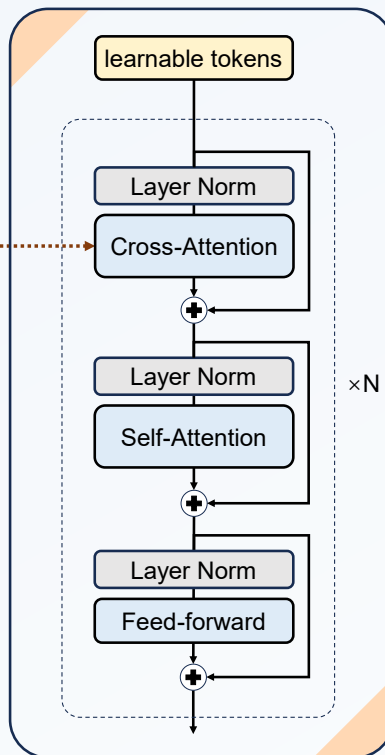
(B) Spectral encoder



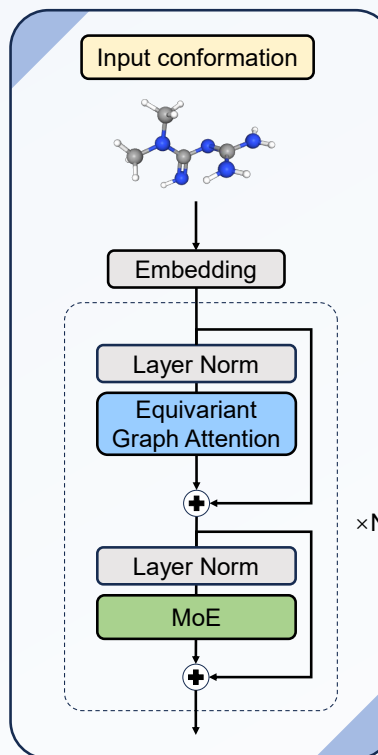
(C) Primary encoder



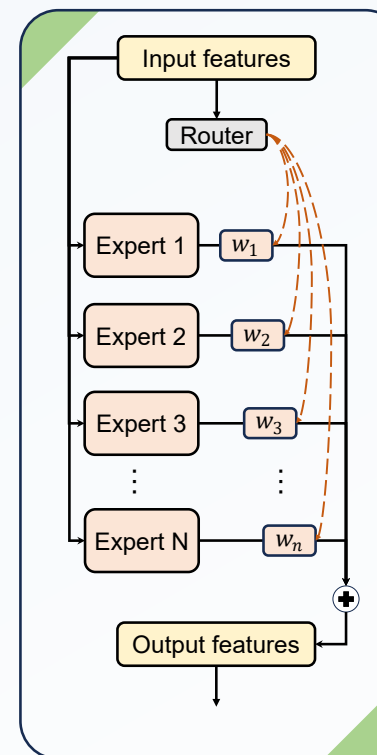
(D) Attentional resampler



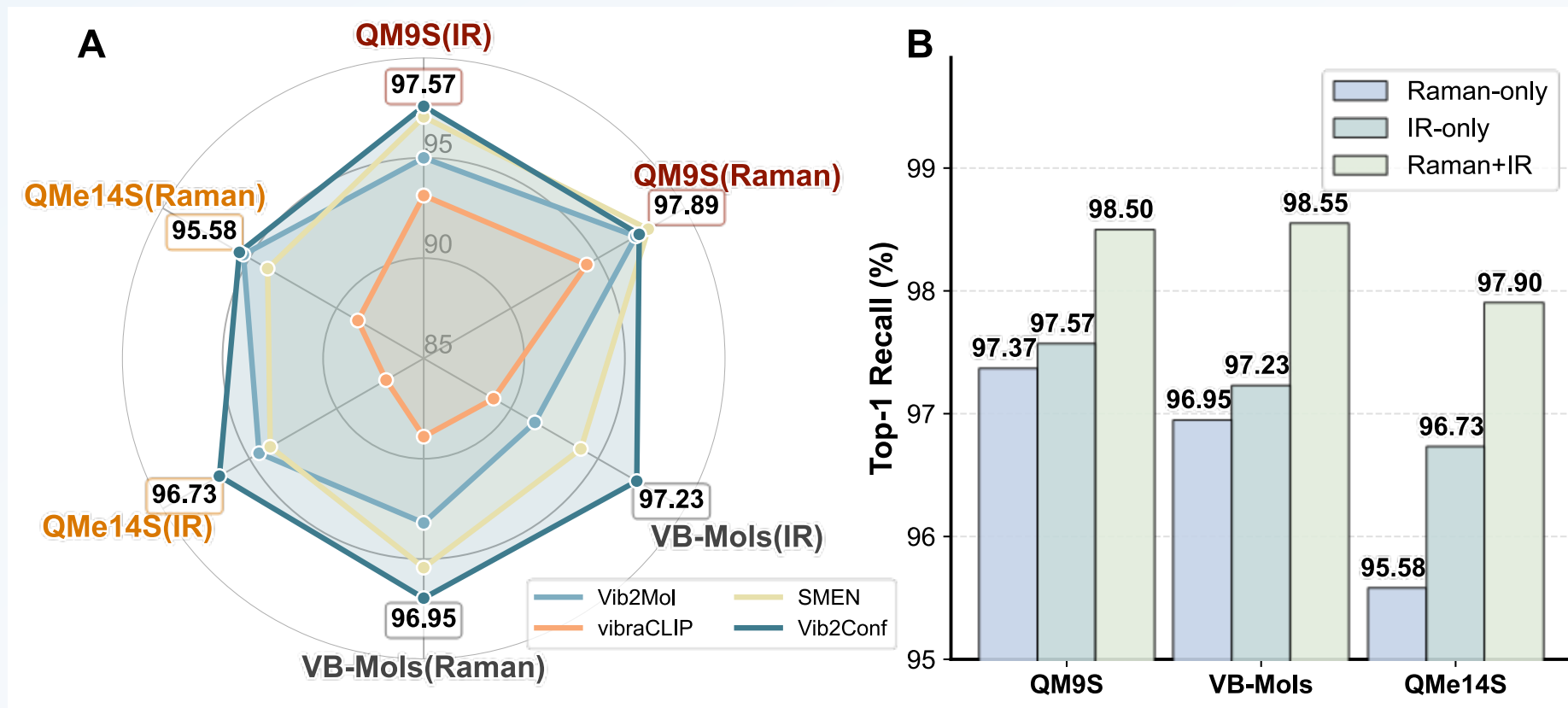
(E) Molecular encoder



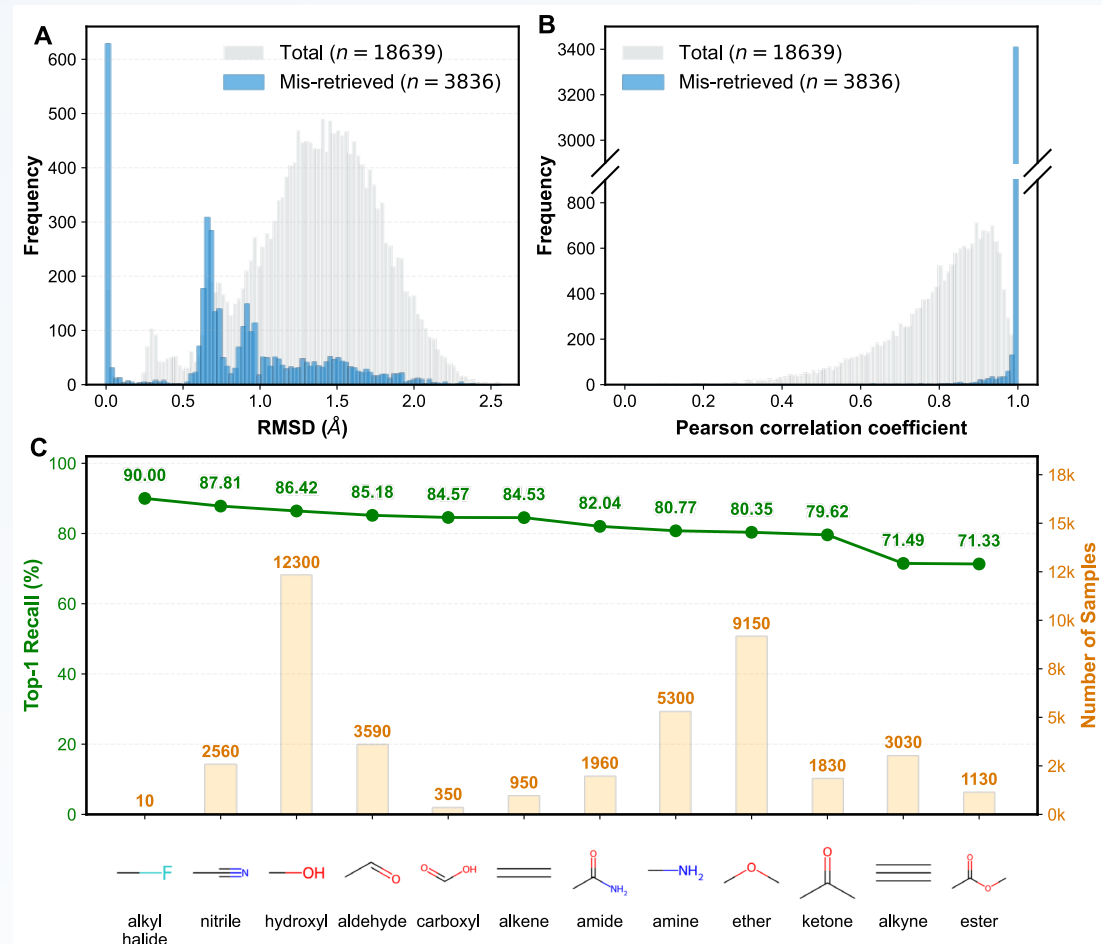
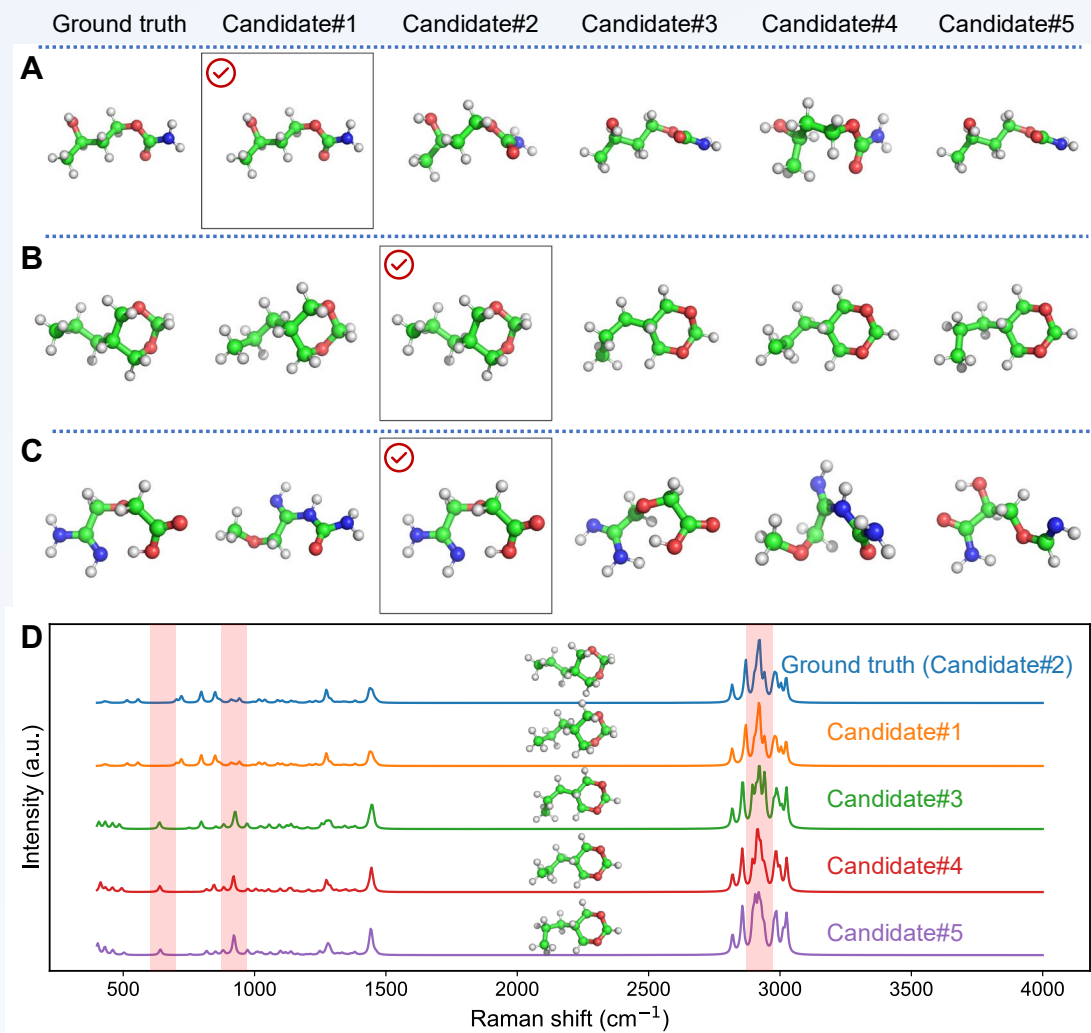
(F) Molecular MoE



UV, IR, Raman...



UV, IR, Raman...



A story of my student about “the important prior knowledge”:

Necessity of Chemometrics

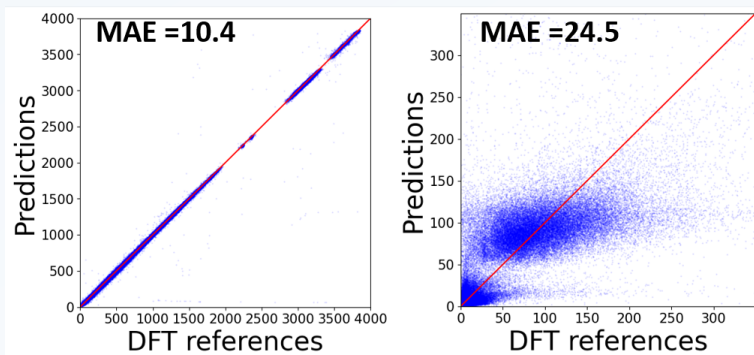
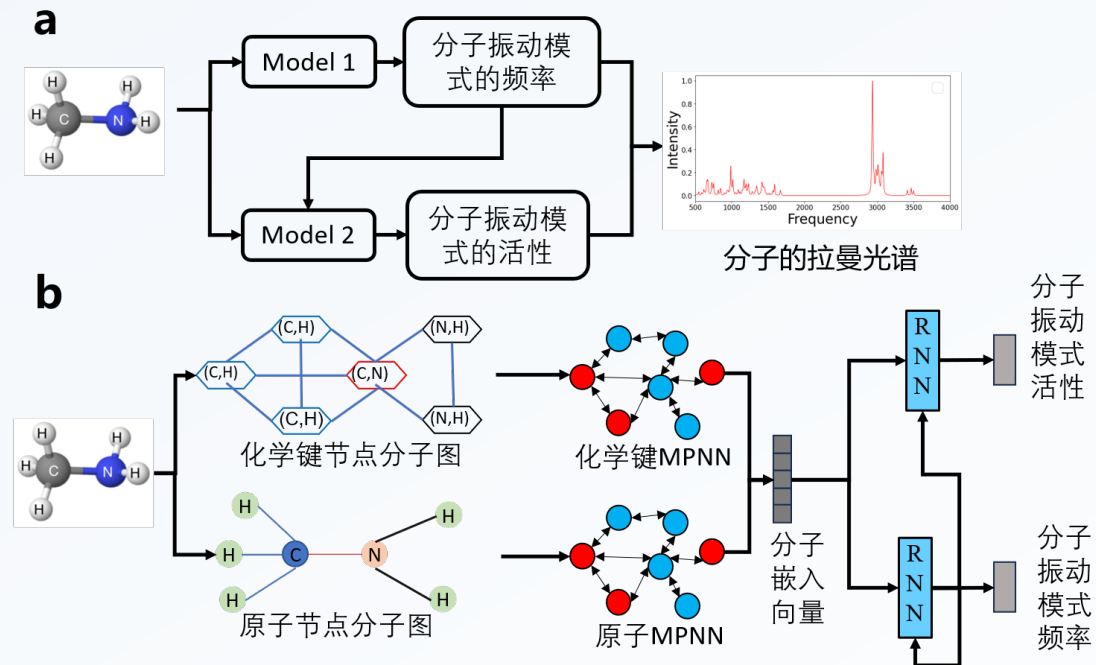
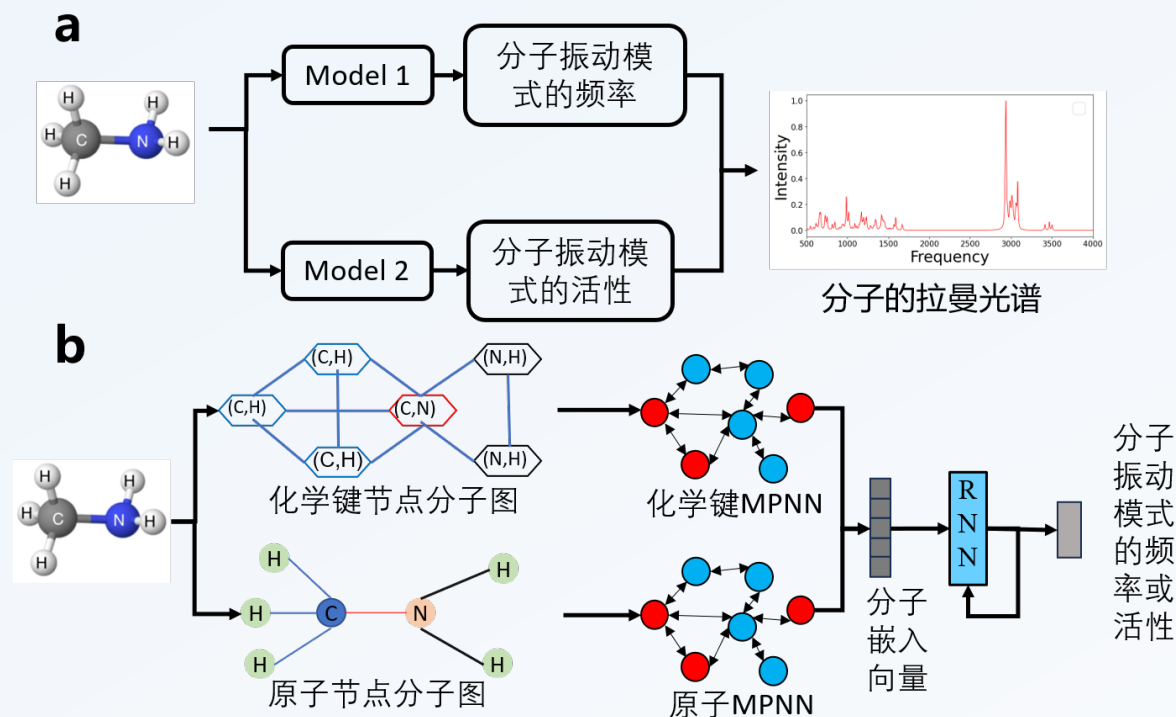
Take home messages

1. When using chemometrics, a deep understanding and mastery of the field involved in the problem or relevant chemical background should be possessed first.

2. When dealing with practical problems, it is necessary to choose the appropriate chemometric method according to the specific case, instead of using the latest or the most complicated method. In fact, some basic chemometric concepts can address many application problems.

UV, IR, Raman...

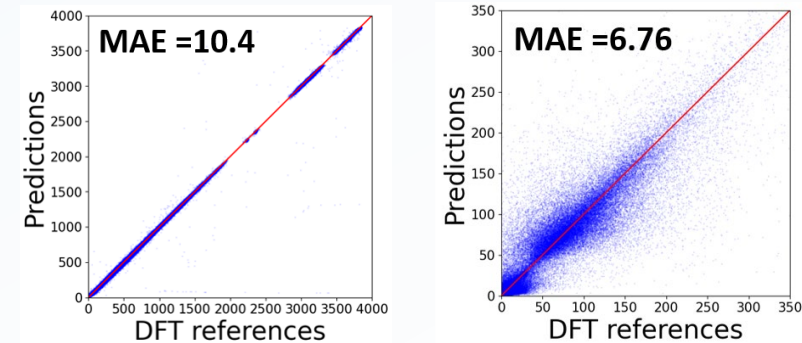
Raman Spectroscopy



Prior knowledge

→

Data Augmentation



UV, IR, Raman...

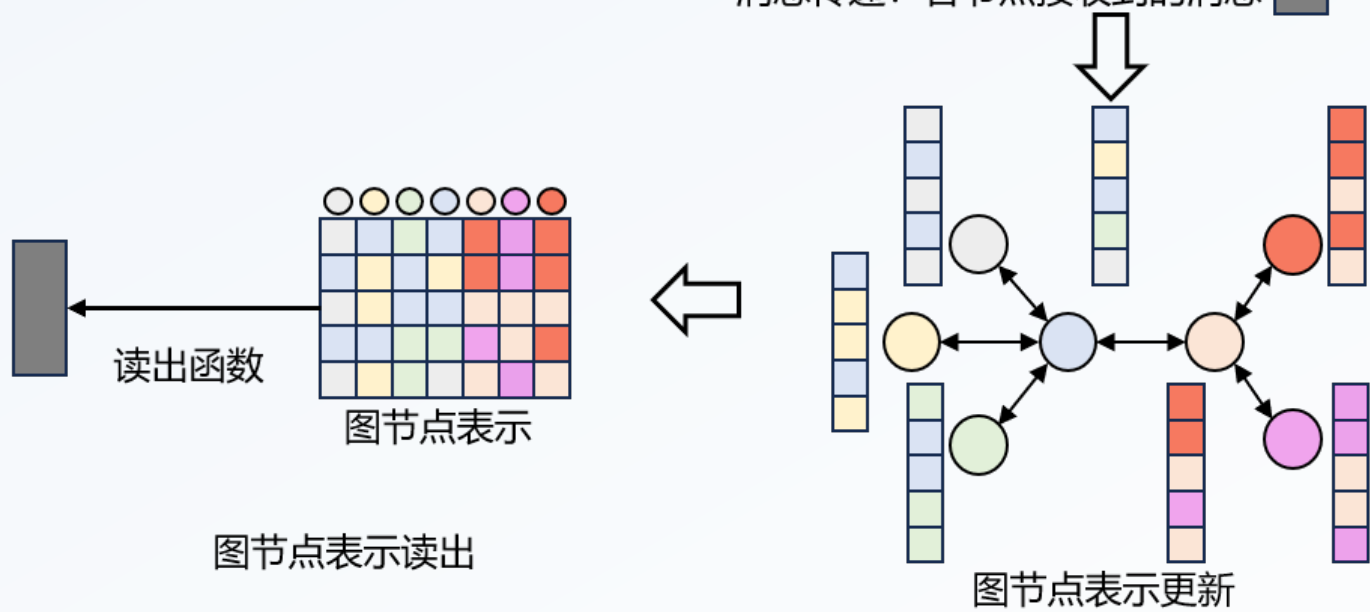
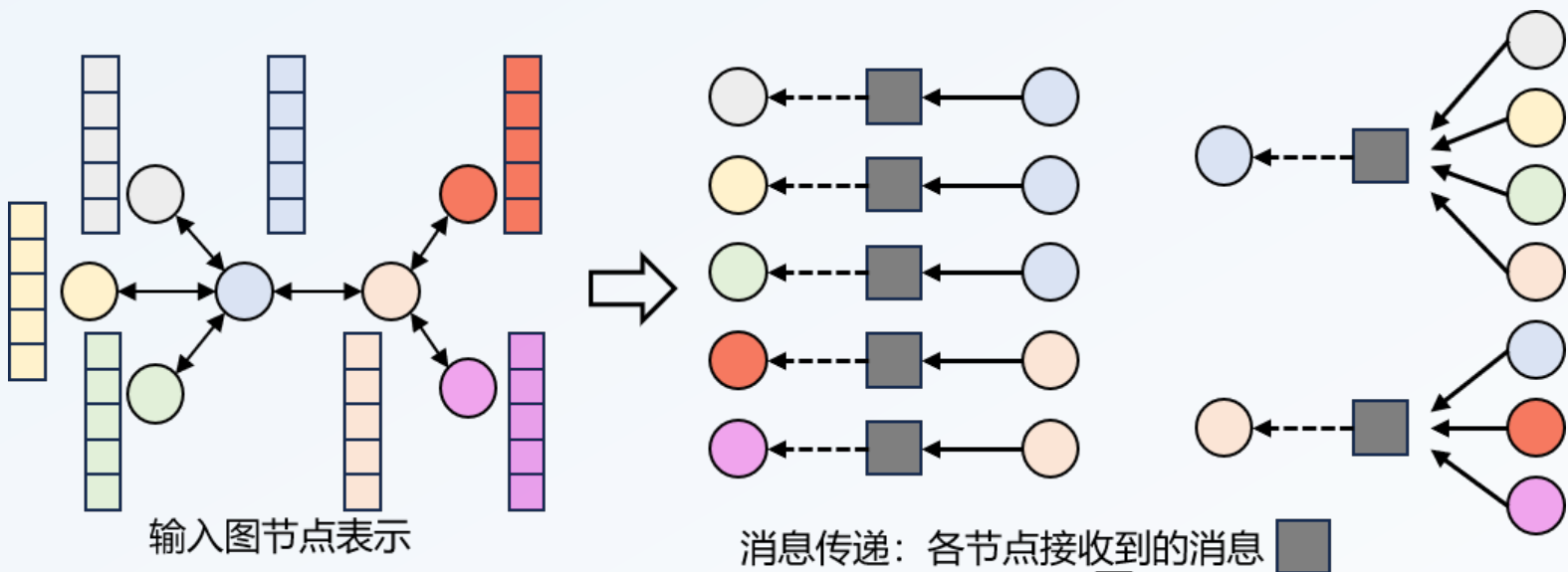
Raman Spectroscopy

Code

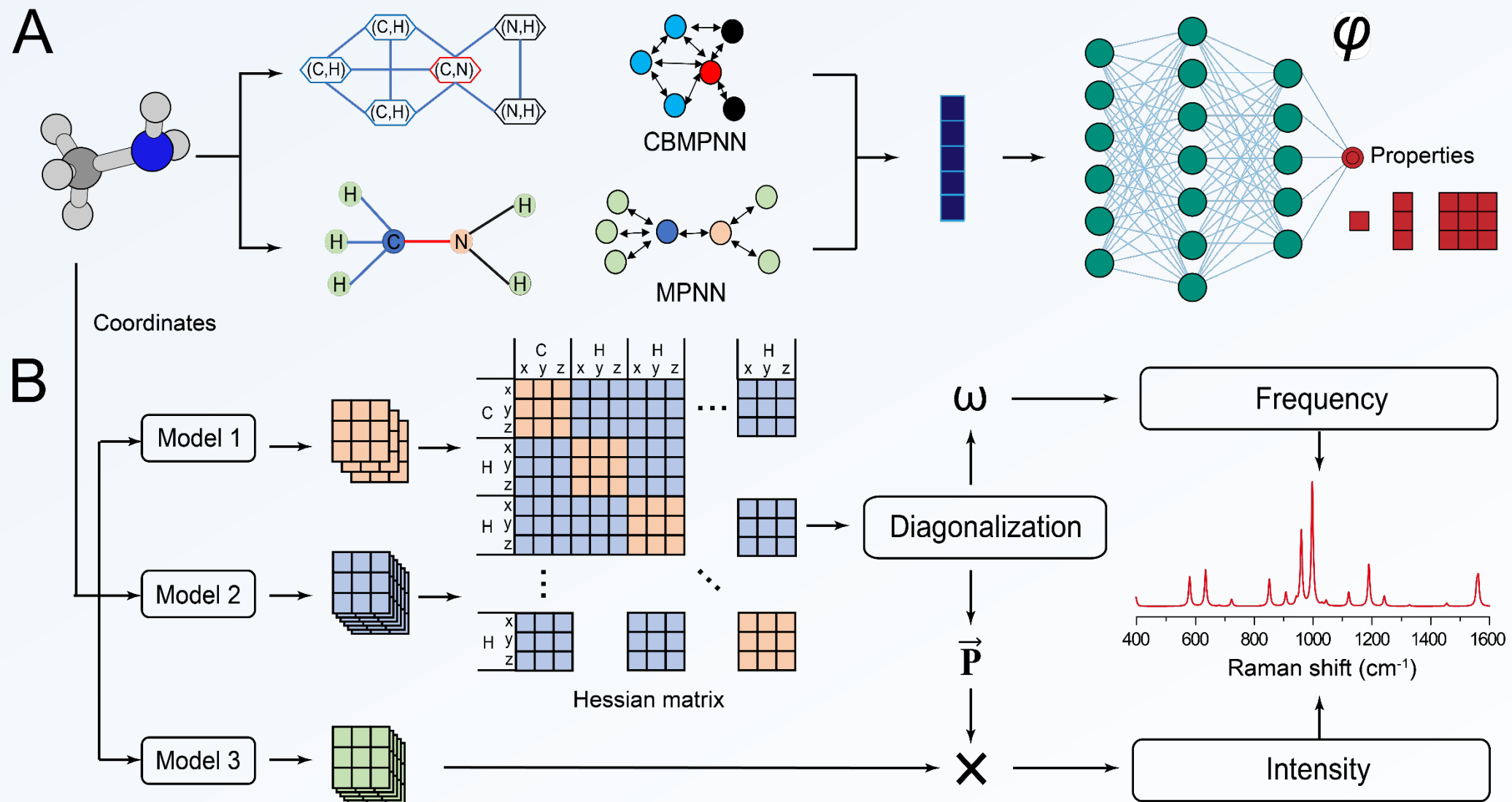
$$POI = \frac{\partial^{nF+nB+nI+nR} E}{\partial F^{nF} \partial B^{nB} \partial I^{nI} \partial R^{nR}}$$

Table 1. 对能量求导可获得的物理量⁶。

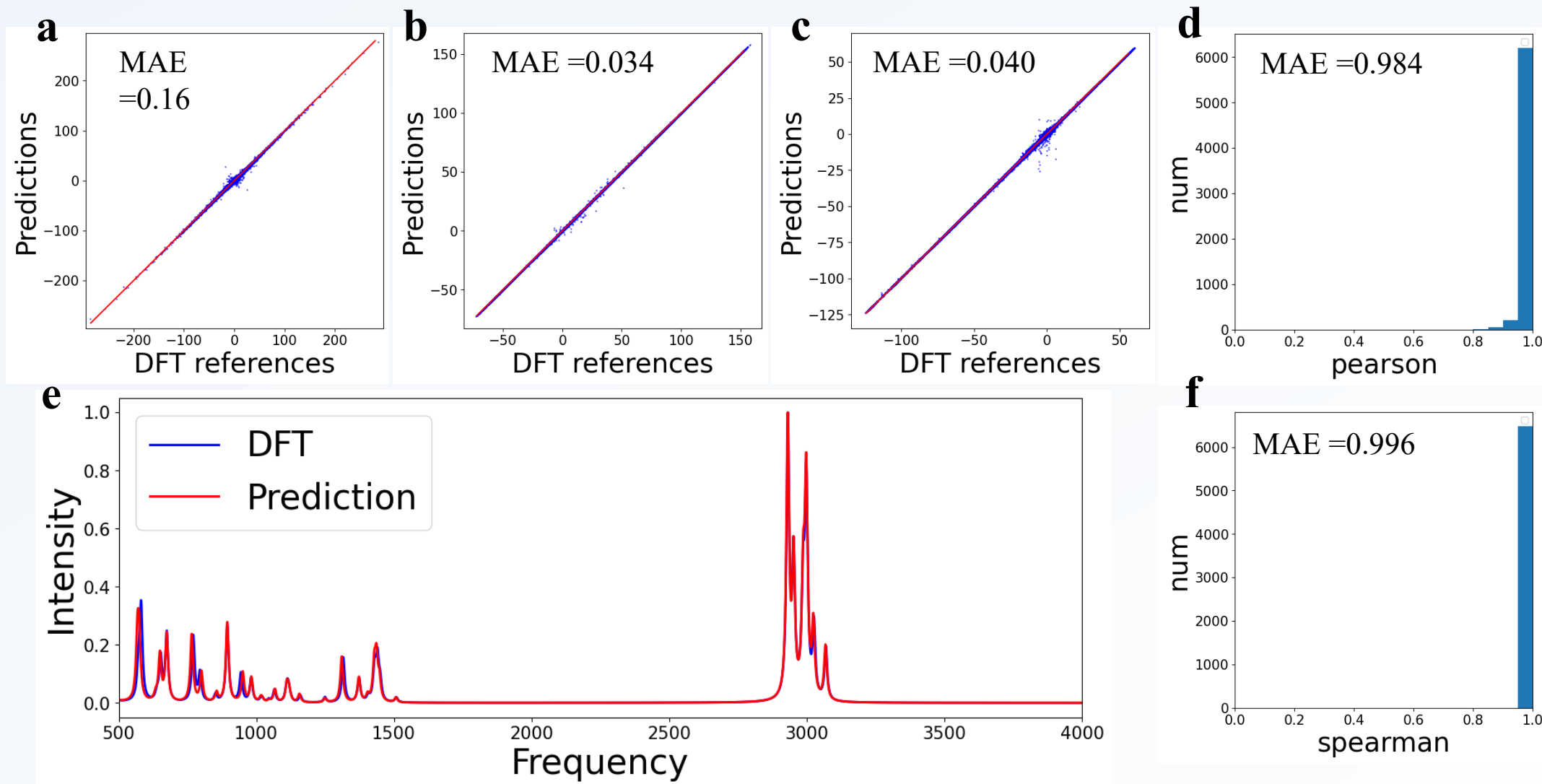
nF	nB	nI	nR	POI
1	0	0	0	电偶极矩
0	1	0	0	磁偶极矩
0	0	1	0	超精细耦合常数
0	0	0	1	分子梯度
2	0	0	0	电极化率
0	2	0	0	磁化率
0	0	2	0	核自旋耦合
0	0	0	2	谐振频率
1	0	0	1	红外吸收强度
1	1	0	0	旋光, 圆二
0	1	1	0	核自旋屏蔽效应
3	0	0	0	第一超极化率
0	3	0	0	第一超磁化率
0	0	0	3	振动非谐振校正
2	0	0	1	拉曼强度
3	0	0	1	超拉曼效应
2	1	0	0	磁性圆二
1	0	0	2	红外倍频等
4	0	0	0	第二超极化率
0	4	0	0	第二超磁化率
0	0	0	0	振动非谐振校正
2	0	0	2	拉曼倍频等
2	2	0	0	科顿-穆顿效应 (磁双折射效应)



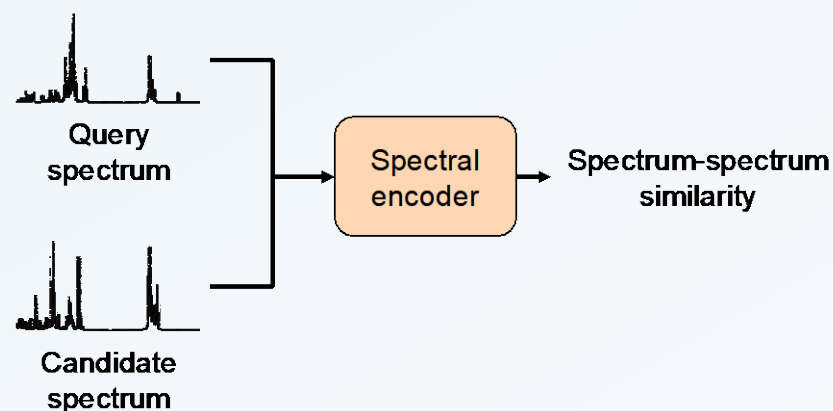
UV, IR, Raman...



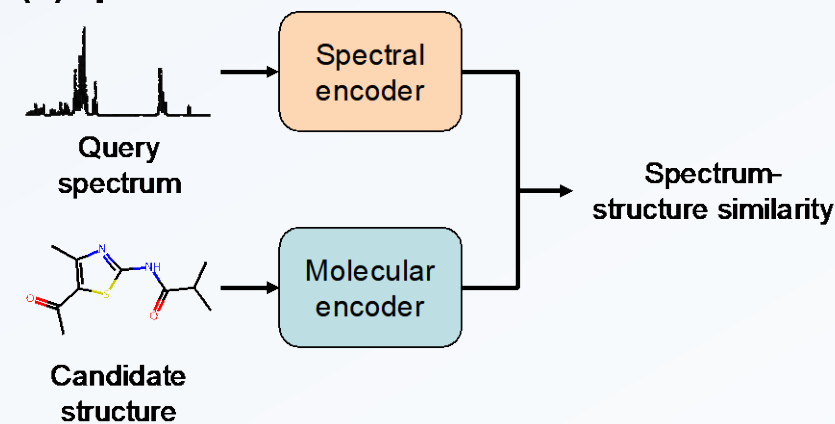
UV, IR, Raman...



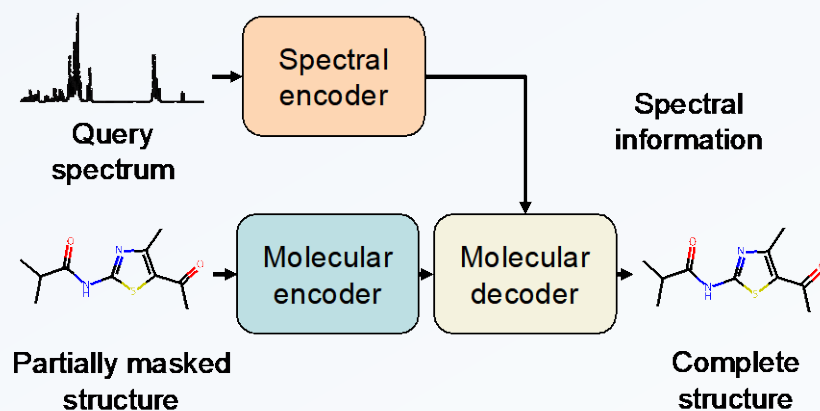
(A) Spectrum-spectrum retrieval



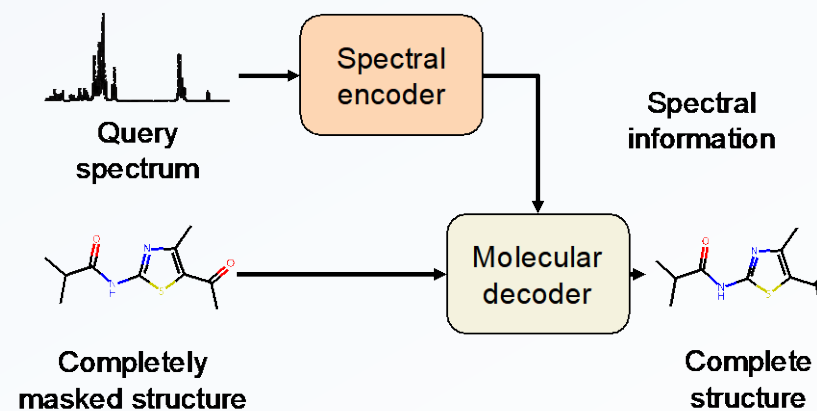
(B) Spectrum-structure retrieval



(C) Conditional generation

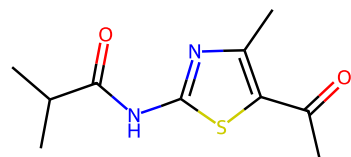
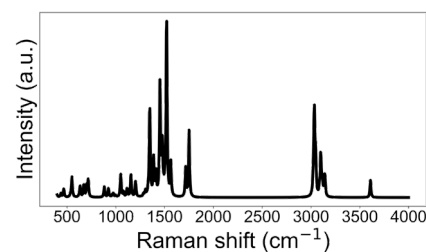


(D) De novo generation

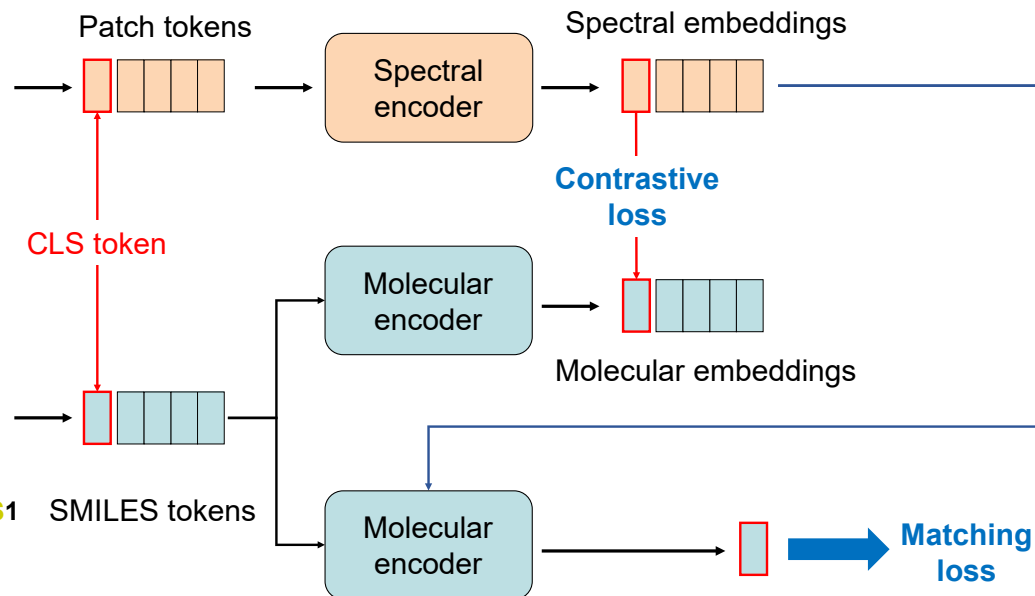


Current approaches can only handle specific tasks and overlook the connections between them. How can we develop more general methods?

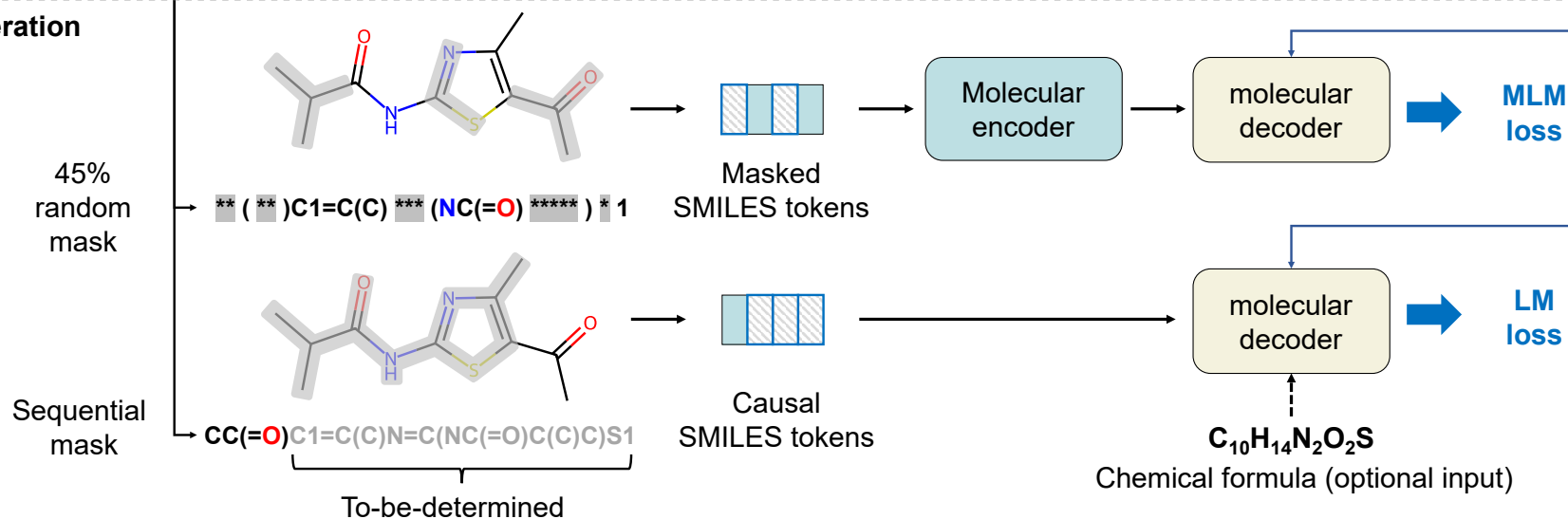
(A) Alignment



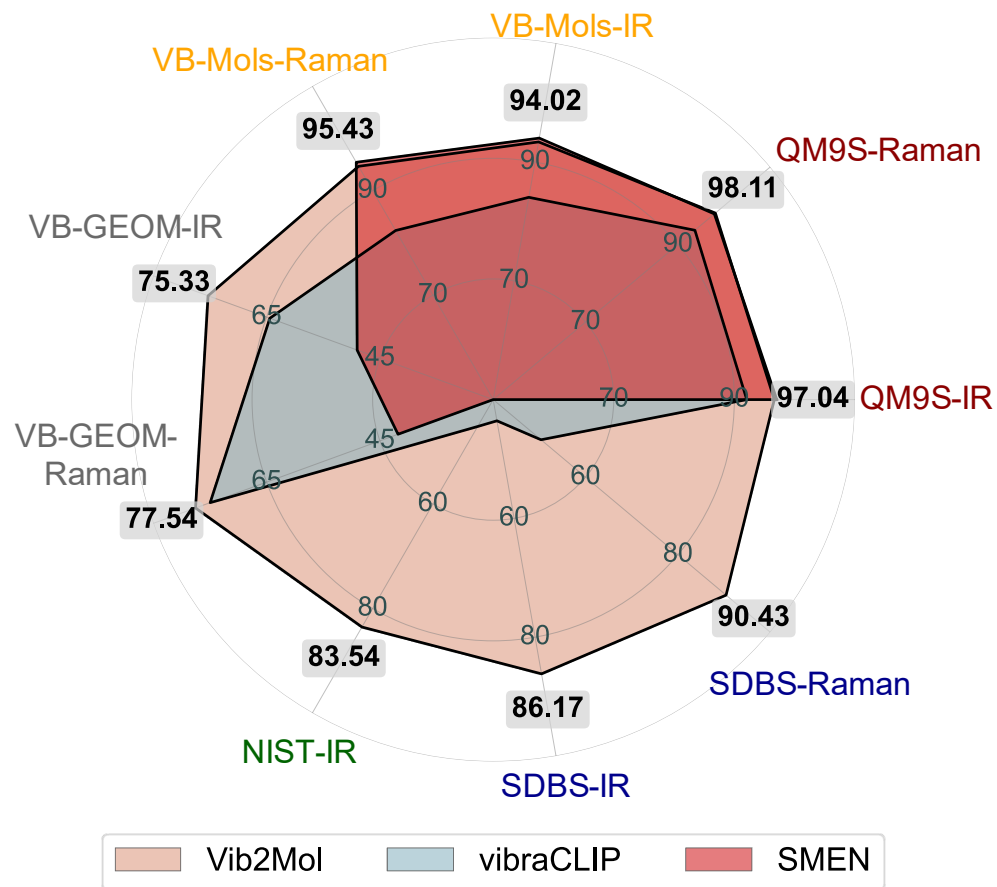
CC(=O)C1=C(C)N=C(NC(=O)C(C)C)S1



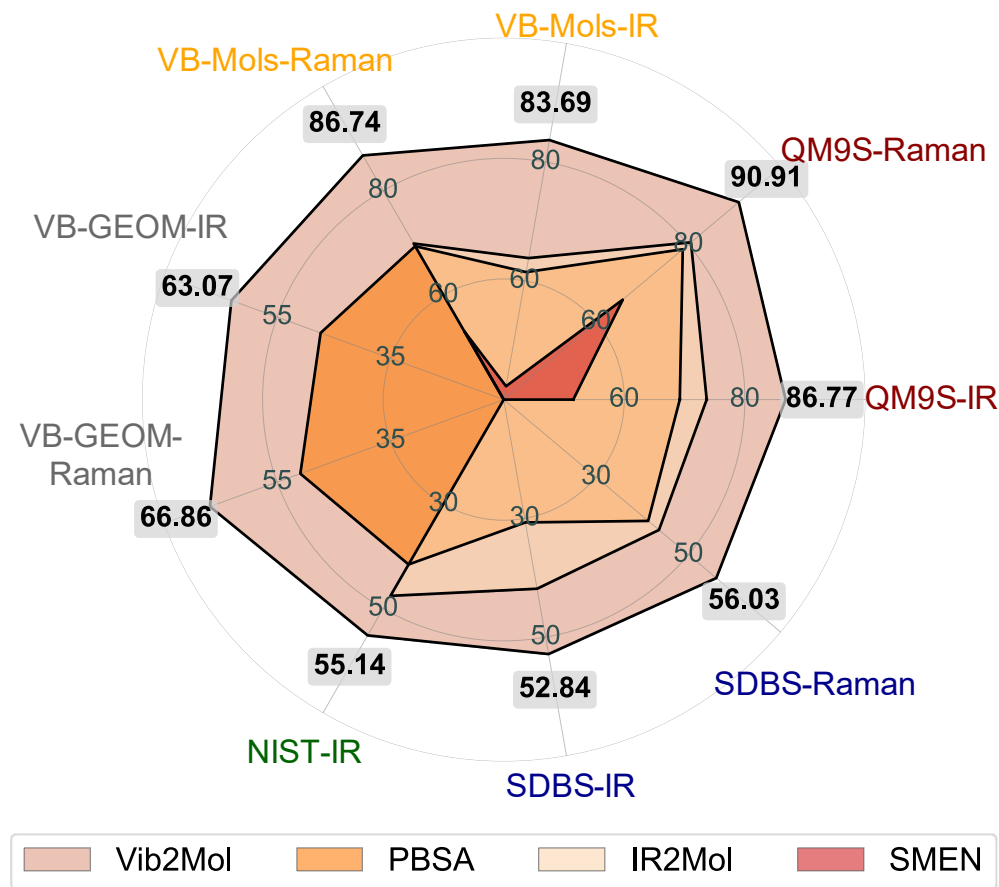
(B) Generation



(A) Performance of retrieval

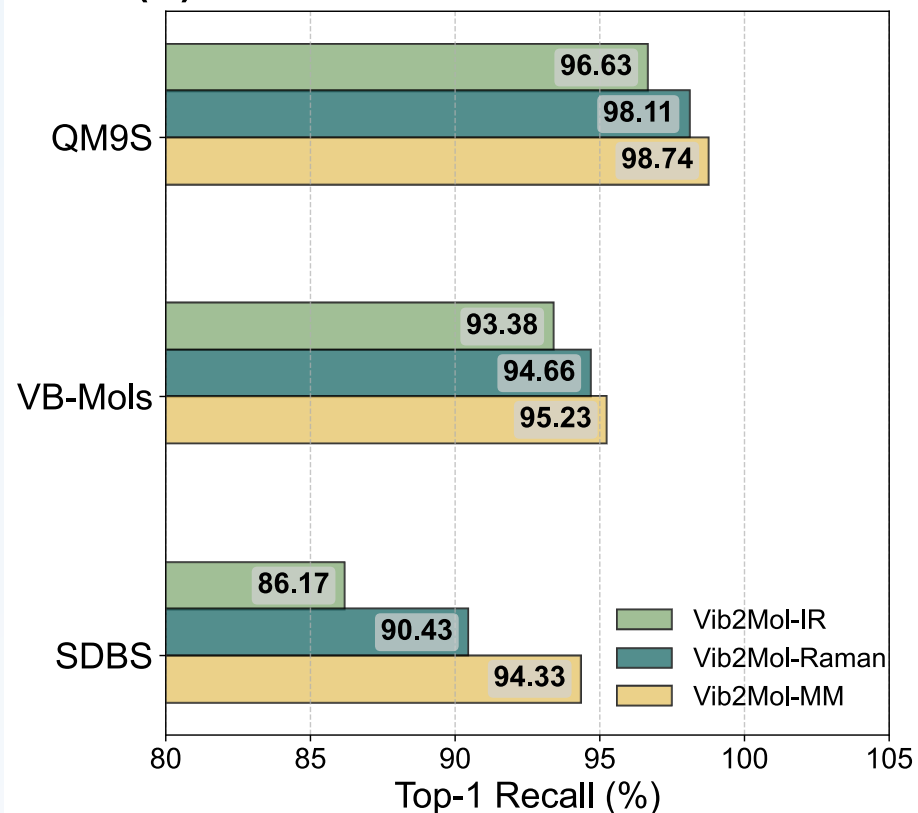


(B) Performance of de novo generation

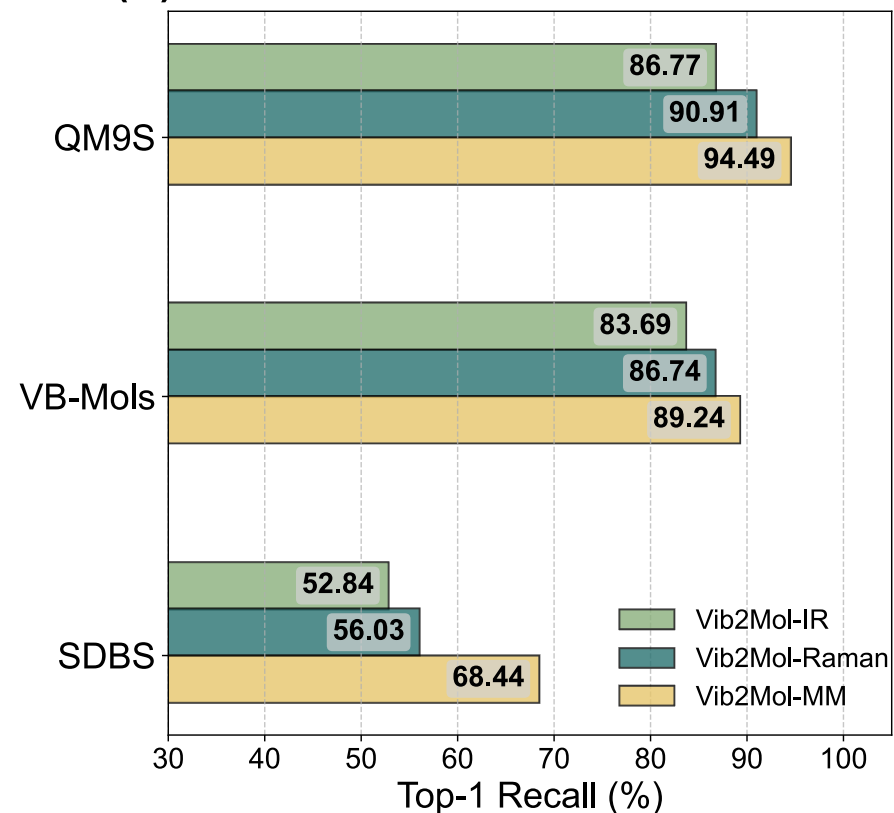


Vib2Mol achieved SOTA performance on theoretical and experimental datasets

(C) Multi-modal Retrieval Performance

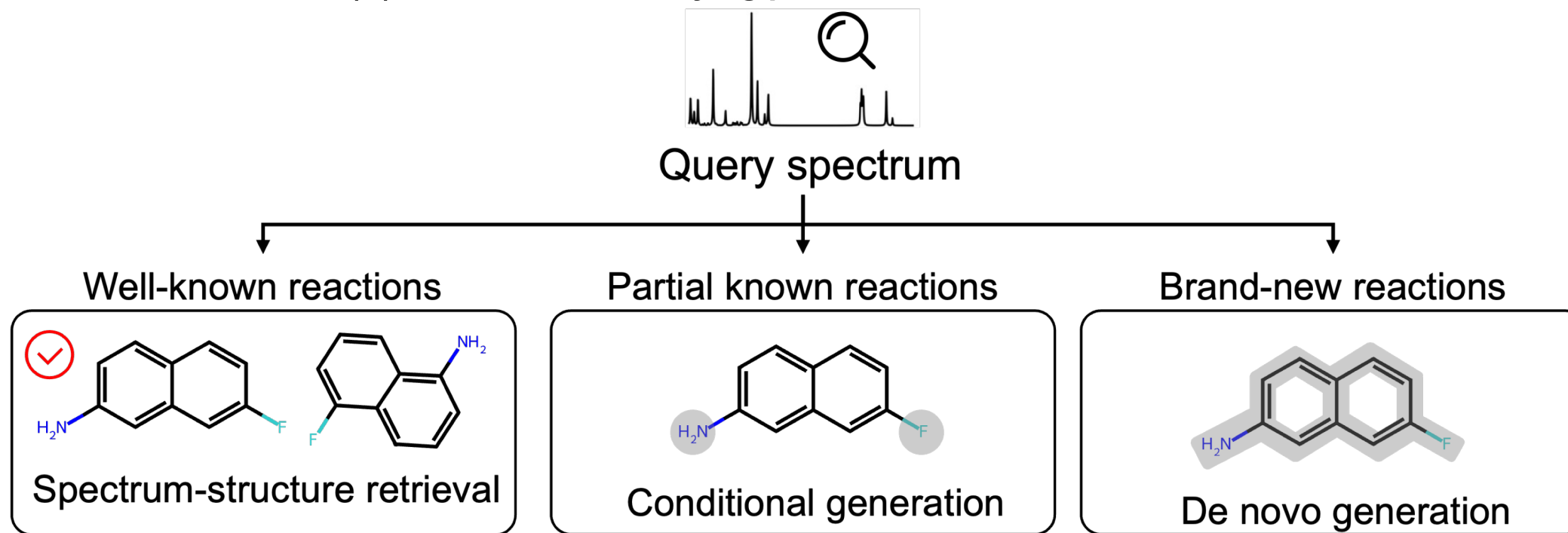


(D) Multi-modal Generation Performance

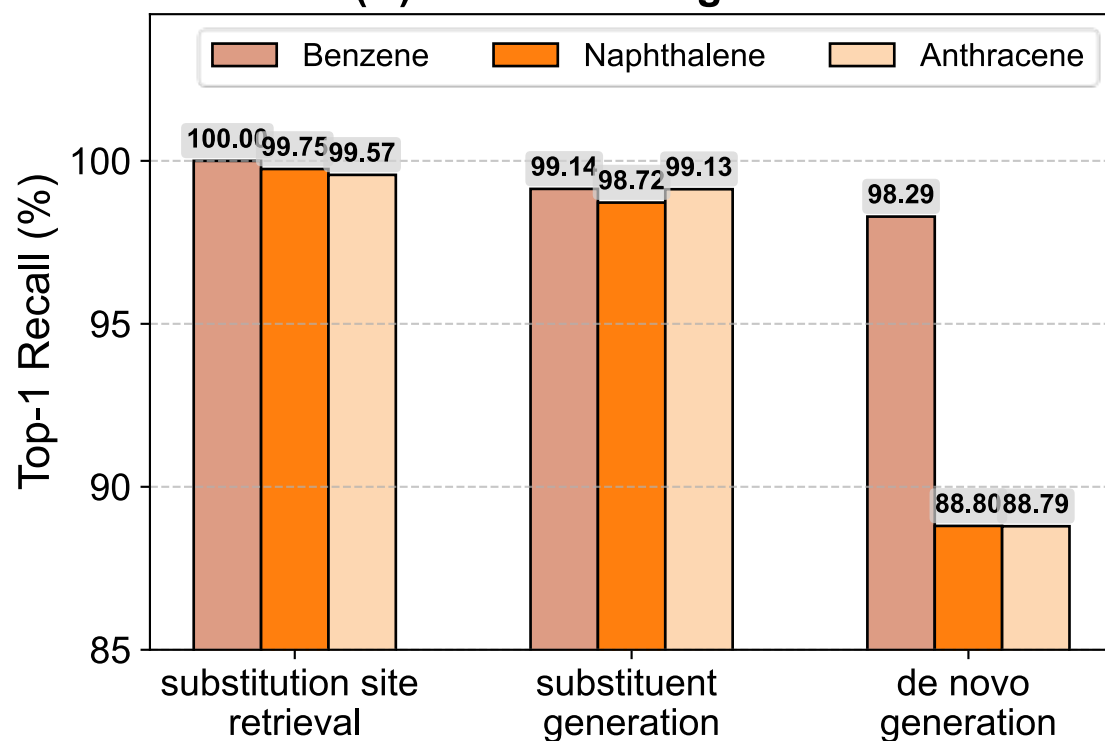


Vib2Mol can handle IR and Raman spectra at the same time

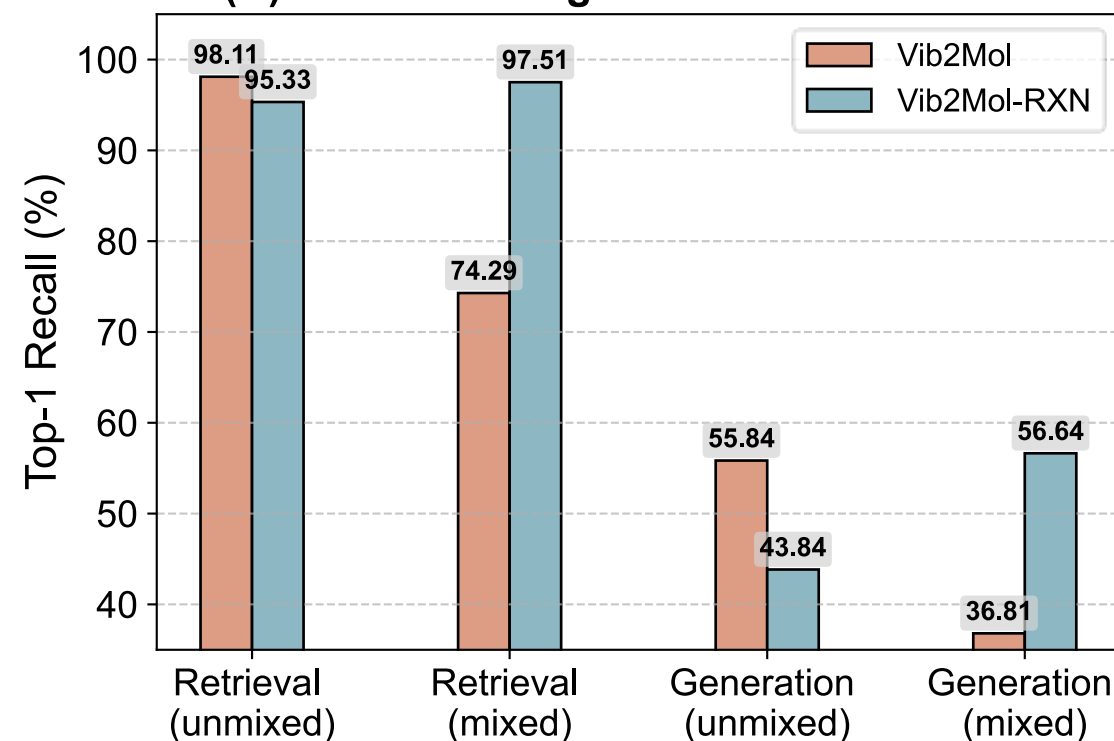
(A) Workflow of identifying products of chemical reactions



(B) Benchmarking on PAHs

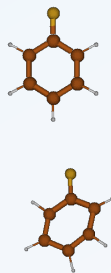


(C) Benchmarking on chemical reactions



Vib2Mol can predict products on the basis of mixed spectra

Raman Spectroscopy



Inherit all merits of Raman

Raman spectroscopy:

Limitation: Weak!

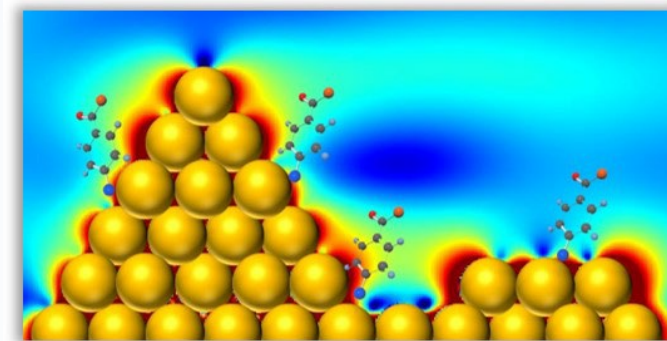
- ◆ Vibrational spectroscopy --- Structural information
- ◆ High spectral resolution --- Multicomponent analysis
- ◆ Undisturbed by water and air --- Suitable for Bio-system
- ◆ light sources are available from ultraviolet to near-infrared --- Resonance effect



Amplify the signal

Surface-enhanced Raman spectroscopy:

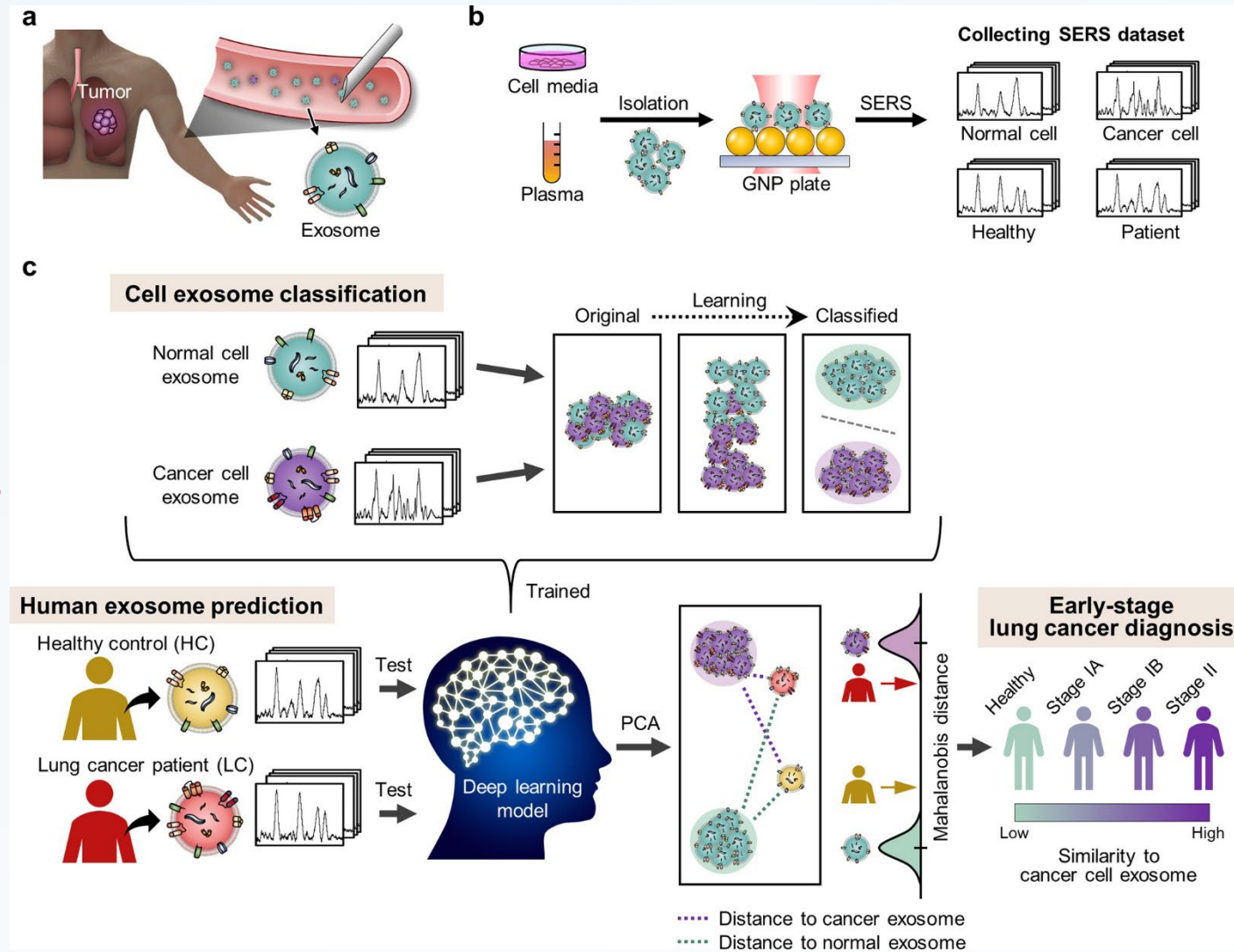
- Ultrahigh sensitivity—Enhancement factor $\sim 10^6$
- Nanostructure needed—Physical enhancement
- Special interactions with nanostructure—Chemical enhancement
- Signal is stable, cannot be quenched
- Non-invasive detection of biomolecules
- No interference from fluorescence



UV, IR, Raman...

Raman Spectroscopy

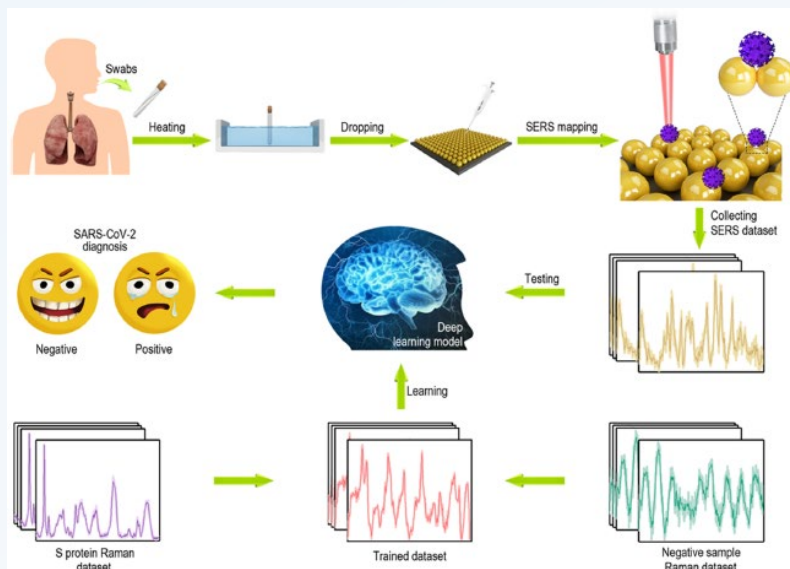
Deconvolution based on big datasets



Yeonho Choi et al., ACS Nano 2020, 14, 5, 5435-5444

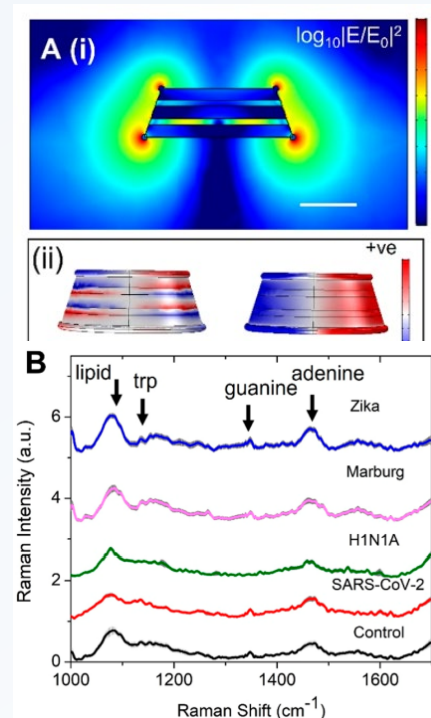
Raman Spectroscopy

20 min 87%



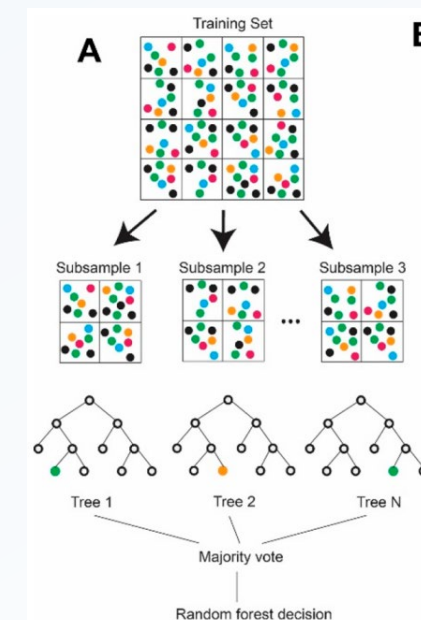
Kai Du et al., Anal. Chem. 2021, 93, 26

Limited sensitivity



Ishan et al., Nano Lett. 2022, 22, 9

Poor spectral reliability



Random forest

25 min 83%

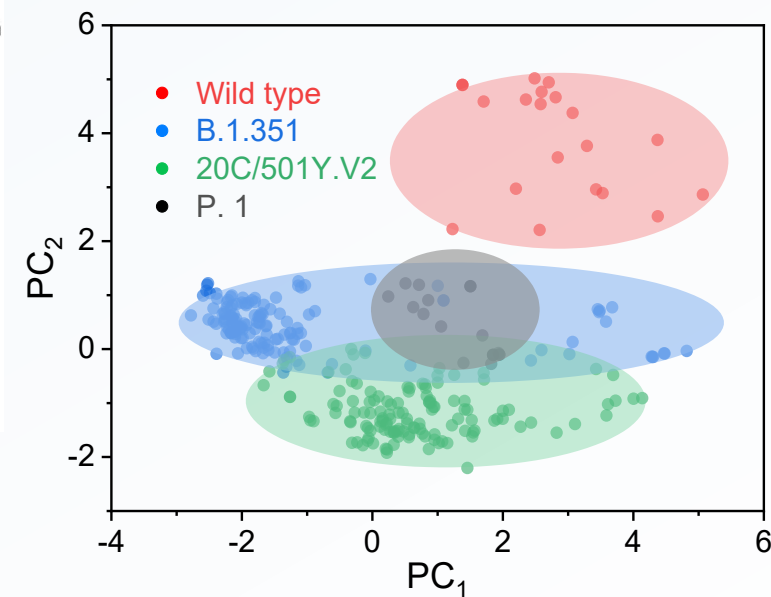
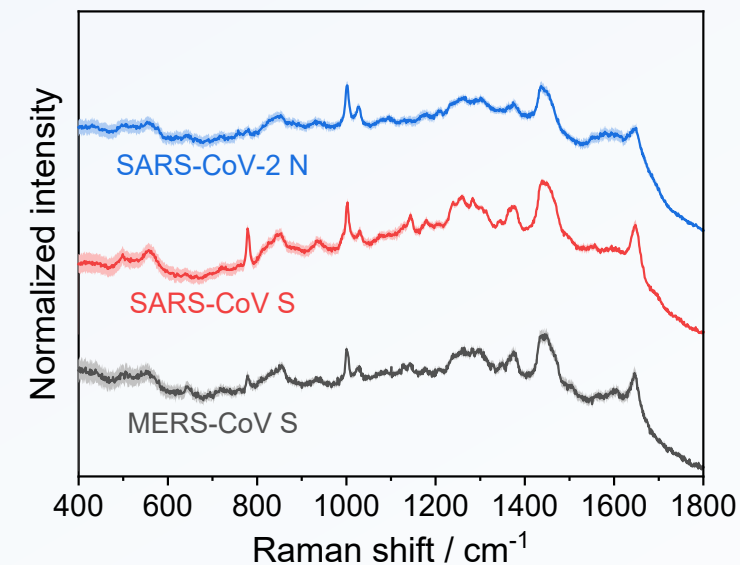
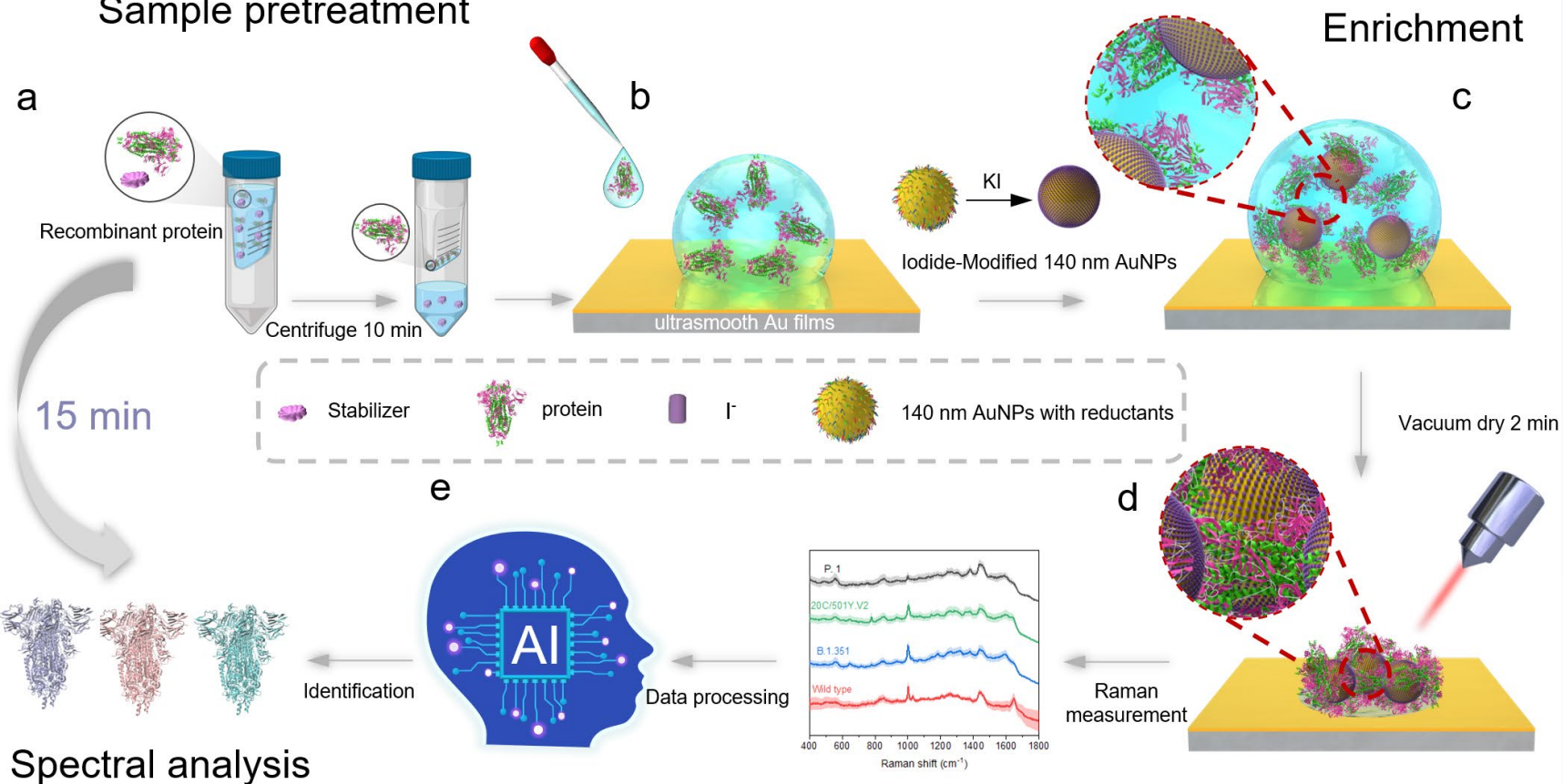
Difficulties and challenges:

- Limited by sensitivity, reproducibility, accuracy <87%
- Rapid and accurate identification of **SARS-CoV-2 variants** has not been achieved
- The **matrix complexity** of the actual COVID sample (human throat swab) was not addressed.

UV, IR, Raman...

Autoglon

Sample pretreatment



 AutoGluon

Q Search

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 Quick Examples

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Tabular

Predict the `class` column in a data table:

```
from autogluon.tabular import TabularDataset, TabularPredictor

data_root = 'https://autogluon.s3.amazonaws.com/datasets/Inc/'
train_data = TabularDataset(data_root + 'train.csv')
test_data = TabularDataset(data_root + 'test.csv')

predictor = TabularPredictor(label='class').fit(train_data=train_data)
predictions = predictor.predict(test_data)
```

Multimodal

Time Series

 Installation

python 3.8 | 3.9 | 3.10 | 3.11 pypi v1.1.0 downloads 85k/month

Install AutoGluon using `pip`:

```
pip install autogluon
```

AutoGluon supports Linux, MacOS, and Windows. See [Installing AutoGluon](#) for detailed instructions.

Homework 1